

Function of One Complex Variable I

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Chapter 1

The Complex Number System

1.1 The real numbers

1.2 The field of complex numbers

6. Let $R(z)$ be a rational function of z . Show that $\overline{R(z)} = R(\bar{z})$ if all the coefficients in $R(z)$ are real.

Solution. Let:

$$R(z) = \frac{P(z)}{Q(z)},$$

where $P(z)$ and $Q(z)$ are polynomials with real coefficients and $Q(z) \neq 0$.

$$P(z) = \sum_{k=0}^n a_k z^k, \quad a_k \in \mathbb{R}.$$

Then

$$\overline{P(z)} = \sum_{k=0}^n \overline{a_k z^k} = \sum_{k=0}^n \overline{a_k} \bar{z}^k = \sum_{k=0}^n a_k (\bar{z})^k = P(\bar{z}).$$

Similarly,

$$\overline{Q(z)} = Q(\bar{z}).$$

Therefore:

$$\overline{R(z)} = \overline{P(z)/Q(z)} = \overline{P(z)}/\overline{Q(z)} = P(\bar{z})/Q(\bar{z}) = R(\bar{z}).$$

■

1.3 The complex plane

1.4 Polar representation and roots of complex numbers

2. Calculate the following:

1. the square roots of i
2. the cube roots of i
3. the square roots of $\sqrt{3} + 3i$

Solution. To solve these questions we need solve $z^2 = i$, $z^3 = i$ and $z^2 = \sqrt{3} + 3i$. In general case we have:

$$z = \sqrt[n]{r} e^{\frac{\theta + 2k\pi}{n}i}, \quad k = 0, 1, \dots, n-1$$

Then:

1. $\sqrt{i} = e^{\frac{\pi}{4}i}, e^{\frac{5\pi}{4}i}, \quad \theta = \frac{\pi}{2}, k = 0, 1$
2. $\sqrt[3]{i} = e^{\frac{\pi}{6}i}, e^{\frac{5\pi}{6}i}, e^{\frac{9\pi}{6}i}, \quad \theta = \frac{\pi}{2}, k = 0, 1, 2$

For the third one we need to write it in polar coordinates:

$$r = 2\sqrt{3}, \quad \theta = \tan^{-1}\left(\frac{3}{\sqrt{3}}\right) = \frac{\pi}{3}.$$

So the square roots are

$$z = \sqrt{2\sqrt{3}} e^{i(\pi/6 + k\pi)}, \quad k = 0, 1.$$

■

5. Let $z = e^{\frac{2\pi}{n}i}$ for an integer $n \geq 2$. Show that $1 + z + \dots + z^{n-1} = 0$.

Solution. The equation $z^n = 1$ has exactly n distinct solutions, namely the n -th roots of unity:

$$z_k = e^{i\frac{2\pi}{n}k}, \quad k = 0, 1, \dots, n-1.$$

On the other hand, by factoring we have

$$z^n - 1 = (z - 1)(1 + z + \dots + z^{n-1}).$$

For $z = e^{i\frac{2\pi}{n}k}$ with $k = 1, \dots, n-1$, it is clear that $z \neq 1$. Hence the factor $z - 1$ is nonzero, which forces

$$1 + z + z^2 + \dots + z^{n-1} = 0.$$

In particular, this holds when $z = e^{\frac{2\pi i}{n}}$, which is the desired case. ■

1.5 Lines and half planes in the complex plane

1. Let C be the circle $\{z : |z - c| = r\}$, $r > 0$; let $a = c + re^{i\alpha}$ and put

$$L_\beta = \left\{ z : \operatorname{Im}\left(\frac{z - a}{b}\right) = 0 \right\}$$

where $b = e^{i\beta}$. Find necessary and sufficient conditions in terms of β that L_β be tangent to C at a .

Proof. The line L_β passes through the point a with slope $\tan \beta$. For L_β to be tangent to the circle C at a , it must be perpendicular to the radius drawn from the center c to the point a . Since:

$$a - c = re^{i\alpha},$$

the direction of the radius vector is given by angle α . For perpendicularity, the direction of the tangent line must differ by $\frac{\pi}{2} \pmod{\pi}$. Therefore:

$$\beta = \alpha + \frac{\pi}{2} + k\pi, \quad k \in \mathbb{Z}.$$
■

1.6 The extended plane and its spherical representation

4. Let Λ be a circle lying in S . Then there is a unique plane P in $\mathbb{R}^3$ such that $P \cap S = \Lambda$. Recall from analytic geometry that

$$P = \{(x_1, x_2, x_3) : x_1\beta_1 + x_2\beta_2 + x_3\beta_3 = \ell\}$$

where $(\beta_1, \beta_2, \beta_3)$ is a vector orthogonal to P and ℓ is some real number. It can be assumed that

$$\beta_1^2 + \beta_2^2 + \beta_3^2 = 1.$$

Use this information to show that if Λ contains the point N then its stereographic projection on $\mathbb{C}$ is a straight line. Otherwise, Λ projects onto a circle in $\mathbb{C}$.

Proof. Let $z = u + vi$ and $Z = (x_1, x_2, x_3)$ its corresponding point on the circle. Then:

$$x_1 = \frac{2u}{u^2 + v^2 + 1}, \quad x_2 = \frac{2v}{u^2 + v^2 + 1}, \quad x_3 = \frac{u^2 + v^2 - 1}{u^2 + v^2 + 1}.$$

Point Z is also belong to the P so:

$$x_1\beta_1 + x_2\beta_2 + x_3\beta_3 = \ell$$

Now substitute into the plane equation and multiply by $u^2 + v^2 + 1$:

$$2\beta_1u + 2\beta_2v + \beta_3(u^2 + v^2 - 1) = \ell(u^2 + v^2 + 1).$$

Rearranging gives

$$(\beta_3 - \ell)(u^2 + v^2) + 2\beta_1u + 2\beta_2v - (\beta_3 + \ell) = 0. \quad (1.1)$$

Case 1: $N \in \Lambda$. Then $N \in P$, so $\ell = \beta_3$, and (1.1) reduces to

$$2\beta_1u + 2\beta_2v - 2\beta_3 = 0 \quad \Longleftrightarrow \quad \beta_1u + \beta_2v = \beta_3,$$

the equation of a straight line in the (u, v) -plane. Thus if $N \in \Lambda$, the image is a line.

Case 2: $N \notin \Lambda$. Then $\ell \neq \beta_3$. Divide (1.1) by $\beta_3 - \ell$ and complete the squares:

$$\begin{aligned} u^2 + v^2 + \frac{2\beta_1}{\beta_3 - \ell}u + \frac{2\beta_2}{\beta_3 - \ell}v - \frac{\beta_3 + \ell}{\beta_3 - \ell} &= 0 \\ \Longleftrightarrow \left(u + \frac{\beta_1}{\beta_3 - \ell}\right)^2 + \left(v + \frac{\beta_2}{\beta_3 - \ell}\right)^2 &= \frac{\beta_1^2 + \beta_2^2 + \beta_3^2 - \ell^2}{(\beta_3 - \ell)^2} = \frac{1 - \ell^2}{(\beta_3 - \ell)^2}. \end{aligned}$$

Hence the image is the circle with radius $\frac{\sqrt{1-\ell^2}}{|\beta_3-\ell|}$ and center

$$\left(-\frac{\beta_1}{\beta_3 - \ell}, -\frac{\beta_2}{\beta_3 - \ell}\right).$$

■

Chapter 2

Metric Spaces and the Topology of $\mathbb{C}$

2.1 Definition and examples of metric spaces

3. If (X, d) is any metric space show that every open ball is an open set. Also, show that every closed ball is a closed set.

Proof. Fix $a \in X$ and $r \geq 0$.

(1) Open balls are open. Let $B(a, r) := \{x \in X : d(x, a) < r\}$. If $r = 0$, then $B(a, 0) = \emptyset$, which is open. Assume $r > 0$ and take any $y \in B(a, r)$, so $d(y, a) < r$. Set

$$\varepsilon := r - d(y, a) > 0.$$

We claim that $B(y, \varepsilon) \subset B(a, r)$. Indeed, if $z \in B(y, \varepsilon)$, then by the triangle inequality,

$$d(z, a) \leq d(z, y) + d(y, a) < \varepsilon + d(y, a) = r,$$

so $z \in B(a, r)$. Thus every $y \in B(a, r)$ has a neighborhood contained in $B(a, r)$, hence $B(a, r)$ is open.

(2) Closed balls are closed. Let $\overline{B}(a, r) := \{x \in X : d(x, a) \leq r\}$ and consider its complement

$$X \setminus \overline{B}(a, r) = \{x \in X : d(x, a) > r\}.$$

Take any y in this complement, so $d(y, a) > r$. Define

$$\varepsilon := \frac{d(y, a) - r}{2} > 0.$$

If $z \in B(y, \varepsilon)$, then by the reverse triangle inequality,

$$d(z, a) \geq d(y, a) - d(z, y) > d(y, a) - \varepsilon = d(y, a) - \frac{d(y, a) - r}{2} = \frac{d(y, a) + r}{2} > r.$$

Hence z also lies in the complement. Therefore $X \setminus \overline{B}(a, r)$ is open, and so $\overline{B}(a, r)$ is closed. ■

8. Let (X, d) be a metric space and $Y \subset X$.

1. If $G \subset X$ is open (in X), show that $G \cap Y$ is open in the subspace (Y, d) .
2. Conversely, if $G_1 \subset Y$ is open in (Y, d) , show there exists an open $G \subset X$ such that $G_1 = G \cap Y$.

Proof. Recall that open balls in the subspace (Y, d) are of the form

$$B_Y(y, r) = \{z \in Y : d(z, y) < r\} = B_X(y, r) \cap Y.$$

(1) Let $G \subset X$ be open. For any $y \in G \cap Y$, since G is open in X , there exists $r > 0$ with $B_X(y, r) \subset G$. Intersecting with Y gives

$$B_Y(y, r) = B_X(y, r) \cap Y \subset G \cap Y.$$

Thus each $y \in G \cap Y$ has a subspace ball contained in $G \cap Y$, so $G \cap Y$ is open in (Y, d) .

(2) Let $G_1 \subset Y$ be open in (Y, d) . For each $y \in G_1$ choose $r_y > 0$ such that $B_Y(y, r_y) \subset G_1$; equivalently, $B_X(y, r_y) \cap Y \subset G_1$. Define

$$G := \bigcup_{y \in G_1} B_X(y, r_y) \subset X.$$

Each $B_X(y, r_y)$ is open in X , hence G is open in X (union of open sets). Moreover,

$$G \cap Y = \bigcup_{y \in G_1} (B_X(y, r_y) \cap Y) = \bigcup_{y \in G_1} B_Y(y, r_y) = G_1,$$

because the last union covers G_1 and is contained in it by construction. Therefore there exists an open $G \subset X$ with $G_1 = G \cap Y$. ■

9. Do Exercise 8 with "closed" in place of "open".

Proof. (1) If F is closed in X , then $X \setminus F$ is open in X . By Exercise 8, $(X \setminus F) \cap Y$ is open in (Y, d) . Since

$$(X \setminus F) \cap Y = Y \setminus (F \cap Y),$$

the complement in Y of $F \cap Y$ is open in Y , hence $F \cap Y$ is closed in (Y, d) .

(2) Suppose $F_1 \subset Y$ is closed in (Y, d) . Then $Y \setminus F_1$ is open in (Y, d) . By

Exercise 8, there exists an open set $G \subset X$ such that

$$Y \setminus F_1 = G \cap Y.$$

Let $F := X \setminus G$. Then F is closed in X , and

$$F \cap Y = (X \setminus G) \cap Y = Y \setminus (G \cap Y) = Y \setminus (Y \setminus F_1) = F_1,$$

as desired. ■

2.2 Connectedness

4. Prove the following generalization of Lemma 2.6. If $\{D_j : j \in J\}$ is a collection of connected subsets of X and if for each $j, k \in J$ we have $D_j \cap D_k \neq \emptyset$, then $D = \bigcup_{j \in J} D_j$ is connected.

Proof. Fix an index $j_0 \in J$ and, for each $k \in J$, set

$$E_k := D_{j_0} \cup D_k.$$

Each E_k is connected because it is the union of two connected sets with nonempty intersection: indeed $D_{j_0} \cap D_k \neq \emptyset$ by hypothesis.

Moreover, the family $\{E_k : k \in J\}$ has a common point (in fact, a common subset):

$$\bigcap_{k \in J} E_k \supset D_{j_0} \neq \emptyset.$$

By Lemma 2.6 (union of connected sets with a point in common is connected), the union $\bigcup_{k \in J} E_k$ is connected. Finally,

$$\bigcup_{k \in J} E_k = \bigcup_{k \in J} (D_{j_0} \cup D_k) = D_{j_0} \cup \bigcup_{k \in J} D_k = \bigcup_{j \in J} D_j = D.$$

Therefore D is connected. ■

5. Show that if $F \subset X$ is closed and connected, then for every pair of points $a, b \in F$ and each $\varepsilon > 0$ there exist points $z_0, z_1, \dots, z_n \in F$ with $z_0 = a$, $z_n = b$, and $d(z_{k-1}, z_k) < \varepsilon$ for $1 \leq k \leq n$. Is the hypothesis that F be closed needed? If a set F satisfies this property, then F is not necessarily connected, even if F is closed. Give an example to illustrate this.

Proof. Fix $\varepsilon > 0$ and for $x \in F$, set

$$C_\varepsilon(x) := \{y \in F : \text{there exist } \varepsilon\text{-chain from } y \text{ to } x\}.$$

We claim that $C_\varepsilon(x)$ is both open and closed in the subspace F .

Open in F : If $y \in C_\varepsilon(x)$ and $y' \in F$ with $d(y, y') < \varepsilon/2$, then concatenating an ε -chain from x to y with the step $y \rightarrow y'$ yields an ε -chain from x to y' . Hence $B(y, \varepsilon/2) \cap F \subset C_\varepsilon(x)$.

Closed in F : If $y \notin C_\varepsilon(x)$ and $y' \in F$ with $d(y, y') < \varepsilon/2$, then any ε -chain from x to y' could be extended by $y' \rightarrow y$, a step of length $< \varepsilon$. This would imply $y \in C_\varepsilon(x)$, a contradiction. Thus $B(y, \varepsilon/2) \cap F \subset F \setminus C_\varepsilon(x)$, so $F \setminus C_\varepsilon(x)$ is open in F .

Therefore $C_\varepsilon(x)$ is clopen in F . Since F is connected, we must have $C_\varepsilon(x) = F$. In particular, for any $a, b \in F$, $b \in C_\varepsilon(a)$, which is exactly the desired ε -chain. The hypothesis that F be closed is *not* needed; the proof uses only connectedness. ■

2.3 Sequences and completeness

3. Show that $\text{diam } A = \text{diam } \overline{A}$.

Proof. Since $A \subset \overline{A}$, we have

$$\text{diam } A \leq \text{diam } \overline{A}.$$

For the reverse inequality, fix $x, y \in \overline{A}$. There exist sequences $(x_n), (y_n) \subset A$ with $x_n \rightarrow x$ and $y_n \rightarrow y$. By continuity of d ,

$$d(x, y) = \lim_{n \rightarrow \infty} d(x_n, y_n) \leq \sup\{d(u, v) : u, v \in A\} = \text{diam } A.$$

Taking the supremum over all $x, y \in \overline{A}$ yields $\text{diam } \overline{A} \leq \text{diam } A$. Hence $\text{diam } A = \text{diam } \overline{A}$. ■

8. Suppose $\{x_n\}$ is a Cauchy sequence and $\{x_{n_k}\}$ is a convergent subsequence. Show that $\{x_n\}$ must be convergent.

Proof. Fix $\varepsilon > 0$. Since (x_n) is Cauchy, there exists N_1 such that $d(x_m, x_n) < \varepsilon/2$ for all $m, n \geq N_1$. Since $x_{n_k} \rightarrow x$, there exists N_2 such that $d(x_{n_k}, x) < \varepsilon/2$ for all $k \geq N_2$. Let $N = \max\{N_1, n_{N_2}\}$. For any $n \geq N$ choose k with $n_k \geq n$ (possible

because (n_k) is strictly increasing and unbounded). Then

$$d(x_n, x) \leq d(x_n, x_{n_k}) + d(x_{n_k}, x) < \varepsilon/2 + \varepsilon/2 = \varepsilon.$$

Hence $x_n \rightarrow x$. Therefore the Cauchy sequence is convergent, with limit equal to the limit of its convergent subsequence. ■

2.4 Compactness

4. Show that the union of a finite number of compact sets is compact.

Proof. Let $K_1, K_2, \dots, K_n$ be compact subsets of a topological space X , and let

$$K = \bigcup_{i=1}^n K_i.$$

Consider an open cover $\mathcal{U}$ of K . Then $\mathcal{U}$ is also an open cover of each K_i . Since each K_i is compact, for every i there exists a finite subcollection $\mathcal{U}_i \subseteq \mathcal{U}$ that covers K_i . Because there are only finitely many sets $K_1, \dots, K_n$, the union

$$\mathcal{V} = \bigcup_{i=1}^n \mathcal{U}_i$$

is a finite union of finite collections, hence itself finite. Furthermore, $\mathcal{V}$ covers all of K , since each K_i is covered by $\mathcal{U}_i$. Therefore, K has a finite subcover of $\mathcal{U}$, and thus K is compact. ■

5. Let X be the set of all bounded sequences of complex numbers. That is, $\{x_n\} \in X \iff \sup\{|x_n| : n \geq 1\} < \infty$. If $x = \{x_n\}$ and $y = \{y_n\}$, define

$$d(x, y) = \sup\{|x_n - y_n| : n \geq 1\}.$$

Show that for each $x \in X$ and $\epsilon > 0$, $\overline{B}(x; \epsilon)$ is not totally bounded although it is complete. (Hint: you might have an easier time of it if you first show that you can assume $x = (0, 0, \dots)$.)

Proof. Reduction to the center 0. The metric is translation-invariant: $d(x + z, y + z) = d(x, y)$ for all $x, y, z \in X$. Hence translation by $-x$ is an isometry sending $\overline{B}(x; \epsilon)$ to $\overline{B}(0; \epsilon)$. Completeness and total boundedness are preserved by isometries, so it suffices to treat $\overline{B}(0; \epsilon)$.

Completeness. First, (X, d) is complete (i.e. ℓ^∞ is a Banach space). Let

$(x^{(m)})_{m \in \mathbb{N}}$ be Cauchy in X , with $x^{(m)} = (x_n^{(m)})_{n \geq 1}$. For each fixed n , $(x_n^{(m)})_m$ is a Cauchy sequence in $\mathbb{C}$, hence converges; define $y_n := \lim_{m \rightarrow \infty} x_n^{(m)}$ and $y = (y_n)_n$.

We claim $y \in X$ and $d(x^{(m)}, y) \rightarrow 0$. Since $(x^{(m)})$ is Cauchy, there exists M and C with $\sup_n |x_n^{(m)} - x_n^{(M)}| \leq 1$ for all $m \geq M$. As $x^{(M)}$ is bounded, so are all $x^{(m)}$ for $m \geq M$, and passing to the limit in m shows y is bounded; thus $y \in X$. Given $\delta > 0$, choose m_0 so that $\sup_n |x_n^{(m)} - x_n^{(k)}| < \delta$ for all $m, k \geq m_0$. Fix $m \geq m_0$; then for every n ,

$$|x_n^{(m)} - y_n| = \lim_{k \rightarrow \infty} |x_n^{(m)} - x_n^{(k)}| \leq \delta,$$

so $\sup_n |x_n^{(m)} - y_n| \leq \delta$. Hence $d(x^{(m)}, y) \leq \delta$ for $m \geq m_0$, proving completeness. Therefore every closed subset, in particular $\overline{B}(0; \epsilon)$, is complete.

Not totally bounded. For $k \in \mathbb{N}$, let $e^{(k)} = (0, 0, \dots, 0, \underbrace{1}_{k\text{th}}, 0, \dots)$. Then $e^{(k)} \in \overline{B}(0; 1)$ and for $i \neq j$,

$$d(e^{(i)}, e^{(j)}) = \sup_{n \geq 1} |e_n^{(i)} - e_n^{(j)}| = 1.$$

Thus the set $\{e^{(k)} : k \in \mathbb{N}\}$ is 1-separated. If $\overline{B}(0; 1)$ were totally bounded, then for $\eta = \frac{1}{2}$ it could be covered by finitely many η -balls; but a ball of radius $< \frac{1}{2}$ contains at most one of the $e^{(k)}$, so infinitely many are needed—a contradiction. Therefore $\overline{B}(0; 1)$ is not totally bounded.

Finally, scaling shows the same for any radius: $\{\epsilon e^{(k)}\} \subset \overline{B}(0; \epsilon)$ is ϵ -separated, so $\overline{B}(0; \epsilon)$ is not totally bounded. By the reduction step, $\overline{B}(x; \epsilon)$ is complete and not totally bounded for every $x \in X$. ■

6. Show that the closure of a totally bounded set is totally bounded.

Proof. Let (X, d) be a metric space and let $A \subset X$ be totally bounded. Fix $\varepsilon > 0$. Since A is totally bounded, there exist points $x_1, \dots, x_N \in X$ such that

$$A \subset \bigcup_{i=1}^N B(x_i; \frac{\varepsilon}{2}).$$

Taking closures and using $\overline{B(x; r)} \subseteq B(x; 2r)$ (or directly, in a metric space, $\overline{B(x; \varepsilon/2)} \subseteq \overline{B(x; \varepsilon/2)} \subset B(x; \varepsilon)$), we get

$$\overline{A} \subset \bigcup_{i=1}^N \overline{B(x_i; \frac{\varepsilon}{2})} \subset \bigcup_{i=1}^N B(x_i; \varepsilon).$$

Thus $\overline{A}$ is covered by finitely many ε -balls. Since $\varepsilon > 0$ was arbitrary, $\overline{A}$ is totally bounded. ■

2.5 Continuity

3. We say that $f : X \rightarrow \mathbb{C}$ is bounded if there is $M > 0$ with $|f(x)| \leq M$ for all $x \in X$. Show that if f and g are bounded uniformly continuous (Lipschitz) functions from X into $\mathbb{C}$, then so is fg .

Proof. Let (X, d) be a metric space and let $f, g : X \rightarrow \mathbb{C}$ be bounded. Set $M_f := \sup_{x \in X} |f(x)| < \infty$ and $M_g := \sup_{x \in X} |g(x)| < \infty$. For all $x, y \in X$ we have

$$\begin{aligned} |f(x)g(x) - f(y)g(y)| &\leq |f(x)| |g(x) - g(y)| + |g(y)| |f(x) - f(y)| \\ &\leq M_f |g(x) - g(y)| + M_g |f(x) - f(y)|. \end{aligned} \quad (*)$$

Uniform continuity. Assume f and g are uniformly continuous. Fix $\varepsilon > 0$ and let $A := \max\{1, M_f\}$, $B := \max\{1, M_g\}$. By uniform continuity, there exist $\delta_1, \delta_2 > 0$ such that

$$d(x, y) < \delta_1 \implies |g(x) - g(y)| < \frac{\varepsilon}{2A}, \quad d(x, y) < \delta_2 \implies |f(x) - f(y)| < \frac{\varepsilon}{2B}.$$

With $\delta := \min\{\delta_1, \delta_2\}$, inequality $(*)$ gives, for $d(x, y) < \delta$,

$$|f(x)g(x) - f(y)g(y)| \leq M_f \frac{\varepsilon}{2A} + M_g \frac{\varepsilon}{2B} \leq \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Hence fg is uniformly continuous.

Lipschitz case. Assume in addition that f and g are Lipschitz with constants L_f, L_g : $|f(x) - f(y)| \leq L_f d(x, y)$ and $|g(x) - g(y)| \leq L_g d(x, y)$. Then by $(*)$,

$$|f(x)g(x) - f(y)g(y)| \leq M_f L_g d(x, y) + M_g L_f d(x, y) = (M_f L_g + M_g L_f) d(x, y),$$

so fg is Lipschitz with constant $M_f L_g + M_g L_f$.

Thus in either case, the product of bounded uniformly continuous (respectively Lipschitz) functions is uniformly continuous (respectively Lipschitz). ■

4. Is the composition of two uniformly continuous (Lipschitz) functions again uniformly continuous (Lipschitz)?

Proof. Let $(X, d_X), (Y, d_Y), (Z, d_Z)$ be metric spaces. Suppose $f : X \rightarrow Y$ and $g : Y \rightarrow Z$.

Uniform continuity is preserved under composition. Assume f and g are uniformly continuous. Given $\varepsilon > 0$, by uniform continuity of g there exists $\eta > 0$

such that

$$d_Y(y_1, y_2) < \eta \implies d_Z(g(y_1), g(y_2)) < \varepsilon.$$

By uniform continuity of f , there exists $\delta > 0$ such that

$$d_X(x_1, x_2) < \delta \implies d_Y(f(x_1), f(x_2)) < \eta.$$

Therefore, for $d_X(x_1, x_2) < \delta$ we have

$$d_Z(g \circ f(x_1), g \circ f(x_2)) = d_Z(g(f(x_1)), g(f(x_2))) < \varepsilon,$$

so $g \circ f$ is uniformly continuous.

Lipschitz constants multiply. Assume f is Lipschitz with constant L_f and g is Lipschitz with constant L_g :

$$d_Y(f(x_1), f(x_2)) \leq L_f d_X(x_1, x_2), \quad d_Z(g(y_1), g(y_2)) \leq L_g d_Y(y_1, y_2).$$

Then for all $x_1, x_2 \in X$,

$$d_Z(g \circ f(x_1), g \circ f(x_2)) \leq L_g d_Y(f(x_1), f(x_2)) \leq L_g L_f d_X(x_1, x_2),$$

so $g \circ f$ is Lipschitz with constant $L_g L_f$ (hence uniformly continuous). ■

6. Recall the definition of a dense set (1.14). Suppose that Ω is a complete metric space and that $f : (D, d) \rightarrow (\Omega, \rho)$ is uniformly continuous, where D is dense in (X, d) . Use Exercise 5 to show that there is a uniformly continuous function $g : X \rightarrow \Omega$ with $g(x) = f(x)$ for every $x \in D$.

Proof. **Step 1: Define g by limits.** Since D is dense in X , for each $x \in X$ there exists a sequence $(x_n) \subset D$ with $x_n \rightarrow x$ in (X, d) . Because f is uniformly continuous, $(f(x_n))$ is a Cauchy sequence in (Ω, ρ) (Exercise 5). By completeness of Ω , the $\lim_{n \rightarrow \infty} f(x_n)$ exists. Define

$$g(x) := \lim_{n \rightarrow \infty} f(x_n).$$

Step 2: Well-definedness (independence of the approximating sequence). Suppose (x_n) and (y_n) are two sequences in D with $x_n \rightarrow x$ and $y_n \rightarrow x$. Then $d(x_n, y_n) \leq d(x_n, x) + d(x, y_n) \rightarrow 0$. By uniform continuity of f , $\rho(f(x_n), f(y_n)) \rightarrow 0$, so the two limits coincide. Thus g is well-defined.

Step 3: Agreement with f on D . If $x \in D$, choose the constant sequence $x_n = x$. Then $g(x) = \lim_n f(x_n) = f(x)$.

Step 4: g is uniformly continuous. Let $\varepsilon > 0$ and choose $\delta > 0$ witnessing uniform continuity of f on D : $d(u, v) < \delta \implies \rho(f(u), f(v)) < \varepsilon$ for all $u, v \in D$. Fix

$x, y \in X$ with $d(x, y) < \delta/3$. Choose sequences $(x_n), (y_n) \subset D$ with $x_n \rightarrow x$ and $y_n \rightarrow y$, and also $d(x_n, x) < \delta/3$, $d(y_n, y) < \delta/3$ for all n large. Then for such n ,

$$d(x_n, y_n) \leq d(x_n, x) + d(x, y) + d(y, y_n) < \frac{\delta}{3} + \frac{\delta}{3} + \frac{\delta}{3} = \delta,$$

hence $\rho(f(x_n), f(y_n)) < \varepsilon$. Passing to the limit in n gives

$$\rho(g(x), g(y)) = \rho(\lim_n f(x_n), \lim_n f(y_n)) \leq \limsup_{n \rightarrow \infty} \rho(f(x_n), f(y_n)) \leq \varepsilon.$$

Since this holds whenever $d(x, y) < \delta/3$, g is uniformly continuous.

(Optional) Uniqueness. If $h : X \rightarrow \Omega$ is uniformly continuous and $h = f$ on D , then for each $x \in X$ and any $x_n \in D$ with $x_n \rightarrow x$, continuity of h gives $h(x) = \lim_n h(x_n) = \lim_n f(x_n) = g(x)$, so $h = g$. Thus f extends uniquely to a uniformly continuous $g : X \rightarrow \Omega$ agreeing with f on D . ■

9. Prove the following converse to Exercise 2.5. Suppose (X, d) is a compact metric space having the property that for every $\epsilon > 0$ and for any points $a, b \in X$, there are points $z_0, z_1, \dots, z_n \in X$ with $z_0 = a, z_n = b$, and $d(z_{k-1}, z_k) < \epsilon$ for $1 \leq k \leq n$. Then (X, d) is connected. (Hint: Use Theorem 5.17.)

Proof. Assume for contradiction that X is disconnected. Then $X = U \cup V$ where $U, V \subseteq X$ are nonempty, disjoint, open sets. Since X is compact, the closed sets $\bar{U}$ and $\bar{V}$ are also compact and disjoint. By Theorem 5.17, the distance

$$\delta := \inf\{d(x, y) : x \in \bar{U}, y \in \bar{V}\}$$

is attained and strictly positive. Thus $\delta > 0$. Set $\varepsilon := \delta/3$. Choose $a \in U$ and $b \in V$. By the ε -chain property, there exists a chain $z_0 = a, z_1, \dots, z_n = b$ with $d(z_{k-1}, z_k) < \varepsilon$. Let k be the least index such that $z_k \in \bar{V}$. Then $z_{k-1} \in \bar{U}$. Hence

$$d(z_{k-1}, z_k) \geq \delta = 3\varepsilon,$$

contradicting $d(z_{k-1}, z_k) < \varepsilon$. This contradiction shows that X cannot be disconnected. Therefore (X, d) is connected. ■

2.6 Uniform convergence

1. Let $\{f_n\}$ be a sequence of uniformly continuous functions from (X, d) into (Ω, ρ) and suppose that $f = u\text{-}\lim f_n$ exists. Prove that f is uniformly continuous. If each f_n is a Lipschitz function with constant M_n and $\sup_n M_n < \infty$, show that f is a Lipschitz function. If $\sup_n M_n = \infty$, show that f may fail to be Lipschitz.

Proof. Uniform limit of uniformly continuous functions is uniformly continuous. Let $\varepsilon > 0$. Since $f_n \rightarrow f$ uniformly, choose N with $\sup_{x \in X} \rho(f_N(x), f(x)) < \varepsilon/3$. Because f_N is uniformly continuous, there exists $\delta > 0$ such that $d(x, y) < \delta \Rightarrow \rho(f_N(x), f_N(y)) < \varepsilon/3$. Then for $d(x, y) < \delta$,

$$\rho(f(x), f(y)) \leq \rho(f(x), f_N(x)) + \rho(f_N(x), f_N(y)) + \rho(f_N(y), f(y)) < \varepsilon,$$

so f is uniformly continuous.

Uniform limit of Lipschitz functions with bounded constants is Lipschitz. Assume each f_n is Lipschitz with constant M_n and set $M := \sup_n M_n < \infty$. For any $x, y \in X$ and any n ,

$$\rho(f(x), f(y)) \leq \rho(f(x), f_n(x)) + \rho(f_n(x), f_n(y)) + \rho(f_n(y), f(y)) \leq 2\|f - f_n\|_\infty + M d(x, y).$$

Letting $n \rightarrow \infty$ (uniform convergence) yields $\rho(f(x), f(y)) \leq M d(x, y)$. Hence f is Lipschitz with constant at most M .

If the Lipschitz constants are unbounded, the limit need not be Lipschitz. Take $X = [0, 1]$ with the usual metric, $\Omega = \mathbb{R}$, and define

$$f_n(x) := \sqrt{x + \frac{1}{n}}, \quad x \in [0, 1].$$

Then $f_n \rightarrow f$ uniformly on $[0, 1]$, where $f(x) = \sqrt{x}$, since $\sup_{x \in [0, 1]} |f_n(x) - f(x)| = \sqrt{1/n} \rightarrow 0$. Each f_n is Lipschitz with constant

$$M_n = \sup_{x \in [0, 1]} \left| \frac{d}{dx} \sqrt{x + \frac{1}{n}} \right| = \sup_{x \in [0, 1]} \frac{1}{2\sqrt{x + \frac{1}{n}}} = \frac{\sqrt{n}}{2} \xrightarrow{n \rightarrow \infty} \infty.$$

However, the limit $f(x) = \sqrt{x}$ is *not* Lipschitz on $[0, 1]$: if it were with constant L , then for $x > 0$,

$$\sqrt{x} = |f(x) - f(0)| \leq L|x - 0| = Lx \implies \frac{\sqrt{x}}{x} \leq L,$$

but $\sqrt{x}/x = 1/\sqrt{x} \rightarrow \infty$ as $x \downarrow 0$, a contradiction. Therefore, boundedness of the Lipschitz constants $\{M_n\}$ is necessary to conclude that the uniform limit f is Lipschitz. ■

Chapter 3

Elementary Properties and Example of Analytic Function

3.1 Power series

6. Find the radius of convergence for each of the following power series:

$$(a) \sum_{n=0}^{\infty} a^n z^n, \quad a \in \mathbb{C};$$

$$(b) \sum_{n=0}^{\infty} a^{n^2} z^n, \quad a \in \mathbb{C};$$

$$(c) \sum_{n=0}^{\infty} k^n z^n, \quad k \text{ an integer } \neq 0;$$

$$(d) \sum_{n=0}^{\infty} z^{n!}.$$

Proof. (a) Here $a_n = a^n$. Then $\limsup |a_n|^{1/n} = |a|$, hence $R = 1/|a|$ if $a \neq 0$, while $R = \infty$ if $a = 0$ (the series is then a constant).

(b) Here $a_n = a^{n^2}$, so $|a_n|^{1/n} = |a|^n$. Therefore

$$R = \begin{cases} \infty, & |a| < 1, \\ 1, & |a| = 1, \\ 0, & |a| > 1. \end{cases}$$

(c) Here $a_n = k^n$ with $k \neq 0$. Then $\limsup |a_n|^{1/n} = |k|$, so $R = 1/|k|$.

(d) The coefficients are $a_{n!} = 1$ and $c_m = 0$ for non-factorial m . Thus

$$|a_n|^{1/n} = \begin{cases} 1, & n = m! \text{ for some } m, \\ 0, & \text{otherwise,} \end{cases}$$

so $\limsup_{n \rightarrow \infty} |a_n|^{1/n} = 1$ (since the value 1 occurs infinitely often). Hence $R = 1$. ■

7. Show that the radius of convergence of the power series

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{n} z^{n(n+1)}$$

is 1, and discuss convergence for $z = 1, -1$, and i . (Hint: The n th coefficient of this series is not $\frac{(-1)^n}{n}$.)

Proof. Let the general term of the power series be $a_m z^m$, where

$$a_m = \begin{cases} \frac{(-1)^n}{n}, & m = n(n+1) \text{ for some } n \in \mathbb{N}, \\ 0, & \text{otherwise.} \end{cases}$$

If $m = n(n+1)$, then $|a_m|^{1/m} = n^{-1/(n(n+1))} \rightarrow 1$ as $n \rightarrow \infty$, while for all other m , $|a_m|^{1/m} = 0$. Therefore,

$$\limsup_{m \rightarrow \infty} |a_m|^{1/m} = 1 \implies R = 1.$$

Convergence on the boundary $|z| = 1$:

- For $z = 1$: the series becomes

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{n},$$

which is the alternating harmonic series, convergent (conditionally) to $-\ln 2$.

- For $z = -1$: since $(-1)^{n(n+1)} = 1$ (because $n(n+1)$ is even), the series again reduces to the alternating harmonic series and hence converges conditionally.
- For $z = i$: note that $n(n+1)$ is even, so we can write $n(n+1) = 2k$. Then $i^{n(n+1)} = i^{2k} = (-1)^k = (-1)^{n(n+1)/2}$. Thus,

$$\frac{(-1)^n}{n} i^{n(n+1)} = \frac{(-1)^{n+n(n+1)/2}}{n}.$$

The factor $(-1)^{n+n(n+1)/2}$ is periodic with period 4 and alternates in sign in a bounded way. Since $\frac{1}{n} \rightarrow 0$ and the partial sums of the periodic sequence are bounded, the Dirichlet test ensures convergence. Hence, the series converges (conditionally) for $z = i$ as well.

Therefore, the radius of convergence is $R = 1$, and the series converges conditionally for $z = 1$, $z = -1$, and $z = i$. ■

3.2 Analytic functions

1. Show that $f(z) = |z|^2 = x^2 + y^2$ has a derivative only at the origin.

Proof. Let $f(z) = |z|^2 = x^2 + y^2$ with $z = x + iy$. Writing $f = u + iv$ gives $u(x, y) = x^2 + y^2$ and $v(x, y) = 0$, so

$$u_x = 2x, \quad u_y = 2y, \quad v_x = 0, \quad v_y = 0.$$

The Cauchy–Riemann equations $u_x = v_y$ and $u_y = -v_x$ force $2x = 0$ and $2y = 0$, hence they hold only at $(x, y) = (0, 0)$. Since the partials are continuous everywhere, f can be complex differentiable only at the origin. At $z = 0$,

$$f'(0) = \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{|h|^2}{h} = \lim_{h \rightarrow 0} \bar{h} = 0.$$

Therefore f has a derivative only at the origin, and $f'(0) = 0$. ■

10. Prove the following generalization of Proposition 2.20. Let G and Ω be open in $\mathbb{C}$ and suppose $f, h : G \rightarrow \mathbb{C}$, $g : \Omega \rightarrow \mathbb{C}$, with $f(G) \subset \Omega$. Assume that g and h are analytic, that $g'(\omega) \neq 0$ for every $\omega \in \Omega$, that f is continuous, h is one-to-one, and that

$$h(z) = g(f(z)) \quad (z \in G).$$

Show that f is analytic. Give a formula for $f'(z)$.

Proof. Fix $a \in G$ and let $h \in \mathbb{C}$ with $h \neq 0$ and $a + h \in G$. Write the given identity $h(z) = g(f(z))$ at the points a and $a + h$:

$$h(a + h) - h(a) = g(f(a + h)) - g(f(a)).$$

Because h is one-to-one, $h(a + h) \neq h(a)$; hence $f(a + h) \neq f(a)$, so we may divide

by $f(a+h) - f(a)$. Then

$$\frac{h(a+h) - h(a)}{h} = \frac{g(f(a+h)) - g(f(a))}{f(a+h) - f(a)} \frac{f(a+h) - f(a)}{h}. \quad (*)$$

Since f is continuous at a and g is analytic, as $h \rightarrow 0$ we have $f(a+h) \rightarrow f(a)$ and therefore

$$\frac{g(f(a+h)) - g(f(a))}{f(a+h) - f(a)} \rightarrow g'(f(a)).$$

Also, because h is analytic, the left side of $(*)$ tends to $h'(a)$. Thus the limit of the right side exists, which shows f is differentiable at a and

$$h'(a) = g'(f(a)) f'(a).$$

Hence

$$f'(a) = \frac{h'(a)}{g'(f(a))}.$$

Since $a \in G$ was arbitrary, f is differentiable on G . Moreover, h' is analytic, g' is continuous and nowhere zero on Ω , and f is continuous, so $z \mapsto \frac{h'(z)}{g'(f(z))}$ is continuous on G . Therefore f is analytic on G with

$$f'(z) = \frac{h'(z)}{g'(f(z))}$$

■

12. Show that the real part of the function $z^{\frac{1}{2}}$ is always positive.

Proof. Let $z = re^{i\theta}$ with $r > 0$ and $-\pi < \theta \leq \pi$. The principal branch of the square root is defined by

$$z^{1/2} = \sqrt{r} e^{i\theta/2} = \sqrt{r} (\cos(\frac{\theta}{2}) + i \sin(\frac{\theta}{2})).$$

Hence the real part of $z^{1/2}$ is

$$\Re(z^{1/2}) = \sqrt{r} \cos\left(\frac{\theta}{2}\right).$$

Since $-\pi < \theta \leq \pi$, we have

$$-\frac{\pi}{2} < \frac{\theta}{2} \leq \frac{\pi}{2},$$

and therefore $\cos(\frac{\theta}{2}) > 0$. As $\sqrt{r} > 0$, it follows that

$$\Re(z^{1/2}) = \sqrt{r} \cos\left(\frac{\theta}{2}\right) > 0.$$

Thus, the real part of $z^{1/2}$ is always positive for the principal branch. ■

14. Suppose $f : G \rightarrow \mathbb{C}$ is analytic and G is connected. Show that if $f(z) \in \mathbb{R}$ for all $z \in G$, then f is constant.

Proof. Write $f = u + iv$ with $u, v : G \rightarrow \mathbb{R}$. The hypothesis $f(G) \subset \mathbb{R}$ means $v \equiv 0$ on G . Since f is analytic, the Cauchy–Riemann equations hold:

$$u_x = v_y, \quad u_y = -v_x.$$

But $v \equiv 0$ implies $v_x = v_y = 0$, hence $u_x = u_y = 0$ on G . So u is constant on each connected component of G . As G is connected, u is constant on G , and thus $f = u + iv = u$ is constant. ■

16. Find an open connected set $G \subset \mathbb{C}$ and two continuous functions f, g defined on G such that

$$f(z)^2 = g(z)^2 = 1 - z^2 \quad (z \in G).$$

Can you make G maximal? Are f and g analytic?

Proof. Define

$$f(z) := \exp\left(\frac{1}{2} \ln(1 - z^2)\right), \quad g(z) := \exp\left(i\pi + \frac{1}{2} \ln(1 - z^2)\right),$$

where $\ln$ denotes the principal branch of the logarithm. Then

$$f(z)^2 = g(z)^2 = 1 - z^2 \quad (z \in G).$$

To determine the domain, let $w = 1 - z^2$. The principal logarithm is analytic on

$$\mathbb{C} \setminus (-\infty, 0] = \mathbb{C} - \{w \in \mathbb{R} : w \leq 0\}.$$

Hence we require

$$1 - z^2 \notin (-\infty, 0].$$

If $1 - z^2 \in (-\infty, 0]$, then $z^2 \in [1, \infty)$, which forces $z \in (-\infty, -1] \cup [1, \infty)$. Thus the preimage of the branch cut $(-\infty, 0]$ under $w = 1 - z^2$ is precisely the two real rays $(-\infty, -1] \cup [1, \infty)$. Therefore

$$G = \mathbb{C} \setminus ((-\infty, -1] \cup [1, \infty)).$$

On this open and connected domain G , the function $\ln(1 - z^2)$ is analytic, and

hence so are f and g . Moreover,

$$f^2 = g^2 = 1 - z^2$$

holds throughout G , as required. ■

19. Let G be a region and define $G^* = \{z : \bar{z} \in G\}$. If $f : G \rightarrow \mathbb{C}$ is analytic, prove that

$$f^* : G^* \rightarrow \mathbb{C}, \quad f^*(z) = \overline{f(\bar{z})},$$

is also analytic.

Proof. First note that G open $\Rightarrow G^* = \{z : \bar{z} \in G\}$ is open. Fix $z_0 \in G^*$, so $\bar{z}_0 \in G$. Consider

$$\frac{f^*(z) - f^*(z_0)}{z - z_0} = \frac{\overline{f(\bar{z})} - \overline{f(\bar{z}_0)}}{z - z_0} = \frac{\overline{f(\bar{z}) - f(\bar{z}_0)}}{\bar{z} - \bar{z}_0} = \overline{\left(\frac{f(\bar{z}) - f(\bar{z}_0)}{\bar{z} - \bar{z}_0} \right)}.$$

As $z \rightarrow z_0$ we have $\bar{z} \rightarrow \bar{z}_0$, and since f is analytic at $\bar{z}_0$ the limit

$$\lim_{w \rightarrow \bar{z}_0} \frac{f(w) - f(\bar{z}_0)}{w - \bar{z}_0} = f'(\bar{z}_0)$$

exists. Therefore

$$\lim_{z \rightarrow z_0} \frac{f^*(z) - f^*(z_0)}{z - z_0} = \overline{f'(\bar{z}_0)}.$$

Hence f^* is differentiable at z_0 . Since $z_0 \in G^*$ was arbitrary, f^* is analytic on G^* , with

$$(f^*)'(z) = \overline{f'(\bar{z})} \quad (z \in G^*).$$
■

21. Prove that there is no branch of the logarithm defined on $G = \mathbb{C} \setminus \{0\}$. (Hint: Suppose such a branch exists and compare this with the principal branch.)

Proof. Suppose, for contradiction, that there exists a branch of the logarithm $L : \mathbb{C} \setminus \{0\} \rightarrow \mathbb{C}$ such that $e^{L(z)} = z$ for all $z \in \mathbb{C} \setminus \{0\}$. Consider the unit circle parametrized by $\gamma(\theta) = e^{i\theta}$, $0 \leq \theta \leq 2\pi$, and define

$$h(\theta) := L(\gamma(\theta)).$$

Since L is continuous, h is continuous on $[0, 2\pi]$. Moreover, because $e^{L(z)} = z$, we have

$$e^{h(\theta)} = e^{i\theta}.$$

Thus, for each θ , there exists an integer $n(\theta) \in \mathbb{Z}$ such that

$$h(\theta) = i\theta + 2\pi i n(\theta).$$

The map $\theta \mapsto n(\theta)$ is continuous because both $h(\theta)$ and $i\theta$ are continuous. Since $n(\theta)$ takes values in the discrete set $\mathbb{Z}$, it must be constant on the connected interval $[0, 2\pi]$. Hence $n(\theta) \equiv k$ for some fixed integer k .

Therefore

$$h(\theta) = i\theta + 2\pi i k.$$

In particular, at $\theta = 0$ and $\theta = 2\pi$,

$$L(1) = h(0) = 2\pi i k \quad \text{and} \quad L(1) = h(2\pi) = i(2\pi) + 2\pi i k.$$

Equating these values yields $2\pi i k = 2\pi i + 2\pi i k$, which is impossible. This contradiction shows that no such branch of the logarithm exists on $\mathbb{C} \setminus \{0\}$. ■

3.3 Analytic functions as mappings, Möbius transformation

4. Discuss the mapping properties of z^n and $z^{1/n}$ for $n \geq 2$. (Hint: use polar coordinates.)

Discussion. **1. The map $z \mapsto z^n$.**

Write $z = re^{i\theta}$ in polar coordinates, with $r \geq 0$ and $\theta \in \mathbb{R}$. Then

$$z^n = r^n e^{in\theta}.$$

- *Modulus:* $|z^n| = r^n$. Circles $|z| = r_0$ map to circles $|w| = r_0^n$.
- *Argument:* $\arg(z^n) = n\theta \pmod{2\pi}$. Rays $\theta = \theta_0$ map to rays of argument $n\theta_0$.
- *Sectors:* A sector of opening α is mapped to one of opening $n\alpha$.
- *Annuli:* $r_1 < |z| < r_2$ maps to $r_1^n < |w| < r_2^n$.
- *Multiplicity:* On $\mathbb{C}^* = \mathbb{C} \setminus \{0\}$, the map is n -to-1: each $w \neq 0$ has n preimages $w^{1/n} e^{2\pi i k/n}$, with $k = 0, \dots, n-1$.
- *Conformality:* The map is holomorphic everywhere and conformal except at 0, where $f'(0) = 0$ and the map is not injective.

In particular, the unit circle $|z| = 1$ is mapped to itself and is wound around n times.

2. The map $z \mapsto z^{1/n}$.

The n th root function is multivalued on $\mathbb{C}^*$. In polar form,

$$z^{1/n} = r^{1/n} e^{i(\theta+2\pi k)/n}, \quad k = 0, \dots, n-1.$$

Thus there are n branches differing by rotation by $2\pi/n$.

To obtain a single-valued branch, one must restrict the argument (e.g. $-\pi < \theta < \pi$). This introduces a branch cut.

- *Principal branch:* Define $\arg z \in (-\pi, \pi)$ (cut along $(-\infty, 0]$). Then

$$z^{1/n} = r^{1/n} e^{i\theta/n}$$

is single-valued and holomorphic on $\mathbb{C} \setminus (-\infty, 0]$.

- *Image of principal branch:* Under this branch, the image is the sector

$$\{ w : \arg w \in (-\pi/n, \pi/n) \}.$$

Radii map as $r \mapsto r^{1/n}$, while angles divide by n .

- *Other branches:* Rotating the branch cut yields n distinct branches whose images are n disjoint sectors of opening $2\pi/n$.
- *Conformality:* Each branch is holomorphic and conformal on its domain, which is simply connected and does not encircle 0. The point 0 is a branch point: one cannot define a single-valued analytic n th root around a loop surrounding 0.

Summary. The map z^n multiplies arguments by n and raises moduli to the n th power; it is n -to-1 on $\mathbb{C}^*$ and wraps the unit circle n times around itself. Each branch of $z^{1/n}$ is single-valued and conformal on a slit plane, dividing arguments by n and taking n th roots of moduli, with image a sector of opening $2\pi/n$. ■

8. If $Tz = \frac{az+b}{cz+d}$, show that $T(\mathbb{R}_\infty) = \mathbb{R}_\infty$ iff we can choose a, b, c, d to be real numbers.

Proof. Let $T(z) = \frac{az+b}{cz+d}$ with $ad-bc \neq 0$ and suppose $T(\mathbb{R}_\infty) = \mathbb{R}_\infty$. Evaluate T at three real points $0, 1, \infty$.

At 0: $T(0) = \frac{b}{d} \in \mathbb{R}_\infty$. Thus either $d = 0$ or $r_1 := \frac{b}{d} \in \mathbb{R}$.

At ∞ : $T(\infty) = \frac{a}{c} \in \mathbb{R}_\infty$. Thus either $c = 0$ or $r_2 := \frac{a}{c} \in \mathbb{R}$.

At 1: $T(1) = \frac{a+b}{c+d} \in \mathbb{R}_\infty$. Thus either $c+d=0$ or $k := \frac{a+b}{c+d} \in \mathbb{R}$.

We treat cases.

Case 1: $c, d \neq 0$ and $c+d \neq 0$. From $b = r_1 d$ and $a = r_2 c$ we have

$$k = \frac{a+b}{c+d} = \frac{r_2 c + r_1 d}{c+d} \in \mathbb{R}.$$

Hence $(r_2 - k)c + (r_1 - k)d = 0$, so

$$\frac{c}{d} = \frac{k - r_1}{r_2 - k} \in \mathbb{R}.$$

Therefore c/d , b/d , and a/c are all real. Choose a unimodular $\lambda = e^{-i \arg d}$ so that $d_0 := \lambda d \in \mathbb{R}$. Then $b_0 := \lambda b = (b/d)d_0 \in \mathbb{R}$, $c_0 := \lambda c = (c/d)d_0 \in \mathbb{R}$, and $a_0 := \lambda a = (a/c)c_0 \in \mathbb{R}$. Scaling the coefficients by the common nonzero factor λ does not change T , so we have represented T with real coefficients.

Case 2: $c = 0$. Then $T(z) = \frac{az+b}{d}$ and $T(\mathbb{R}_\infty) \subset \mathbb{R}_\infty$ gives $\frac{b}{d} = T(0) \in \mathbb{R}_\infty$ and $\frac{a+b}{d} = T(1) \in \mathbb{R}_\infty$, so $\frac{a}{d}$ and $\frac{b}{d}$ are real (allowing $d=0$ would force $b=0$ and then $\frac{a}{d} = \infty$ real). Choose λ with $\lambda d \in \mathbb{R}$; then $\lambda a, \lambda b, \lambda d \in \mathbb{R}$, and $c_0 = \lambda c = 0$ is real.

Case 3: $d = 0$. Then $T(0) = \infty \in \mathbb{R}_\infty$ and from $T(\infty) = \frac{a}{c} \in \mathbb{R}_\infty$ and $T(1) = \frac{a+b}{c} \in \mathbb{R}_\infty$ we get $\frac{a}{c}, \frac{b}{c} \in \mathbb{R}$. Choose λ with $\lambda c \in \mathbb{R}$; then $\lambda a, \lambda b, \lambda c \in \mathbb{R}$, and $d_0 = \lambda d = 0$ is real.

In all cases we can multiply (a, b, c, d) by a nonzero constant to obtain real coefficients.

Conversely, if $a, b, c, d \in \mathbb{R}$, then for $x \in \mathbb{R}$, $T(x) = \frac{ax+b}{cx+d} \in \mathbb{R}_\infty$ and $T(\infty) = a/c \in \mathbb{R}_\infty$, so $T(\mathbb{R}_\infty) = \mathbb{R}_\infty$.

Thus $T(\mathbb{R}_\infty) = \mathbb{R}_\infty$ iff the coefficients can be chosen real. ■

10. Let $D = \{z : |z| < 1\}$ and find all Möbius transformations T such that $T(D) = D$.

Proof. Let $\phi_a(z) = \frac{z-a}{1-\bar{a}z}$ for $|a| < 1$. Then ϕ_a is a Möbius map with $\phi_a(a) = 0$ and $\phi_a(\mathbb{D}) = \mathbb{D}$, where $\mathbb{D} = D = \{z : |z| < 1\}$; moreover $\phi_a^{-1} = \phi_{-a}$.

Claim. A Möbius map T satisfies $T(\mathbb{D}) = \mathbb{D}$ iff

$$T(z) = e^{i\theta} \phi_a(z) = e^{i\theta} \frac{z-a}{1-\bar{a}z} \quad \text{for some } a \in \mathbb{D}, \theta \in \mathbb{R}.$$

($\subseteq$) Suppose $T(\mathbb{D}) = \mathbb{D}$. Set $a := T^{-1}(0) \in \mathbb{D}$ and define

$$F := T \circ \phi_a^{-1} = T \circ \phi_{-a}.$$

Then $F : \mathbb{D} \rightarrow \mathbb{D}$ is holomorphic and $F(0) = 0$. By Schwarz's lemma, $F(z) = e^{i\theta} z$ for some $\theta \in \mathbb{R}$. Hence

$$T = F \circ \phi_a = e^{i\theta} \phi_a.$$

($\supseteq$) Conversely, for any $|a| < 1$ and $\theta \in \mathbb{R}$, both ϕ_a and the rotation $R_\theta(z) = e^{i\theta} z$ map $\mathbb{D}$ onto $\mathbb{D}$, so their composition $T = R_\theta \circ \phi_a$ also satisfies $T(\mathbb{D}) = \mathbb{D}$.

Therefore,

$$\text{Aut}(\mathbb{D}) = \left\{ z \mapsto e^{i\theta} \frac{z - a}{1 - \bar{a}z} : a \in \mathbb{D}, \theta \in \mathbb{R} \right\}.$$

■

14. Suppose that one circle is contained inside another and that they are tangent at the point a . Let G be the region between the two circles and map G conformally onto the open unit disk. (Hint: first try $(z - a)^{-1}$.)

Proof. Normalization. By translating we may assume the tangency point is $a = 0$. By a rotation (precomposition by a unimodular constant) we may assume both circles are centered on the positive real axis. Thus there exist radii $0 < r_1 < r_2$ such that the two circles are

$$C_j : |z - r_j| = r_j \quad (j = 1, 2),$$

which are internally tangent at 0. Let G be the region between C_1 and C_2 .

Step 1: Inversion sends tangent circles to parallel lines. Set

$$w = \Phi_1(z) := \frac{1}{z}.$$

A circle $|z - a| = |a|$ through the origin maps to the line $\Re(aw) = \frac{1}{2}$. Here $a = r_j \in \mathbb{R}_+$, so

$$\Phi_1(C_j) = \left\{ \Re w = \frac{1}{2r_j} \right\}.$$

Hence $\Phi_1(G)$ is the open vertical strip

$$S := \left\{ w : \alpha < \Re w < \beta \right\}, \quad \alpha := \frac{1}{2r_2} < \beta := \frac{1}{2r_1}.$$

Step 2: Rotate to a horizontal strip. Let

$$u = \Phi_2(w) := i w.$$

Then $\Im u = \Re w$, so $\Phi_2(S)$ is the horizontal strip

$$S_h := \{u : \alpha < \Im u < \beta\}.$$

Set its width and midline as

$$d := \beta - \alpha = \frac{r_2 - r_1}{2r_1r_2}, \quad m := \frac{\alpha + \beta}{2} = \frac{r_1 + r_2}{4r_1r_2}.$$

Step 3: Strip $\rightarrow$ half-plane $\rightarrow$ disk. Define

$$\xi = \Phi_3(u) := \exp\left(\frac{\pi}{d}(u - im)\right),$$

which maps S_h conformally onto the right half-plane $\{\Re \xi > 0\}$. Then apply the Cayley map

$$\Phi_4(\xi) := \frac{\xi - 1}{\xi + 1},$$

which sends $\{\Re \xi > 0\}$ conformally onto the unit disk $\mathbb{D}$.

Conclusion. The composition

$$F := \Phi_4 \circ \Phi_3 \circ \Phi_2 \circ \Phi_1$$

is a conformal bijection $F : G \rightarrow \mathbb{D}$. Writing it explicitly,

$$F(z) = \frac{\exp\left(\frac{\pi}{d}\left(\frac{i}{z} - im\right)\right) - 1}{\exp\left(\frac{\pi}{d}\left(\frac{i}{z} - im\right)\right) + 1} \quad \text{with} \quad d = \frac{r_2 - r_1}{2r_1r_2}, \quad m = \frac{r_1 + r_2}{4r_1r_2}.$$

Equivalently, using $\frac{e^t - 1}{e^t + 1} = \tanh(t/2)$,

$$F(z) = \tanh\left(\frac{\pi}{2d}\left(\frac{i}{z} - im\right)\right).$$

Any other configuration (arbitrary $a \neq 0$ and non-horizontal tangent) is obtained by pre/post-composing with a rotation/translation: replace z by $z - a$ in Φ_1 , and possibly rotate by a unimodular constant before Step 2. In all cases, the boundary circles map to $|w| = 1$, and F is one-to-one and onto $\mathbb{D}$. ■

15. *Can you map the open unit disk conformally onto $\{z : 0 < |z| < 1\}$?*

Proof. I am unsure how to prove this without using Cauchy's integral theorem, singularity theory, or the fact that conformal maps preserve topological properties such as simple connectedness. I know the answer is negative, since $\mathbb{D}$ is simply connected whereas $\mathbb{D} \setminus \{0\}$ is not, so there cannot exist a surjective conformal map between them. However, I would appreciate guidance on how to prove this using only the material from the first three chapters. ■

16. *Map $G = \mathbb{C} \setminus \{z \in \mathbb{R} : -1 \leq z \leq 1\}$ onto the open unit disk by an analytic function f . Can f be one-to-one?*

Proof. Since G is not simply connected, any surjective analytic function from G onto the unit disk cannot be one-to-one. ■

17. Let G be a region and suppose that $f : G \rightarrow \mathbb{C}$ is analytic such that $f(G)$ is a subset of a circle. Show that f is constant.

Proof. Without loss of generality assume $f(G) \subset \{w \in \mathbb{C} : |w| = 1\}$. Then $|f(z)| = 1$ for all $z \in G$, so $\overline{f(z)} = 1/f(z)$ and $f(z) \neq 0$.

Define

$$F(z) = f(z) + \frac{1}{f(z)}.$$

Since f is analytic and never zero, $1/f$ is analytic; hence F is analytic on G . Moreover,

$$F(z) = f(z) + \overline{f(z)} = 2\Re f(z) \in \mathbb{R} \quad (z \in G).$$

By Exercise 14 (Section 2), any real-valued analytic function on a connected domain is constant; therefore $F \equiv c \in \mathbb{R}$. Thus

$$f(z) + \frac{1}{f(z)} = c \iff f(z)^2 - cf(z) + 1 = 0 \quad (z \in G).$$

Let α, β be the (constant) roots of $w^2 - cw + 1 = 0$. Then

$$(f(z) - \alpha)(f(z) - \beta) = 0 \quad \text{for all } z \in G,$$

so $f(G) \subset \{\alpha, \beta\}$. Since f is continuous and G is connected, $f(G)$ is connected. The only connected subsets of the two-point set $\{\alpha, \beta\}$ are singletons; hence $f(G) = \{\alpha\}$ or $f(G) = \{\beta\}$. Therefore f is constant on G . ■

Chapter 4

Complex Integration

4.1 Riemann-Stieltjes integral

6. Show that if $\gamma : [a, b] \rightarrow \mathbb{C}$ is a Lipschitz function, then γ is of bounded variation.

Proof. Let $\gamma : [a, b] \rightarrow \mathbb{C}$ be Lipschitz with constant $L \geq 0$, i.e.,

$$|\gamma(s) - \gamma(t)| \leq L |s - t| \quad \text{for all } s, t \in [a, b].$$

For any partition $P = \{a = t_0 < t_1 < \cdots < t_n = b\}$ of $[a, b]$, the variation sum satisfies

$$\sum_{i=1}^n |\gamma(t_i) - \gamma(t_{i-1})| \leq \sum_{i=1}^n L |t_i - t_{i-1}| = L(b - a).$$

Taking the supremum over all partitions P gives

$$V(\gamma, [a, b]) = \sup_P \sum_{i=1}^n |\gamma(t_i) - \gamma(t_{i-1})| \leq L(b - a) < \infty.$$

Hence γ is of bounded variation on $[a, b]$. ■

8. Let γ and σ be the two polygons $[1, i]$ and $[1, 1 + i, i]$. Express γ and σ as paths and calculate $\int_{\gamma} f$ and $\int_{\sigma} f$ where $f(z) = |z|^2$.

Proof. A straight segment from 1 to i can be parametrized by

$$\gamma(t) = 1 + t(i - 1) = (1 - t) + it, \quad t \in [0, 1], \quad \gamma'(t) = i - 1.$$

Hence $|\gamma(t)|^2 = (1 - t)^2 + t^2 = 1 - 2t + 2t^2$, and

$$\int_{\gamma} |z|^2 dz = \int_0^1 (1 - 2t + 2t^2) (i - 1) dt = (i - 1) \left[t - t^2 + \frac{2}{3} t^3 \right]_0^1 = \frac{2}{3} (i - 1).$$

For the polygon $[1, 1 + i, i]$, write $\sigma = \sigma_1 * \sigma_2$ with

$$\sigma_1(t) = 1 + it, \quad t \in [0, 1], \quad \sigma_1'(t) = i,$$

$$\sigma_2(t) = 1 - t + i, \quad t \in [0, 1], \quad \sigma_2'(t) = -1.$$

Then $|\sigma_1(t)|^2 = 1 + t^2$ and $|\sigma_2(t)|^2 = (1 - t)^2 + 1 = 2 - 2t + t^2$. Thus

$$\int_{\sigma_1} |z|^2 dz = \int_0^1 (1 + t^2) i dt = i \left[t + \frac{1}{3} t^3 \right]_0^1 = \frac{4}{3} i,$$

$$\int_{\sigma_2} |z|^2 dz = \int_0^1 (2 - 2t + t^2)(-1) dt = - \left[2t - t^2 + \frac{1}{3} t^3 \right]_0^1 = -\frac{4}{3}.$$

Therefore

$$\int_{\sigma} |z|^2 dz = \int_{\sigma_1} |z|^2 dz + \int_{\sigma_2} |z|^2 dz = \frac{4}{3} i - \frac{4}{3} = \frac{4}{3}(i - 1).$$

In particular, $\int_{\gamma} |z|^2 dz = \frac{2}{3}(i - 1)$ and $\int_{\sigma} |z|^2 dz = \frac{4}{3}(i - 1)$, showing dependence on the path since $|z|^2$ is not holomorphic. ■

12. Let $I(r) = \int_{\gamma} \frac{e^{iz}}{z} dz$ where $\gamma : [0, \pi] \rightarrow \mathbb{C}$ is defined by $\gamma(t) = re^{it}$. Show that $\lim_{r \rightarrow \infty} I(r) = 0$.

Proof. Let $\gamma(t) = re^{it}$ for $t \in [0, \pi]$. Then $\gamma'(t) = ire^{it}$ and

$$I(r) = \int_{\gamma} \frac{e^{iz}}{z} dz = \int_0^{\pi} \frac{e^{ire^{it}}}{re^{it}} (ire^{it}) dt = i \int_0^{\pi} e^{ire^{it}} dt.$$

Hence

$$|I(r)| \leq \int_0^{\pi} |e^{ire^{it}}| dt = \int_0^{\pi} e^{\operatorname{Re}(ire^{it})} dt = \int_0^{\pi} e^{-r \sin t} dt.$$

Fix $0 < \varepsilon < \frac{\pi}{2}$ and split the integral:

$$\int_0^{\pi} e^{-r \sin t} dt = \int_0^{\varepsilon} e^{-r \sin t} dt + \int_{\varepsilon}^{\pi - \varepsilon} e^{-r \sin t} dt + \int_{\pi - \varepsilon}^{\pi} e^{-r \sin t} dt.$$

On $[\varepsilon, \pi - \varepsilon]$, $\sin t \geq \sin \varepsilon > 0$, so

$$\int_{\varepsilon}^{\pi - \varepsilon} e^{-r \sin t} dt \leq \pi e^{-r \sin \varepsilon} \xrightarrow{r \rightarrow \infty} 0.$$

For $t \in [0, \frac{\pi}{2}]$, $\sin t \geq \frac{2}{\pi}t$, hence

$$\int_0^\varepsilon e^{-r \sin t} dt \leq \int_0^\varepsilon e^{-\frac{2r}{\pi}t} dt = \frac{\pi}{2r} \left(1 - e^{-\frac{2r}{\pi}\varepsilon}\right) \leq \frac{\pi}{2r} \xrightarrow{r \rightarrow \infty} 0.$$

Similarly, by symmetry $\sin t = \sin(\pi - t)$,

$$\int_{\pi-\varepsilon}^\pi e^{-r \sin t} dt = \int_0^\varepsilon e^{-r \sin s} ds \leq \frac{\pi}{2r} \xrightarrow{r \rightarrow \infty} 0.$$

Combining the estimates gives $\int_0^\pi e^{-r \sin t} dt \rightarrow 0$, hence $|I(r)| \rightarrow 0$. Therefore

$$\lim_{r \rightarrow \infty} I(r) = 0.$$

■

13. Find $\int_\gamma z^{-1/2} dz$ where:

- (a) γ is the upper half of the unit circle from +1 to -1;
- (b) γ is the lower half of the unit circle from +1 to -1.

Proof. Take the principal branch of the square root, i.e. $\arg z \in (-\pi, \pi)$, so that

$$z^{-1/2} = \frac{1}{\sqrt{z}} = e^{-\frac{i}{2} \arg z} \quad \text{for } z \neq 0.$$

(a) **Upper semicircle from +1 to -1.**

Parametrize

$$\gamma(t) = e^{it}, \quad t \in [0, \pi], \quad \gamma'(t) = ie^{it}.$$

Along this arc, $\arg \gamma(t) = t$, hence $\gamma(t)^{-1/2} = e^{-it/2}$. Thus

$$\int_\gamma z^{-1/2} dz = \int_0^\pi e^{-it/2} ie^{it} dt = \int_0^\pi i e^{it/2} dt = 2(e^{i\pi/2} - 1) = 2(i - 1).$$

(b) **Lower semicircle from +1 to -1.**

Parametrize

$$\gamma(t) = e^{it}, \quad t \in [0, -\pi], \quad \gamma'(t) = ie^{it}.$$

Now $\arg \gamma(t) = t \in (-\pi, 0]$, so $\gamma(t)^{-1/2} = e^{-it/2}$. Hence

$$\int_\gamma z^{-1/2} dz = \int_0^{-\pi} e^{-it/2} ie^{it} dt = \int_0^{-\pi} i e^{it/2} dt = 2(e^{-i\pi/2} - 1) = -2(1 + i).$$

Therefore,

$$\int_{\text{upper arc}} z^{-1/2} dz = 2(i - 1), \quad \int_{\text{lower arc}} z^{-1/2} dz = -2(1 + i).$$

■

21. Show that if F_1 and F_2 are primitives for $f : G \rightarrow \mathbb{C}$ and G is open and connected, then there is a constant c such that $F_1(z) = c + F_2(z)$ for each $z \in G$.

Proof. Let $H := F_1 - F_2$. Then H is holomorphic on G and

$$H'(z) = F_1'(z) - F_2'(z) = f(z) - f(z) = 0 \quad (z \in G).$$

Fix $z_0 \in G$. Since G is open and connected (hence path-connected), for any $z \in G$ there exists a piecewise C^1 path γ in G from z_0 to z . By the fundamental theorem of calculus for line integrals of holomorphic functions,

$$H(z) - H(z_0) = \int_{\gamma} H'(w) dw = \int_{\gamma} 0 dw = 0.$$

Thus $H(z) = H(z_0) =: c$ for all $z \in G$. Therefore,

$$F_1(z) = F_2(z) + c \quad \text{for all } z \in G.$$

■

22. Let γ be a closed rectifiable curve in an open set G and $a \notin G$. Show that for $n \geq 2$, $\int_{\gamma} (z - a)^{-n} dz = 0$.

Proof. Define

$$F(z) := -\frac{(z - a)^{-n+1}}{n - 1}, \quad z \in G.$$

Then F is analytic on G and

$$F'(z) = (z - a)^{-n} \quad (z \in G).$$

Thus $(z - a)^{-n}$ has a primitive F on G . By the fundamental theorem of calculus for line integrals (or Cauchy's theorem), for any closed rectifiable curve $\gamma \subset G$,

$$\int_{\gamma} (z - a)^{-n} dz = \int_{\gamma} F'(z) dz = 0.$$

■

23. Prove the following integration by parts formula. Let f and g be analytic in G and let γ be a rectifiable curve from a to b in G . Then

$$\int_{\gamma} f g' dz = f(b)g(b) - f(a)g(a) - \int_{\gamma} f' g dz.$$

Proof. Let $\gamma : [\alpha, \beta] \rightarrow G$ be a rectifiable parametrization with $\gamma(\alpha) = a$ and $\gamma(\beta) = b$. Since f and g are analytic, the product rule gives

$$(fg)' = f'g + fg' \quad \text{on } G.$$

Hence, by the fundamental theorem of calculus for line integrals,

$$\int_{\gamma} (fg)' dz = (fg)(b) - (fg)(a) = f(b)g(b) - f(a)g(a).$$

But

$$\int_{\gamma} (fg)' dz = \int_{\gamma} f'g dz + \int_{\gamma} fg' dz.$$

Rearranging gives

$$\int_{\gamma} fg' dz = f(b)g(b) - f(a)g(a) - \int_{\gamma} f'g dz,$$

which is the desired integration by parts formula. ■

4.2 Power series representation of analytic function

2. Prove the following analogue of Leibniz's rule. Let G be an open set and let γ be a rectifiable curve in $\mathbb{C}$. Suppose that $\varphi : \{\gamma\} \times G \rightarrow \mathbb{C}$ is a continuous function and define

$$g(z) = \int_{\gamma} \varphi(w, z) dw.$$

Then g is continuous. If $\frac{\partial \varphi}{\partial z}$ exists for each $(w, z) \in \{\gamma\} \times G$ and is continuous, then g is analytic and

$$g'(z) = \int_{\gamma} \frac{\partial \varphi}{\partial z}(w, z) dw.$$

Proof. Let $\gamma : [a, b] \rightarrow \mathbb{C}$ be a rectifiable parametrization of the curve, and let $\{\gamma\} = \gamma([a, b])$ denote its image. We are given that

$$\varphi : \{\gamma\} \times G \rightarrow \mathbb{C}$$

is continuous, and we define

$$g(z) = \int_{\gamma} \varphi(w, z) dw = \int_a^b \varphi(\gamma(t), z) \gamma'(t) dt.$$

1. Continuity of g .

Fix $z_0 \in G$. Since φ is continuous on $\{\gamma\} \times G$, and $\{\gamma\}$ is compact (the continuous image of the compact interval $[a, b]$), it follows that φ is *uniformly* continuous on $\{\gamma\} \times K$ for any compact $K \subset G$. In particular, choose a closed disk D around z_0 contained in G . Given $\varepsilon > 0$, there exists $\delta > 0$ such that

$$|z - z_0| < \delta, \quad w \in \{\gamma\} \quad \implies \quad |\varphi(w, z) - \varphi(w, z_0)| < \frac{\varepsilon}{\ell(\gamma)},$$

where $\ell(\gamma)$ denotes the length of the curve γ .

Then for $|z - z_0| < \delta$,

$$\begin{aligned} |g(z) - g(z_0)| &= \left| \int_{\gamma} (\varphi(w, z) - \varphi(w, z_0)) dw \right| \\ &\leq \int_{\gamma} |\varphi(w, z) - \varphi(w, z_0)| |dw| \\ &\leq \sup_{w \in \{\gamma\}} |\varphi(w, z) - \varphi(w, z_0)| \cdot \ell(\gamma) \\ &< \frac{\varepsilon}{\ell(\gamma)} \cdot \ell(\gamma) = \varepsilon. \end{aligned}$$

Hence g is continuous at z_0 , and since $z_0 \in G$ was arbitrary, g is continuous on G .

2. Existence of $g'(z)$ and formula for the derivative.

Assume now that $\partial\varphi/\partial z$ exists for each $(w, z) \in \{\gamma\} \times G$ and is continuous there. Fix $z_0 \in G$ and choose a closed disk K centered at z_0 such that $K \subset G$. The continuity of φ and $\partial\varphi/\partial z$ implies that these functions are uniformly continuous on the compact set $\{\gamma\} \times K$. We will show that g is complex differentiable at z_0 and that

$$g'(z_0) = \int_{\gamma} \frac{\partial\varphi}{\partial z}(w, z_0) dw.$$

For $h \neq 0$ with $z_0 + h \in K$, write

$$\frac{g(z_0 + h) - g(z_0)}{h} - \int_{\gamma} \frac{\partial\varphi}{\partial z}(w, z_0) dw = \int_{\gamma} R_h(w) dw,$$

where

$$R_h(w) = \frac{\varphi(w, z_0 + h) - \varphi(w, z_0)}{h} - \frac{\partial\varphi}{\partial z}(w, z_0).$$

Thus

$$\left| \frac{g(z_0 + h) - g(z_0)}{h} - \int_{\gamma} \frac{\partial \varphi}{\partial z}(w, z_0) dw \right| \leq \int_{\gamma} |R_h(w)| |dw| \leq \ell(\gamma) \sup_{w \in \{\gamma\}} |R_h(w)|.$$

It remains to show that

$$\sup_{w \in \{\gamma\}} |R_h(w)| \longrightarrow 0 \quad \text{as } h \rightarrow 0.$$

For each fixed $w \in \{\gamma\}$, the function $z \mapsto \varphi(w, z)$ is complex differentiable at z_0 with derivative $\partial \varphi / \partial z(w, z_0)$. Thus, by definition of the complex derivative,

$$\lim_{h \rightarrow 0} R_h(w) = 0 \quad \text{for each } w \in \{\gamma\}.$$

Define

$$F(w, z) = \begin{cases} \frac{\varphi(w, z) - \varphi(w, z_0)}{z - z_0} - \frac{\partial \varphi}{\partial z}(w, z_0), & z \neq z_0, \\ 0, & z = z_0. \end{cases}$$

The function F is continuous on the compact set $\{\gamma\} \times K$, because the difference quotient converges to the derivative and $\partial \varphi / \partial z$ is continuous. Moreover,

$$R_h(w) = F(w, z_0 + h).$$

By continuity of F on the compact set, we have

$$\sup_{w \in \{\gamma\}} |F(w, z_0 + h)| \longrightarrow 0 \quad \text{as } h \rightarrow 0.$$

Therefore

$$\sup_{w \in \{\gamma\}} |R_h(w)| = \sup_{w \in \{\gamma\}} |F(w, z_0 + h)| \longrightarrow 0,$$

and hence

$$\left| \frac{g(z_0 + h) - g(z_0)}{h} - \int_{\gamma} \frac{\partial \varphi}{\partial z}(w, z_0) dw \right| \leq \ell(\gamma) \sup_w |R_h(w)| \longrightarrow 0.$$

Thus the complex derivative $g'(z_0)$ exists and

$$g'(z_0) = \int_{\gamma} \frac{\partial \varphi}{\partial z}(w, z_0) dw.$$

Since $z_0 \in G$ was arbitrary and the integrand is continuous in z , it follows that g is analytic on G and

$$g'(z) = \int_{\gamma} \frac{\partial \varphi}{\partial z}(w, z) dw \quad \text{for all } z \in G.$$

■

3. Suppose that γ is a rectifiable curve in $\mathbb{C}$ and φ is defined and continuous on $\{\gamma\}$. Use Exercise 2 to show that

$$g(z) = \int_{\gamma} \frac{\varphi(w)}{w - z} dw$$

is analytic on $\mathbb{C} - \{\gamma\}$ and

$$g^{(n)}(z) = n! \int_{\gamma} \frac{\varphi(w)}{(w - z)^{n+1}} dw.$$

Proof. Let γ be a rectifiable curve in $\mathbb{C}$ and denote its image by $\{\gamma\}$. Since γ is continuous on a compact interval, $\{\gamma\}$ is compact and therefore closed. Thus $\mathbb{C} \setminus \{\gamma\}$ is an open set. Define

$$G := \mathbb{C} \setminus \{\gamma\}.$$

For $(w, z) \in \{\gamma\} \times G$ define

$$\Phi(w, z) := \frac{\varphi(w)}{w - z}.$$

Note that for each such pair, $w \neq z$ by definition of G , so Φ is well-defined. Since φ is continuous on $\{\gamma\}$ and the map

$$(w, z) \mapsto \frac{1}{w - z}$$

is continuous on $\{(w, z) \in \mathbb{C}^2 : w \neq z\}$, it follows that Φ is continuous on $\{\gamma\} \times G$. Moreover, for fixed $w \in \{\gamma\}$, as a function of z we have

$$\Phi(w, z) = \varphi(w)(w - z)^{-1}.$$

Thus

$$\frac{\partial \Phi}{\partial z}(w, z) = \varphi(w) \frac{d}{dz}(w - z)^{-1} = \varphi(w)(w - z)^{-2},$$

since

$$\frac{d}{dz}(w - z)^{-1} = -1 \cdot (-(w - z)^{-2}) = (w - z)^{-2}.$$

This partial derivative is continuous on $\{\gamma\} \times G$. Define

$$g(z) := \int_{\gamma} \frac{\varphi(w)}{w - z} dw = \int_{\gamma} \Phi(w, z) dw.$$

By Exercise 2 applied to Φ on $\{\gamma\} \times G$, the function g is analytic on G and

$$g'(z) = \int_{\gamma} \frac{\partial \Phi}{\partial z}(w, z) dw = \int_{\gamma} \frac{\varphi(w)}{(w - z)^2} dw.$$

More generally by induction on $n \in \mathbb{N}$ we have:

$$g^{(n)}(z) = n! \int_{\gamma} \frac{\varphi(w)}{(w - z)^{n+1}} dw \quad \text{for all } z \in G.$$

■

5. Give the power series expansion of $\log z$ about $z = i$ and find its radius of convergence.

Proof. We expand $f(z) = \log z$ in a Taylor series about $z = i$ using Theorem 2.8. The derivatives of $\log z$ are

$$f^{(n)}(z) = (-1)^{n-1} \frac{(n-1)!}{z^n}, \quad n \geq 1.$$

Thus

$$f^{(n)}(i) = (-1)^{n-1} \frac{(n-1)!}{i^n}.$$

The Taylor coefficients are

$$a_n = \frac{f^{(n)}(i)}{n!} = (-1)^{n-1} \frac{1}{n} i^{-n}.$$

Since $i^{-n} = (-1)^n i^n$, we obtain the simpler formula

$$a_n = -\frac{i^n}{n}, \quad n \geq 1.$$

Together with $a_0 = f(i) = \log i$, this gives the Taylor series

$$\log z = \log i - \sum_{n=1}^{\infty} \frac{i^n}{n} (z - i)^n$$

The nearest singularity of $\log z$ to i is 0, at distance 1, so the radius of convergence is $R = 1$. ■

7. Use the results of this section to evaluate the integral

$$(a) \int_{\gamma} \frac{e^{iz}}{z^2} dz, \quad \gamma(t) = e^{it}, \quad 0 \leq t \leq 2\pi.$$

$$(c) \int_{\gamma} \frac{\sin z}{z^3} dz, \quad \gamma(t) = e^{it}, \quad 0 \leq t \leq 2\pi.$$

Proof. We use Cauchy's integral formula for derivatives:

$$f^{(n)}(0) = \frac{n!}{2\pi i} \int_{\gamma} \frac{f(z)}{z^{n+1}} dz, \quad \gamma(t) = e^{it}, \quad 0 \leq t \leq 2\pi.$$

(a) Evaluate

$$\int_{\gamma} \frac{e^{iz}}{z^2} dz.$$

Here the integrand has the form

$$\frac{f(z)}{z^2}, \quad f(z) = e^{iz}.$$

Thus

$$\int_{\gamma} \frac{e^{iz}}{z^2} dz = \frac{2\pi i}{1!} f'(0) = 2\pi i \cdot (ie^{i \cdot 0}) = 2\pi i \cdot i = -2\pi.$$

(c) Evaluate

$$\int_{\gamma} \frac{\sin z}{z^3} dz.$$

Now we have

$$\frac{f(z)}{z^3}, \quad f(z) = \sin z.$$

We use the second derivative:

$$\int_{\gamma} \frac{\sin z}{z^3} dz = \frac{2\pi i}{2!} f''(0).$$

we compute

$$f''(z) = \sin z'' = -\sin z$$

Thus

$$f''(0) = -\sin 0 = 0.$$

Therefore,

$$\int_{\gamma} \frac{\sin z}{z^3} dz = \frac{2\pi i}{2} \cdot 0 = 0.$$

■

9. Evaluate the following integrals:

$$(a) \int_{\gamma} \frac{e^z - e^{-z}}{z^n} dz \text{ where } n \text{ is a positive integer and } \gamma(t) = e^{it}, \quad 0 \leq t \leq 2\pi.$$

(c) $\int_{\gamma} \frac{dz}{z^2 + 1}$ where $\gamma(t) = 2e^{it}$, $0 \leq t \leq 2\pi$. (Hint: expand $(z^2 + 1)^{-1}$ by partial fractions.)

(e) $\int_{\gamma} \frac{z^{1/m}}{(z-1)^m} dz$ where $\gamma(t) = 1 + \frac{1}{2}e^{it}$, $0 \leq t \leq 2\pi$.

Proof. (a) Evaluate

$$\int_{\gamma} \frac{e^z - e^{-z}}{z^n} dz, \quad \gamma(t) = e^{it}.$$

Write the integrand as

$$\frac{f(z)}{z^n}, \quad f(z) = e^z - e^{-z}.$$

By Cauchy's integral formula for derivatives,

$$\int_{\gamma} \frac{f(z)}{z^n} dz = \frac{2\pi i}{(n-1)!} f^{(n-1)}(0).$$

Since

$$e^z = \sum_{k=0}^{\infty} \frac{z^k}{k!}, \quad e^{-z} = \sum_{k=0}^{\infty} \frac{(-1)^k z^k}{k!},$$

we have

$$f(z) = e^z - e^{-z} = \sum_{k=0}^{\infty} (1 - (-1)^k) \frac{z^k}{k!}.$$

Thus

$$f^{(n-1)}(0) = \begin{cases} \frac{2}{(n-1)!}, & n-1 \text{ odd}, \\ 0, & n-1 \text{ even}. \end{cases}$$

Therefore

$$\int_{\gamma} \frac{e^z - e^{-z}}{z^n} dz = \begin{cases} 2\pi i \cdot \frac{2}{(n-1)!} = \frac{4\pi i}{(n-1)!}, & n \text{ even}, \\ 0, & n \text{ odd}. \end{cases}$$

(c) Evaluate

$$\int_{\gamma} \frac{dz}{z^2 + 1}, \quad \gamma(t) = 2e^{it}.$$

Factor:

$$\frac{1}{z^2 + 1} = \frac{1}{(z-i)(z+i)} = \frac{1}{2i} \left(\frac{1}{z-i} - \frac{1}{z+i} \right).$$

The circle $|z| = 2$ encloses both poles $\pm i$, so

$$\int_{\gamma} \frac{dz}{z-i} = 2\pi i, \quad \int_{\gamma} \frac{dz}{z+i} = 2\pi i.$$

Thus

$$\int_{\gamma} \frac{dz}{z^2 + 1} = \frac{1}{2i} (2\pi i - 2\pi i) = 0.$$

(e) Evaluate

$$\int_{\gamma} \frac{z^{1/m}}{(z-1)^m} dz, \quad \gamma(t) = 1 + \frac{1}{2}e^{it}.$$

The path γ is a small circle centered at $z = 1$. Let

$$f(z) = z^{1/m}, \quad \text{analytic near } z = 1.$$

The integrand has the form

$$\frac{f(z)}{(z-1)^m}.$$

By Cauchy's integral formula for derivatives,

$$\int_{\gamma} \frac{f(z)}{(z-1)^m} dz = \frac{2\pi i}{(m-1)!} f^{(m-1)}(1).$$

Compute the derivatives:

$$f(z) = z^{1/m}, \quad f^{(k)}(z) = \frac{1}{m} \left(\frac{1}{m} - 1 \right) \cdots \left(\frac{1}{m} - k + 1 \right) z^{1/m-k}.$$

Thus

$$f^{(m-1)}(1) = \frac{1}{m} \left(\frac{1}{m} - 1 \right) \cdots \left(\frac{1}{m} - m + 2 \right).$$

Therefore

$$\int_{\gamma} \frac{z^{1/m}}{(z-1)^m} dz = \frac{2\pi i}{(m-1)!} \frac{1}{m} \left(\frac{1}{m} - 1 \right) \cdots \left(\frac{1}{m} - m + 2 \right).$$

■

4.3 Zeros of an analytic function

1. Let f be an entire function and suppose there is a constant M , an $R > 0$, and an integer $n \geq 1$ such that

$$|f(z)| \leq M |z|^n \quad \text{for } |z| > R.$$

Show that f is a polynomial of degree $\leq n$.

Proof. Let f be entire and suppose there exist constants $M > 0$, $R > 0$, and an integer $n \geq 1$ such that

$$|f(z)| \leq M|z|^n \quad \text{for all } |z| > R.$$

We will show that $f^{(k)}(0) = 0$ for every $k > n$, which implies that the Taylor series of f terminates after order n , making f a polynomial of degree at most n .

Let $k > n$ and let $r > R$. By Cauchy's integral formula for derivatives,

$$f^{(k)}(0) = \frac{k!}{2\pi i} \int_{\gamma} \frac{f(z)}{z^{k+1}} dz.$$

Using the estimation lemma, we obtain

$$|f^{(k)}(0)| \leq \frac{k!}{r^k} \cdot \max_{|z|=r} |f(z)|.$$

Since $|f(z)| \leq Mr^n$ on the circle $|z| = r$, this gives

$$|f^{(k)}(0)| \leq k! M r^{n-k}.$$

Because $k > n$, the exponent $n - k$ is negative, and hence

$$r^{n-k} \rightarrow 0 \quad \text{as } r \rightarrow \infty.$$

Thus letting $r \rightarrow \infty$ yields

$$f^{(k)}(0) = 0 \quad \text{for all } k > n.$$

Therefore all terms of order $> n$ in the Taylor expansion vanish, and we conclude that

$$f(z) = a_0 + a_1 z + \cdots + a_n z^n,$$

so f is a polynomial of degree at most n . ■

6. Let G be a region and suppose that $f : G \rightarrow \mathbb{C}$ is analytic and $a \in G$ such that

$$|f(a)| \leq |f(z)| \quad \text{for all } z \in G.$$

Show that either $f(a) = 0$ or f is constant.

Proof. Suppose $f : G \rightarrow \mathbb{C}$ is analytic on a region G and that

$$|f(a)| \leq |f(z)| \quad \text{for all } z \in G,$$

for some point $a \in G$. We will show that either $f(a) = 0$ or f is constant.

Assume first that $f(a) \neq 0$. Then f has no zeros at a , and therefore $1/f$ is analytic in some neighborhood of a , and in fact analytic on all of G because f never vanishes on G (otherwise $|f(a)| \leq |f(z_0)| = 0$ would force $f(a) = 0$, contradicting the assumption).

For the analytic function $1/f$, we compute

$$\left| \frac{1}{f(a)} \right| = \frac{1}{|f(a)|} \geq \frac{1}{|f(z)|} = \left| \frac{1}{f(z)} \right| \quad \text{for all } z \in G.$$

Thus $|1/f|$ attains its *maximum* at the interior point a . By the Maximum Modulus Principle, an analytic function that attains its maximum modulus at an interior point must be constant. Hence $1/f$ is constant on G , and therefore f is constant on G . Thus we conclude that either $f(a) = 0$ or f is constant. ■

8. Let G be a region and let f and g be analytic functions on G such that

$$f(z)g(z) = 0 \quad \text{for all } z \in G.$$

Show that either $f \equiv 0$ or $g \equiv 0$.

Proof. Suppose f and g are analytic on a region G and satisfy

$$f(z)g(z) = 0 \quad \text{for all } z \in G.$$

Assume first that f is not identically zero. By the Identity Theorem, the zeros of a nontrivial analytic function cannot have a limit point in G ; hence there exists a point $a \in G$ and a radius $r > 0$ such that $f(z) \neq 0$ for all $z \in B(a, r)$. From the identity

$$f(z)g(z) = 0,$$

it follows that $g(z) = 0$ whenever $f(z) \neq 0$. Therefore g vanishes on the open set $B(a, r)$. Since g is analytic and vanishes on an open set, the Identity Theorem implies that $g \equiv 0$. ■

9. Let $U: \mathbb{C} \rightarrow \mathbb{R}$ be a harmonic function such that

$$U(z) \geq 0 \quad \text{for all } z \in \mathbb{C}.$$

Prove that U is constant.

Proof. Let $U: \mathbb{C} \rightarrow \mathbb{R}$ be harmonic and assume that

$$U(z) \geq 0 \quad \text{for all } z \in \mathbb{C}.$$

Since U is harmonic on all of $\mathbb{C}$, it admits a harmonic conjugate V on $\mathbb{C}$ (because $\mathbb{C}$ is simply connected). Define

$$F(z) = U(z) + iV(z).$$

Then F is an entire function. Now consider the entire function

$$G(z) = e^{-F(z)}.$$

We compute its modulus:

$$|G(z)| = |e^{-F(z)}| = e^{-\Re(F(z))} = e^{-U(z)}.$$

Since $U(z) \geq 0$ for all z , we have

$$|G(z)| = e^{-U(z)} \leq 1.$$

Thus G is an entire function bounded in modulus by 1. By Liouville's theorem, every bounded entire function is constant, so G is constant. Therefore,

$$e^{-F(z)} = \text{constant},$$

which implies that $F(z)$ itself is constant. Since $F = U + iV$, its real part U must also be constant. Hence U is constant on $\mathbb{C}$. ■

10. Show that if f and g are analytic functions on a region G such that $\overline{f}g$ is analytic, then either f is constant or $g \equiv 0$.

Proof. Let f and g be analytic on a region G and suppose that the function

$$h(z) = \overline{f(z)}g(z)$$

is analytic on G . We will show that either f is constant or $g \equiv 0$. First observe that h is continuous and satisfies the Cauchy–Riemann equations. Let $z \in G$ be a point where $g(z) \neq 0$. Since g is analytic, there exists a neighborhood U of z on which $g(w) \neq 0$ for all $w \in U$. On this set U we may write

$$\overline{f(w)} = \frac{h(w)}{g(w)}, \quad w \in U,$$

which expresses $\overline{f}$ as the quotient of two analytic functions. Thus $\overline{f}$ is analytic on U . But if a function is analytic and conjugate-analytic on a region, it must be constant. To see this, note that if $\overline{f}$ is analytic, then f satisfies both the Cauchy–Riemann equations and their conjugates, which forces all first derivatives of f to

vanish. Hence f is constant on U . Since U is nonempty and f is analytic on the connected region G , the Identity Theorem implies that f is constant on all of G . ■

4.4 The index of a closed curve

3. Let $p(z)$ be a polynomial of degree n and let $R > 0$ be sufficiently large so that p never vanishes in $\{z : |z| \geq R\}$. If $\gamma(t) = Re^{it}$, $0 \leq t \leq 2\pi$, show that

$$\int_{\gamma} \frac{p'(z)}{p(z)} dz = 2\pi i n.$$

Proof. Let p be a polynomial of degree n . Then we can factor it as

$$p(z) = a_n \prod_{k=1}^n (z - z_k),$$

where $a_n \neq 0$ and the z_k are the zeros of p , listed with multiplicity.

By hypothesis, we choose $R > 0$ so large that

$$p(z) \neq 0 \quad \text{for all } |z| \geq R.$$

This means that all zeros z_k of p satisfy $|z_k| < R$, so they all lie inside the circle

$$\gamma(t) = Re^{it}, \quad 0 \leq t \leq 2\pi.$$

Differentiating the factorization of p , we have

$$\frac{p'(z)}{p(z)} = \frac{d}{dz} \log p(z) = \frac{d}{dz} \left(\log a_n + \sum_{k=1}^n \log(z - z_k) \right) = \sum_{k=1}^n \frac{1}{z - z_k},$$

where the last equality is valid for z not equal to any z_k . Therefore, on and inside γ ,

$$\frac{p'(z)}{p(z)} = \sum_{k=1}^n \frac{1}{z - z_k}.$$

Now integrate around γ :

$$\int_{\gamma} \frac{p'(z)}{p(z)} dz = \int_{\gamma} \sum_{k=1}^n \frac{1}{z - z_k} dz = \sum_{k=1}^n \int_{\gamma} \frac{1}{z - z_k} dz.$$

Each z_k lies inside the circle $|z| = R$, so by Cauchy's integral formula (or directly

by the standard winding number integral),

$$\int_{\gamma} \frac{1}{z - z_k} dz = 2\pi i.$$

Hence

$$\int_{\gamma} \frac{p'(z)}{p(z)} dz = \sum_{k=1}^n 2\pi i = 2\pi i n.$$

This completes the proof. ■

4. Fix $w = re^{i\theta} \neq 0$ and let γ be a rectifiable path in $\mathbb{C} - \{0\}$ from 1 to w . Show that there is an integer k such that

$$\int_{\gamma} z^{-1} dz = \log r + i\theta + 2\pi i k.$$

Proof. Let $w = re^{i\theta} \neq 0$ and let γ be any rectifiable path in $\mathbb{C} \setminus \{0\}$ from 1 to w .

Step 1: Compare γ to a reference path.

Define a specific path σ from 1 to w by first going radially on the real axis from 1 to r , then along the circle of radius r from angle 0 to θ :

$$\sigma = \sigma_1 * \sigma_2,$$

where

$$\sigma_1(t) = 1 + (r - 1)t, \quad 0 \leq t \leq 1, \quad \sigma_2(t) = re^{it\theta}, \quad 0 \leq t \leq 1.$$

Compute the integral along σ_1 :

$$\int_{\sigma_1} \frac{dz}{z} = \int_0^1 \frac{(r - 1) dt}{1 + (r - 1)t} = \int_1^r \frac{du}{u} = \log r.$$

Compute the integral along σ_2 :

$$\int_{\sigma_2} \frac{dz}{z} = \int_0^1 \frac{ri\theta e^{it\theta}}{re^{it\theta}} dt = \int_0^1 i\theta dt = i\theta.$$

Thus along the reference path σ we have

$$\int_{\sigma} \frac{dz}{z} = \log r + i\theta.$$

Step 2: Any other path differs by a loop around 0.

Now let γ be any rectifiable path in $\mathbb{C} \setminus \{0\}$ from 1 to w . Consider the closed

curve

$$\Gamma = \gamma - \sigma,$$

that is, follow γ from 1 to w and then follow σ backwards from w to 1. Then Γ is a closed rectifiable curve in $\mathbb{C} \setminus \{0\}$. By the winding number formula,

$$\int_{\Gamma} \frac{dz}{z} = 2\pi i n(\Gamma; 0),$$

for some integer $n(\Gamma; 0) = k \in \mathbb{Z}$. On the other hand,

$$\int_{\Gamma} \frac{dz}{z} = \int_{\gamma} \frac{dz}{z} - \int_{\sigma} \frac{dz}{z} = \int_{\gamma} \frac{dz}{z} - (\log r + i\theta).$$

Equating the two expressions for $\int_{\Gamma} \frac{dz}{z}$, we obtain

$$\int_{\gamma} \frac{dz}{z} - (\log r + i\theta) = 2\pi i k$$

for some integer k . Rewriting,

$$\int_{\gamma} z^{-1} dz = \log r + i\theta + 2\pi i k.$$

■

4.5 Cauchy's Theorem and Integral Formula

1. Suppose $f : G \rightarrow \mathbb{C}$ is analytic and define $\varphi : G \times G \rightarrow \mathbb{C}$ by

$$\varphi(z, w) = \frac{f(z) - f(w)}{z - w} \quad \text{if } z \neq w, \quad \varphi(z, z) = f'(z).$$

Prove that φ is continuous and for each fixed w , $z \rightarrow \varphi(z, w)$ is analytic.

Proof. Let $f : G \rightarrow \mathbb{C}$ be analytic and define

$$\varphi(z, w) = \begin{cases} \frac{f(z) - f(w)}{z - w}, & z \neq w, \\ f'(z), & z = w. \end{cases}$$

We will prove that φ is continuous on $G \times G$, and that for each fixed w , the map $z \mapsto \varphi(z, w)$ is analytic.

Step 1: Taylor-series representation.

For each fixed $w \in G$, the Taylor expansion of f about w is

$$f(z) = \sum_{k=0}^{\infty} \frac{f^{(k)}(w)}{k!} (z - w)^k,$$

valid for z sufficiently near w . Subtracting $f(w)$ and dividing by $(z - w)$ gives, for $z \neq w$,

$$\frac{f(z) - f(w)}{z - w} = \sum_{k=1}^{\infty} \frac{f^{(k)}(w)}{k!} (z - w)^{k-1}.$$

This motivates defining

$$\varphi(z, w) = \sum_{k=1}^{\infty} \frac{f^{(k)}(w)}{k!} (z - w)^{k-1}. \quad (*)$$

At $z = w$, all terms with $(k - 1) \geq 1$ vanish and $(*)$ gives $\varphi(w, w) = f'(w)$, so the series representation agrees with the given definition of φ .

Step 2: Analyticity in z .

Fix $w \in G$. Then $(*)$ is a power series in $(z - w)$:

$$\varphi(z, w) = f'(w) + \frac{f''(w)}{2!} (z - w) + \frac{f^{(3)}(w)}{3!} (z - w)^2 + \cdots.$$

A power series defines an analytic function in z on its disk of convergence. Thus $z \mapsto \varphi(z, w)$ is analytic in a neighborhood of w , and by analytic continuation is analytic on all of G .

Step 3: Continuity of φ on $G \times G$.

Each term

$$(z, w) \mapsto \frac{f^{(k)}(w)}{k!} (z - w)^{k-1}$$

is continuous on $G \times G$. On compact subsets the series $(*)$ converges uniformly (because it is the Taylor series of f), so the sum of the series is continuous. Therefore φ is continuous on $G \times G$. Finally, as $z \rightarrow w$, the power series $(*)$ satisfies

$$\lim_{z \rightarrow w} \varphi(z, w) = f'(w) = \varphi(w, w),$$

so φ is continuous on the diagonal as well. Hence φ is continuous on $G \times G$, and analytic in z for each fixed w . ■

5. Let γ be a closed rectifiable curve in $\mathbb{C}$ and $a \notin \gamma$. Show that for $n \geq 2$,

$$\int_{\gamma} (z - a)^{-n} dz = 0.$$

Proof. Let γ be a closed rectifiable curve in $\mathbb{C}$ and let $a \notin \gamma$. Fix an integer $n \geq 2$. We use Corollary 5.9 with the analytic function $f(z) = 1$ and $k = n - 1$. Since $f(z) = 1$ is constant, all of its derivatives of order ≥ 1 vanish, and in particular

$$f^{(n-1)}(a) = 0.$$

Corollary 5.9 states that for any analytic f ,

$$f^{(k)}(a) n(\gamma; a) = \frac{k!}{2\pi i} \int_{\gamma} \frac{f(z)}{(z-a)^{k+1}} dz.$$

Applying this with $f(z) = 1$ and $k = n - 1$ gives

$$0 \cdot n(\gamma; a) = \frac{(n-1)!}{2\pi i} \int_{\gamma} \frac{1}{(z-a)^n} dz.$$

Thus

$$\int_{\gamma} (z-a)^{-n} dz = 0.$$

This completes the proof. ■

6. Let f be analytic on $D = B(0; 1)$ and suppose $|f(z)| \leq 1$ for $|z| < 1$. Show that $|f'(0)| \leq 1$.

Proof. Let f be analytic on $D = B(0; 1)$ and assume $|f(z)| \leq 1$ for all $|z| < 1$. We wish to prove that $|f'(0)| \leq 1$.

Fix $0 < r < 1$ and consider the circle $\gamma_r(t) = re^{it}$, $0 \leq t \leq 2\pi$. Since γ_r is a closed rectifiable curve in D and 0 lies in the interior of the circle, we have $n(\gamma_r; 0) = 1$.

Apply Corollary 5.9 with $k = 1$ and $a = 0$:

$$f'(0) n(\gamma_r; 0) = \frac{1!}{2\pi i} \int_{\gamma_r} \frac{f(z)}{(z-0)^2} dz = \frac{1}{2\pi i} \int_{\gamma_r} \frac{f(z)}{z^2} dz.$$

Since $n(\gamma_r; 0) = 1$, we obtain

$$f'(0) = \frac{1}{2\pi i} \int_{\gamma_r} \frac{f(z)}{z^2} dz.$$

Parametrize γ_r by $z = re^{it}$. Then $dz = ire^{it}dt$, and so

$$f'(0) = \frac{1}{2\pi i} \int_0^{2\pi} \frac{f(re^{it})}{r^2 e^{2it}} (ire^{it}) dt = \frac{1}{2\pi r} \int_0^{2\pi} f(re^{it}) e^{-it} dt.$$

Taking absolute values,

$$|f'(0)| \leq \frac{1}{2\pi r} \int_0^{2\pi} |f(re^{it})| dt.$$

But $|f(re^{it})| \leq 1$ for all t , so

$$|f'(0)| \leq \frac{1}{2\pi r} \int_0^{2\pi} 1 dt = \frac{1}{2\pi r} (2\pi) = \frac{1}{r}.$$

Since this holds for every $0 < r < 1$, giving

$$|f'(0)| \leq 1.$$

This completes the proof. ■

7. Let $\gamma(t) = 1 + e^{it}$ for $0 \leq t \leq 2\pi$. Find

$$\int_{\gamma} \left(\frac{z}{z-1} \right)^n dz$$

for all positive integers n .

Proof. Let $\gamma(t) = 1 + e^{it}$ for $0 \leq t \leq 2\pi$. Then γ is the circle $|z-1| = 1$, which winds once counterclockwise around $z = 1$. Hence

$$n(\gamma; 1) = 1.$$

We wish to compute

$$I_n = \int_{\gamma} \left(\frac{z}{z-1} \right)^n dz.$$

Rewrite the integrand in the form required by Corollary 5.9:

$$\left(\frac{z}{z-1} \right)^n = z^n (z-1)^{-n} = \frac{f(z)}{(z-1)^n}, \quad f(z) = z^n.$$

Thus in Corollary 5.9 we take $k = n-1$, since $k+1 = n$. Corollary 5.9 states:

$$f^{(k)}(1) n(\gamma; 1) = \frac{k!}{2\pi i} \int_{\gamma} \frac{f(z)}{(z-1)^{k+1}} dz.$$

Substituting $k = n-1$ and $f(z) = z^n$ gives

$$f^{(n-1)}(1) = \frac{(n-1)!}{2\pi i} \int_{\gamma} \left(\frac{z}{z-1} \right)^n dz.$$

Next compute $f^{(n-1)}(1)$. Since $f(z) = z^n$,

$$f^{(n-1)}(z) = n! z,$$

and therefore

$$f^{(n-1)}(1) = n!.$$

Putting this into the previous equation yields

$$n! = \frac{(n-1)!}{2\pi i} \int_{\gamma} \left(\frac{z}{z-1} \right)^n dz.$$

Thus

$$\int_{\gamma} \left(\frac{z}{z-1} \right)^n dz = 2\pi i n.$$

This holds for all positive integers n . ■

8. Let G be a region and suppose $f_n : G \rightarrow \mathbb{C}$ is analytic for each $n \geq 1$. Suppose that $\{f_n\}$ converges uniformly to a function $f : G \rightarrow \mathbb{C}$. Show that f is analytic.

Proof. Since each f_n is analytic on G , it is continuous on G . Because $f_n \rightarrow f$ uniformly on G , the limit function f is also continuous on G . Let T be any triangular path in G . Since T is a compact curve, uniform convergence implies uniform convergence on ∂T as well. Therefore

$$\int_T f_n(z) dz \rightarrow \int_T f(z) dz. \quad (1)$$

But each f_n is analytic, so by Cauchy's theorem,

$$\int_T f_n(z) dz = 0 \quad \text{for all } n. \quad (2)$$

Taking the limit of (2) and using (1),

$$\int_T f(z) dz = \lim_{n \rightarrow \infty} \int_T f_n(z) dz = \lim_{n \rightarrow \infty} 0 = 0.$$

Thus for every triangular path $T \subset G$,

$$\int_T f(z) dz = 0.$$

Since f is continuous and its integral around every triangle vanishes, Morera's Theorem applies and implies that f is analytic on G . ■

9. Show that if $f : \mathbb{C} \rightarrow \mathbb{C}$ is a continuous function such that f is analytic off

$[-1, 1]$, then f is an entire function.

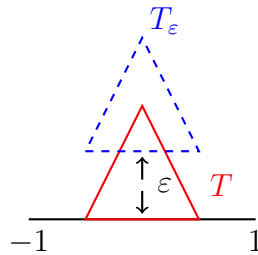


Figure 4.1: The triangle T touches the segment $[-1, 1]$ on the real axis. Shifting it vertically by $\varepsilon > 0$ produces the translated triangle $T_\varepsilon = T + i\varepsilon$, which lies entirely in $\mathbb{C} \setminus [-1, 1]$.

Proof. We assume $f : \mathbb{C} \rightarrow \mathbb{C}$ is continuous on $\mathbb{C}$ and analytic on $\mathbb{C} \setminus [-1, 1]$. By Morera's theorem, it suffices to prove that

$$\int_T f(z) dz = 0$$

for every triangular path T in $\mathbb{C}$.

Case 1: T and its interior avoid $[-1, 1]$. If

$$T \cup \text{Int}(T) \subset \mathbb{C} \setminus [-1, 1],$$

then f is analytic in a neighborhood of T , and by Cauchy's theorem,

$$\int_T f(z) dz = 0.$$

Case 2: One side of T lies on the segment $[-1, 1]$.

This is the configuration shown in Figure 4.1: the base of the triangle lies on $[-1, 1]$ while the remaining two sides lie in the upper half-plane. Let T denote this triangle. For $\varepsilon > 0$ define its vertical translate

$$T_\varepsilon = T + i\varepsilon.$$

As illustrated in Figure 4.1, when $\varepsilon > 0$ is small, the entire triangle T_ε (together with its interior) lies strictly above the real axis and therefore satisfies

$$T_\varepsilon \cup \text{Int}(T_\varepsilon) \subset \mathbb{C} \setminus [-1, 1].$$

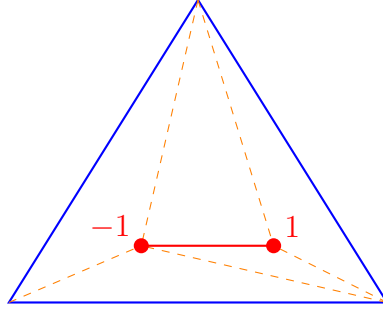


Figure 4.2: Subdivision of a triangle by two interior points 1 and -1 and the segment joining them.

Thus, by Case 1,

$$\int_{T_\varepsilon} f(z) dz = 0 \quad \text{for all sufficiently small } \varepsilon > 0.$$

Let $\gamma : [0, 1] \rightarrow \mathbb{C}$ parametrize T . Then

$$\gamma_\varepsilon(t) = \gamma(t) + i\varepsilon$$

parametrizes T_ε , and clearly $\gamma_\varepsilon \rightarrow \gamma$ *uniformly* on $[0, 1]$ as $\varepsilon \rightarrow 0$.

Since f is continuous on $\mathbb{C}$, it is uniformly continuous on the compact set

$$K := \bigcup_{0 \leq \varepsilon \leq \varepsilon_0} T_\varepsilon$$

for some small ε_0 . Therefore $f(\gamma_\varepsilon(t))$ converges uniformly to $f(\gamma(t))$. Hence we may pass the limit inside the integral:

$$\int_T f(z) dz = \int_0^1 f(\gamma(t)) \gamma'(t) dt = \lim_{\varepsilon \rightarrow 0} \int_0^1 f(\gamma_\varepsilon(t)) \gamma'_\varepsilon(t) dt = \lim_{\varepsilon \rightarrow 0} \int_{T_\varepsilon} f(z) dz = 0.$$

General case.

If T intersects $[-1, 1]$ in any other way, we divide T into finitely many smaller triangles by inserting straight segments between the points of intersection with $[-1, 1]$. A typical example of such a configuration is illustrated in Figure 4.2. Each subtriangle is either of type 1 or type 2. Applying the two cases above to each subtriangle and summing the resulting integrals yields

$$\int_T f(z) dz = 0.$$

Conclusion. Since $\int_T f dz = 0$ for every triangular path T and f is continuous on $\mathbb{C}$, Morera's theorem implies that f is analytic on all of $\mathbb{C}$. Thus f is entire. ■

4.6 The homotopic version of Cauchy's Theorem and simple connectivity

5. Evaluate the integral $\int_{\gamma} \frac{dz}{z^2 + 1}$ where $\gamma(\theta) = 2|\cos 2\theta| e^{i\theta}$ for $0 \leq \theta \leq 2\pi$.

Proof. We are given the curve

$$\gamma(\theta) = 2|\cos 2\theta| e^{i\theta}, \quad 0 \leq \theta \leq 2\pi,$$

and we wish to evaluate

$$\int_{\gamma} \frac{dz}{z^2 + 1}.$$

Step 1: Homotope γ to the circle $|z| = 2$. Define a homotopy

$$H(s, \theta) = [(1-s)2|\cos 2\theta| + 2s]e^{i\theta}, \quad 0 \leq s \leq 1, \quad 0 \leq \theta \leq 2\pi.$$

Then $H(0, \theta) = \gamma(\theta)$ and $H(1, \theta) = 2e^{i\theta}$. We claim $H(s, \theta) \neq \pm i$ for all (s, θ) . Indeed, $\pm i$ have modulus 1, whereas

$$|H(s, \theta)| = (1-s)2|\cos 2\theta| + 2s \geq 2s \geq 0,$$

and for $\theta = \frac{\pi}{2}, \frac{3\pi}{2}$ one has $|H(s, \theta)| = 2$. Thus H never hits $\pm i$, and therefore gives a homotopy in $\mathbb{C} \setminus \{\pm i\}$ from γ to the circle

$$\delta(\theta) = 2e^{i\theta}.$$

Since $f(z) = \frac{1}{z^2+1}$ is analytic on $\mathbb{C} \setminus \{\pm i\}$, it follows from homotopy invariance of contour integrals that

$$\int_{\gamma} \frac{dz}{z^2 + 1} = \int_{\delta} \frac{dz}{z^2 + 1}.$$

Step 2: Partial fractions and winding numbers. We write

$$\frac{1}{z^2 + 1} = \frac{1}{(z-i)(z+i)} = \frac{1}{2i} \left(\frac{1}{z-i} - \frac{1}{z+i} \right).$$

Thus

$$\int_{\delta} \frac{dz}{z^2 + 1} = \frac{1}{2i} \left(\int_{\delta} \frac{dz}{z-i} - \int_{\delta} \frac{dz}{z+i} \right).$$

Now $\delta(\theta) = 2e^{i\theta}$ winds once counterclockwise about both i and $-i$, since both satisfy $|\pm i| = 1 < 2$. Hence

$$n(\delta; i) = 1, \quad n(\delta; -i) = 1.$$

By the winding number formula,

$$\int_{\delta} \frac{dz}{z-i} = 2\pi i \cdot n(\delta; i) = 2\pi i, \quad \int_{\delta} \frac{dz}{z+i} = 2\pi i \cdot n(\delta; -i) = 2\pi i.$$

Therefore

$$\int_{\delta} \frac{dz}{z^2+1} = \frac{1}{2i}(2\pi i - 2\pi i) = 0.$$

Hence

$$\int_{\gamma} \frac{dz}{z^2+1} = 0.$$

■

6. Let $\gamma(\theta) = \theta e^{i\theta}$ for $0 \leq \theta \leq 2\pi$ and $\gamma(\theta) = 4\pi - \theta$ for $2\pi \leq \theta \leq 4\pi$. Evaluate $\int_{\gamma} \frac{dz}{z^2 + \pi^2}$.

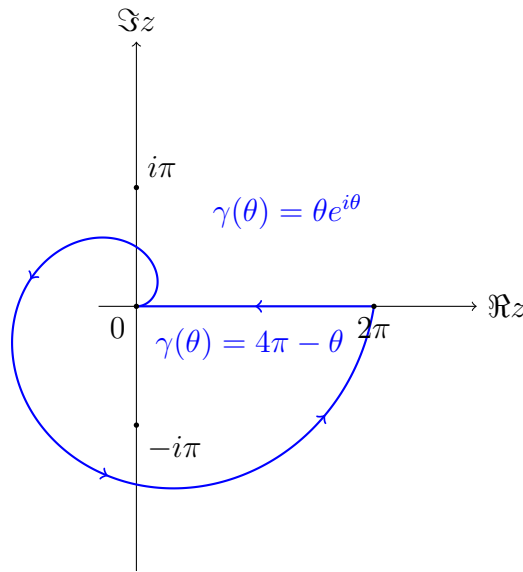


Figure 4.3: The path γ : a spiral from 0 to 2π followed by a straight line back to 0. The spiral encloses the point $-i\pi$ but not $i\pi$.

Proof. The curve γ is defined by

$$\gamma(\theta) = \begin{cases} \theta e^{i\theta}, & 0 \leq \theta \leq 2\pi, \\ 4\pi - \theta, & 2\pi \leq \theta \leq 4\pi. \end{cases}$$

For $0 \leq \theta \leq 2\pi$ the point $\theta e^{i\theta}$ traces a spiral beginning at 0, making one full revolution around the origin, and reaching the point 2π on the real axis. For $2\pi \leq$

$\theta \leq 4\pi$ the path travels along the real axis from 2π back to 0. The resulting curve is shown in Figure 4.3.

The curve winds once counterclockwise around the point $-i\pi$, which lies inside the spiral, and does not wind around the point $i\pi$, which lies outside the spiral. Thus the winding numbers are

$$n(\gamma; -i\pi) = 1, \quad n(\gamma; i\pi) = 0.$$

We compute the integral

$$\int_{\gamma} \frac{dz}{z^2 + \pi^2}.$$

Using the partial fraction decomposition

$$\frac{1}{z^2 + \pi^2} = \frac{1}{(z - i\pi)(z + i\pi)} = \frac{1}{2\pi i} \left(\frac{1}{z - i\pi} - \frac{1}{z + i\pi} \right),$$

we obtain

$$\int_{\gamma} \frac{dz}{z^2 + \pi^2} = \frac{1}{2\pi i} \left(\int_{\gamma} \frac{dz}{z - i\pi} - \int_{\gamma} \frac{dz}{z + i\pi} \right).$$

By the winding number formula,

$$\int_{\gamma} \frac{dz}{z - a} = 2\pi i n(\gamma; a),$$

so using the winding numbers read off from Figure 4.3,

$$\int_{\gamma} \frac{dz}{z - i\pi} = 2\pi i n(\gamma; i\pi) = 0, \quad \int_{\gamma} \frac{dz}{z + i\pi} = 2\pi i n(\gamma; -i\pi) = 2\pi i.$$

Substituting these into the previous expression gives

$$\int_{\gamma} \frac{dz}{z^2 + \pi^2} = \frac{1}{2\pi i} (0 - 2\pi i) = -1.$$

■

7. Let $f(z) = [(z - \frac{1}{2} - i)(z - 1 - \frac{3}{2}i)(z - 1 - i)(z - \frac{3}{2} - i)]^{-1}$ and let γ be the polygon $[0, 2, 2 + 2i, 2i, 0]$. Find $\int_{\gamma} f$.

Proof. Let

$$f(z) = \frac{1}{(z - z_1)(z - z_2)(z - z_3)(z - z_4)},$$

where

$$z_1 = \frac{1}{2} + i, \quad z_2 = 1 + \frac{3}{2}i, \quad z_3 = 1 + i, \quad z_4 = \frac{3}{2} + i.$$

These are four distinct points, so the partial fraction expansion has the form

$$f(z) = \sum_{k=1}^4 \frac{A_k}{z - z_k},$$

where

$$A_k = \frac{1}{\prod_{j \neq k} (z_k - z_j)}.$$

Thus

$$\int_{\gamma} f(z) dz = \sum_{k=1}^4 A_k \int_{\gamma} \frac{dz}{z - z_k}.$$

The path

$$\gamma = [0, 2, 2 + 2i, 2i, 0]$$

is the positively oriented boundary of the square

$$0 < \Re z < 2, \quad 0 < \Im z < 2.$$

A simple check of real and imaginary parts shows that all four points z_1, z_2, z_3, z_4 lie inside this square. Therefore

$$n(\gamma; z_k) = 1 \quad \text{for } k = 1, 2, 3, 4,$$

and hence

$$\int_{\gamma} \frac{dz}{z - z_k} = 2\pi i.$$

Therefore

$$\int_{\gamma} f(z) dz = \sum_{k=1}^4 A_k \cdot 2\pi i = 2\pi i \sum_{k=1}^4 A_k.$$

It remains to compute the four coefficients A_k . A direct computation gives

$$A_1 = -2 + 2i, \quad A_2 = -4i, \quad A_3 = -8i, \quad A_4 = 2 + 2i.$$

Adding them,

$$A_1 + A_2 + A_3 + A_4 = (-2 + 2) + (2i - 4i - 8i + 2i) = -8i.$$

Thus

$$\int_{\gamma} f(z) dz = 2\pi i (-8i) = 16\pi.$$

■

10. Find all possible values of $\int_{\gamma} \frac{dz}{1+z^2}$ where γ is any closed rectifiable curve.

Proof. We use the partial fraction decomposition

$$\frac{1}{1+z^2} = \frac{1}{(z-i)(z+i)} = \frac{1}{2i} \left(\frac{1}{z-i} - \frac{1}{z+i} \right).$$

For any closed rectifiable curve γ and any $a \in \mathbb{C}$, we have

$$\int_{\gamma} \frac{dz}{z-a} = 2\pi i n(\gamma; a),$$

where $n(\gamma; a)$ is the winding number of γ about a . Therefore

$$\int_{\gamma} \frac{dz}{1+z^2} = \frac{1}{2i} (2\pi i n(\gamma; i) - 2\pi i n(\gamma; -i)) = \pi (n(\gamma; i) - n(\gamma; -i)).$$

Since $n(\gamma; i)$ and $n(\gamma; -i)$ may be any integers (positive, negative, or zero) depending on how the curve winds around i and $-i$, the integral may take any value of the form

$$\{\pi k : k \in \mathbb{Z}\}.$$

■

4.7 Counting zeros; the Open Mapping Theorem

4. Suppose that $f : G \rightarrow \mathbb{C}$ is analytic and one-one; show that $f'(z) \neq 0$ for any z in G .

Proof. Let $f : G \rightarrow \mathbb{C}$ be analytic and one-one, and fix a point $a \in G$. We show that $f'(a) \neq 0$. Suppose for contradiction that $f'(a) = 0$. Then $f(z) - f(a)$ has a zero at $z = a$ of multiplicity $m \geq 2$. By Theorem 7.4, there exist $\varepsilon > 0$ and $\delta > 0$ such that whenever $0 < |\zeta - f(a)| < \delta$, the equation

$$f(z) = \zeta$$

has *exactly* m distinct solutions in the disk $B(a; \varepsilon)$. Since $m \geq 2$, every ζ sufficiently close to $f(a)$ is taken at least twice by f . This contradicts the assumption that f is one-one on G . Therefore our assumption was impossible, and we must have $f'(a) \neq 0$. As a was arbitrary, it follows that

$$f'(z) \neq 0 \quad \text{for all } z \in G.$$



Chapter 5

Singularity

5.1 Classification of singularities

4. Let

$$f(z) = \frac{1}{z(z-1)(z-2)};$$

give the Laurent expansion of $f(z)$ in each of the following annuli: (a) $\text{ann}(0; 0, 1)$; (b) $\text{ann}(0; 1, 2)$; (c) $\text{ann}(0; 2, \infty)$.

Solution. We start from the partial fraction decomposition

$$f(z) = \frac{1}{z(z-1)(z-2)} = \frac{1}{2z} - \frac{1}{z-1} + \frac{1}{2(z-2)}.$$

We expand in powers of z using geometric series formula.

(a) The annulus $0 < |z| < 1$.

$$\frac{1}{z-1} = -\sum_{n=0}^{\infty} z^n, \quad \frac{1}{z-2} = -\frac{1}{2} \sum_{n=0}^{\infty} \left(\frac{z}{2}\right)^n.$$

Thus

$$f(z) = \frac{1}{2}z^{-1} + \sum_{n=0}^{\infty} z^n - \frac{1}{4} \sum_{n=0}^{\infty} 2^{-n} z^n.$$

Therefore the Laurent coefficients are

$$a_{-1} = \frac{1}{2}, \quad a_n = 1 - 2^{-(n+2)} \quad (n \geq 0), \quad a_n = 0 \quad (n \leq -2).$$

So

$$f(z) = \frac{1}{2}z^{-1} + \sum_{n=0}^{\infty} (1 - 2^{-(n+2)}) z^n, \quad 0 < |z| < 1.$$

(b) The annulus $1 < |z| < 2$.

$$\frac{1}{z-1} = \frac{1}{z} \frac{1}{1-\frac{1}{z}} = \sum_{n=1}^{\infty} z^{-n}, \quad \frac{1}{z-2} = -\frac{1}{2} \frac{1}{1-\frac{z}{2}} = -\frac{1}{2} \sum_{n=0}^{\infty} \left(\frac{z}{2}\right)^n.$$

Hence

$$f(z) = \frac{1}{2}z^{-1} - \sum_{n=1}^{\infty} z^{-n} - \frac{1}{4} \sum_{n=0}^{\infty} 2^{-n} z^n.$$

The coefficients are

$$a_{-1} = -\frac{1}{2}, \quad a_{-n} = -1 \quad (n \geq 2), \quad a_n = -2^{-(n+2)} \quad (n \geq 0).$$

Thus,

$$f(z) = -\frac{1}{2}z^{-1} - \sum_{n=2}^{\infty} z^{-n} - \sum_{n=0}^{\infty} 2^{-(n+2)} z^n, \quad 1 < |z| < 2.$$

(c) The annulus $|z| > 2$.

To apply the geometric series here, we factor out z so that $|1/z|$ and $|2/z|$ are less than 1 in the annuli.

$$\frac{1}{z-1} = \frac{1}{z} \frac{1}{1-\frac{1}{z}} = \sum_{n=1}^{\infty} z^{-n}, \quad \frac{1}{z-2} = \frac{1}{z} \frac{1}{1-\frac{2}{z}} = \sum_{n=1}^{\infty} \frac{2^{n-1}}{z^n}.$$

Hence

$$f(z) = \frac{1}{2}z^{-1} - \sum_{n=1}^{\infty} z^{-n} + \sum_{n=1}^{\infty} 2^{n-2} z^{-n}.$$

The coefficients are

$$a_{-1} = 0, \quad a_{-n} = 2^{n-2} - 1 \quad (n \geq 2), \quad a_n = 0 \quad (n \geq 0).$$

Thus

$$f(z) = \sum_{n=2}^{\infty} (2^{n-2} - 1) z^{-n}, \quad |z| > 2.$$

■

15. Let f be analytic in

$$G = \{z : 0 < |z - a| < r\}$$

except that there is a sequence of poles $\{a_n\}$ in G with $a_n \rightarrow a$. Show that for any $\omega \in \mathbb{C}$ there exists a sequence $\{z_n\} \subset G$ with

$$a = \lim z_n \quad \text{and} \quad \omega = \lim f(z_n).$$

Proof. Fix $\omega \in \mathbb{C}$. We show that there exists a sequence $z_n \in G$ with $z_n \rightarrow a$ and

$f(z_n) \rightarrow \omega$.

Suppose for contradiction that no such sequence exists. We claim that there is $\delta > 0$ such that

$$0 < |z - a| < \delta \implies f(z) \neq \omega. \quad (5.1)$$

Indeed, if (5.1) were false, then for each n we could choose $z_n \in G$ with $0 < |z_n - a| < 1/n$ and $f(z_n) = \omega$, which would imply $z_n \rightarrow a$ and $f(z_n) \rightarrow \omega$, contradicting the assumption. Hence (5.1) holds for some $\delta > 0$.

On the punctured disk $D^* := \{z : 0 < |z - a| < \delta\}$ define

$$g(z) := \frac{1}{f(z) - \omega}$$

whenever $z \neq a_n$. Then g is analytic on $D^* \setminus \{a_n\}$.

Step 1: Each a_n is a removable singularity of g . Let $a_n \in D^*$ be a pole of f , say of order $m \geq 1$. Then there is a holomorphic h near a_n with $h(a_n) \neq 0$ such that

$$f(z) = \frac{h(z)}{(z - a_n)^m}.$$

Hence

$$f(z) - \omega = \frac{h(z) - \omega(z - a_n)^m}{(z - a_n)^m},$$

and the numerator satisfies $(h - \omega(\cdot - a_n)^m)(a_n) = h(a_n) \neq 0$. Therefore $f(z) - \omega$ has a pole of order m at a_n , so $g(z)$ has a zero of order m at a_n and extends holomorphically across a_n by setting $g(a_n) = 0$.

Doing this for each $a_n \in D^*$, we obtain a holomorphic function (still denoted g) on D^* whose zeros include all a_n in D^* . Since $a_n \rightarrow a$, g has infinitely many zeros accumulating at a .

Step 2: The singularity of g at a is essential. The point a is an isolated singularity of g (because g is holomorphic on D^*). It cannot be removable: if g extended holomorphically to a , then the extension would be holomorphic on a neighborhood of a and vanish at infinitely many points $a_n \rightarrow a$, hence would vanish identically by the identity theorem, impossible since $g(z) = 1/(f(z) - \omega)$ is not identically zero.

It also cannot be a pole: if a were a pole of order $m \geq 1$, then $g(z) = h(z)/(z - a)^m$ near a with h holomorphic and $h(a) \neq 0$. But for n large, a_n lies in this neighborhood and $0 = g(a_n) = h(a_n)/(a_n - a)^m$, so $h(a_n) = 0$ for infinitely many $a_n \rightarrow a$, forcing $h \equiv 0$ near a by the identity theorem, contradicting $h(a) \neq 0$.

Thus a is neither removable nor a pole, hence it is an essential singularity of g .

Step 3: Produce a sequence with $f(z_n) \rightarrow \omega$. By Casorati–Weierstrass (equivalently, by the fact that a function with an essential singularity is unbounded on every punctured neighborhood), g is unbounded on each $\{0 < |z - a| < 1/n\}$. Hence

we can choose z_n such that

$$0 < |z_n - a| < \frac{1}{n} \quad \text{and} \quad |g(z_n)| > n.$$

Then $z_n \rightarrow a$ and $|g(z_n)| \rightarrow \infty$, so $1/g(z_n) \rightarrow 0$. But

$$\frac{1}{g(z_n)} = f(z_n) - \omega,$$

so $f(z_n) \rightarrow \omega$, contradicting the initial assumption.

Therefore for this ω there exists a sequence $z_n \in G$ with $z_n \rightarrow a$ and $f(z_n) \rightarrow \omega$. Since $\omega \in \mathbb{C}$ was arbitrary, the conclusion holds for every $\omega \in \mathbb{C}$. ■

5.2 Residues

2 (c). *Verify the following equation:*

$$\int_0^\infty \frac{\cos(ax)}{(1+x^2)^2} dx = \frac{\pi(a+1)e^{-a}}{4}, \quad a > 0.$$

Solution. We want to show that for $a > 0$,

$$\int_0^\infty \frac{\cos(ax)}{(1+x^2)^2} dx = \frac{\pi(a+1)e^{-a}}{4}.$$

Consider the complex-valued function

$$f(z) = \frac{e^{iaz}}{(1+z^2)^2}, \quad a > 0,$$

and for $R > 1$ integrate it over the semicircular contour C_R in the upper half-plane, consisting of the segment $[-R, R]$ on the real axis and the semicircle

$$\gamma_R : z = Re^{it}, \quad t \in [0, \pi].$$

The singularities of f are at $z = i$ and $z = -i$. In the upper half-plane only $z = i$ lies inside the contour, and it is a pole of order two. By the residue theorem,

$$\int_{C_R} f(z) dz = 2\pi i \operatorname{Res}(f; i).$$

The integral over the semicircle. We claim that

$$\int_{\gamma_R} f(z) dz \longrightarrow 0 \quad \text{as } R \rightarrow \infty.$$

On γ_R we have $z = Re^{it}$, $t \in [0, \pi]$, hence

$$|e^{iaz}| = |e^{iaRe^{it}}| = e^{\Re(iaRe^{it})} = e^{-aR \sin t} \leq 1.$$

Moreover, for $R > 1$,

$$|1 + z^2| = |1 + R^2 e^{2it}| \geq ||z|^2 - 1| = R^2 - 1,$$

so

$$|(1 + z^2)^2| \geq (R^2 - 1)^2.$$

Therefore, for $R > 1$,

$$\sup_{z \in \gamma_R} |f(z)| \leq \frac{1}{(R^2 - 1)^2}.$$

The length of γ_R is πR , so

$$\left| \int_{\gamma_R} f(z) dz \right| \leq (\text{length of } \gamma_R) \cdot \sup_{z \in \gamma_R} |f(z)| \leq \pi R \cdot \frac{1}{(R^2 - 1)^2} \longrightarrow 0 \quad \text{as } R \rightarrow \infty.$$

Thus

$$\int_{-\infty}^{\infty} \frac{e^{iax}}{(1 + x^2)^2} dx = 2\pi i \operatorname{Res}(f; i).$$

Residue at $z = i$. We write

$$f(z) = \frac{e^{iaz}}{(z^2 + 1)^2} = \frac{e^{iaz}}{(z - i)^2(z + i)^2}.$$

Thus $z = i$ is a double pole. For a double pole at $z = i$ we have

$$\operatorname{Res}(f; i) = \frac{d}{dz} [(z - i)^2 f(z)] \Big|_{z=i} = \frac{d}{dz} \left(\frac{e^{iaz}}{(z + i)^2} \right) \Big|_{z=i}.$$

Set

$$g(z) = \frac{e^{iaz}}{(z + i)^2}.$$

Then

$$g'(z) = \frac{e^{iaz}(ia(z + i)^2 - 2(z + i))}{(z + i)^4}.$$

So

$$g'(i) = \frac{e^{iai}(ia(-4) - 2(2i))}{16} = -\frac{i}{4}(a+1)e^{-a}.$$

Hence

$$\text{Res}(f; i) = -\frac{i}{4}(a+1)e^{-a}.$$

Thus

$$\int_{-\infty}^{\infty} \frac{e^{iax}}{(1+x^2)^2} dx = 2\pi i \left(-\frac{i}{4}(a+1)e^{-a} \right) = \frac{\pi}{2}(a+1)e^{-a}.$$

Extracting the cosine integral. Write

$$\int_{-\infty}^{\infty} \frac{e^{iax}}{(1+x^2)^2} dx = \int_{-\infty}^{\infty} \frac{\cos(ax)}{(1+x^2)^2} dx + i \int_{-\infty}^{\infty} \frac{\sin(ax)}{(1+x^2)^2} dx.$$

The integrand of the sine term is odd, so

$$\int_{-\infty}^{\infty} \frac{\sin(ax)}{(1+x^2)^2} dx = 0.$$

The integrand of the cosine term is even, hence

$$\int_{-\infty}^{\infty} \frac{\cos(ax)}{(1+x^2)^2} dx = 2 \int_0^{\infty} \frac{\cos(ax)}{(1+x^2)^2} dx.$$

Therefore,

$$2 \int_0^{\infty} \frac{\cos(ax)}{(1+x^2)^2} dx = \frac{\pi}{2}(a+1)e^{-a},$$

which implies

$$\int_0^{\infty} \frac{\cos(ax)}{(1+x^2)^2} dx = \frac{\pi}{4}(a+1)e^{-a},$$

as required. ■

5.3 The Argument Principle

2. Suppose f is analytic on $\overline{B(0;1)}$ and satisfies $|f(z)| < 1$ for $|z| = 1$. Find the number of solutions (counting multiplicities) of the equation

$$f(z) = z^n$$

where n is an integer larger than or equal to 1.

Solution. We seek the number of solutions of

$$f(z) = z^n \quad (n \geq 1).$$

Rewrite this as

$$f(z) - z^n = 0.$$

Define

$$h(z) = z^n, \quad g(z) = f(z) - z^n.$$

Both g and h are analytic on and inside the unit circle $\gamma = \{z : |z| = 1\}$.

On γ we have $|h(z)| = |z^n| = 1$, and by hypothesis

$$|h(z) + g(z)| = |f(z)| < 1 = |h(z)|.$$

Thus the inequality required by Rouché's theorem holds on γ . Therefore h and g have the same number of zeros inside γ . Since $h(z) = z^n$ has exactly n zeros in $|z| < 1$, it follows that $g(z) = f(z) - z^n$ has exactly n zeros in $|z| < 1$. Thus the equation $f(z) = z^n$ has exactly n solutions in the unit disk. ■

Chapter 6

The Maximum Modulus Theorem

6.1 The Maximum Principle

2. Let G be a bounded region and suppose f is continuous on $\overline{G}$ and analytic on G . Show that if there is a constant $c \geq 0$ such that $|f(z)| = c$ for all z on the boundary of G , then either f is a constant function or f has a zero in G .

Solution. Let G be a bounded region, and let f be continuous on $\overline{G}$ and analytic on G . Assume that

$$|f(z)| = c \quad \text{for all } z \in \partial G,$$

where $c \geq 0$.

Case 1: $c = 0$. Then $f = 0$ on ∂G . By the maximum modulus principle,

$$\max_{\overline{G}} |f| = \max_{\partial G} |f| = 0,$$

so $f \equiv 0$ on G , hence f is constant.

Case 2: $c > 0$. Suppose that f has no zeros in G . Then $|f|$ is a positive continuous function on $\overline{G}$ and is the modulus of a nonvanishing analytic function on G . Hence, by the minimum modulus principle,

$$\min_{\overline{G}} |f| = \min_{\partial G} |f|.$$

Since $|f| = c$ on ∂G , we get

$$\min_{\overline{G}} |f| = c,$$

and therefore $|f(z)| \geq c$ for all $z \in G$.

On the other hand, by the maximum modulus principle applied to f ,

$$\max_{\overline{G}} |f| = \max_{\partial G} |f| = c,$$

so $|f(z)| \leq c$ for all $z \in G$.

Combining the two inequalities yields $|f(z)| = c$ for all $z \in G$. By the maximum modulus principle (in its strong form), an analytic function whose modulus is constant on a region must be constant. Hence f is constant.

Therefore, either f is constant or f has a zero in G . ■

6. Suppose that both f and g are analytic on $\overline{B(0; R)}$ with

$$|f(z)| = |g(z)| \quad \text{for } |z| = R.$$

Show that if neither f nor g vanishes in $B(0; R)$, then there is a constant λ , with $|\lambda| = 1$, such that $f = \lambda g$.

Solution. Assume f and g are analytic on $\overline{B(0; R)}$ and satisfy $|f(z)| = |g(z)|$ for $|z| = R$. Also assume neither f nor g vanishes in $B(0; R)$.

Since g has no zeros in $B(0; R)$, the quotient

$$h(z) := \frac{f(z)}{g(z)}$$

is analytic on $B(0; R)$. On the boundary $|z| = R$ we have

$$|h(z)| = \frac{|f(z)|}{|g(z)|} = 1.$$

Because h is continuous on $\overline{B(0; R)}$ and $|h|$ attains both its maximum and minimum on the boundary, the maximum modulus principle gives

$$|h(z)| \leq 1 \quad \text{for } |z| < R,$$

and since h is nonvanishing (as f is also nonvanishing in $B(0; R)$), the minimum modulus principle gives

$$|h(z)| \geq 1 \quad \text{for } |z| < R.$$

Hence $|h(z)| \equiv 1$ throughout $B(0; R)$.

An analytic function with constant modulus on a region must be constant. Therefore h is constant, say $h \equiv \lambda$, and then $|\lambda| = 1$ (because $|h| = 1$). Thus

$$f = \lambda g \quad \text{on } B(0; R),$$

and by continuity the same holds on $\overline{B(0; R)}$. ■

7. Let f be analytic in the disk $B(0; R)$ and for $0 \leq r < R$ define

$$A(r) = \max\{\Re f(z) : |z| = r\}.$$

Show that unless f is a constant, $A(r)$ is a strictly increasing function of r .

Solution. Define

$$g(z) = e^{f(z)}.$$

Since f is analytic in $B(0; R)$, the function g is also analytic there. Moreover,

$$|g(z)| = |e^{f(z)}| = e^{\Re f(z)}.$$

Hence for each $0 \leq r < R$,

$$\max_{|z|=r} |g(z)| = \max_{|z|=r} e^{\Re f(z)} = e^{\max_{|z|=r} \Re f(z)} = e^{A(r)}.$$

Now let

$$M(r) = \max_{|z|=r} |g(z)|.$$

By the maximum modulus principle, $M(r)$ is an increasing function of r . Therefore $e^{A(r)}$ is increasing, and since the exponential function is strictly increasing on $\mathbb{R}$, it follows that $A(r)$ is increasing.

It remains to prove strict increase unless f is constant. Suppose there exist $0 \leq r_1 < r_2 < R$ such that

$$A(r_1) = A(r_2).$$

Then

$$M(r_1) = e^{A(r_1)} = e^{A(r_2)} = M(r_2).$$

So the maximum modulus of the analytic function g is the same on the circles $|z| = r_1$ and $|z| = r_2$. Since $r_1 < r_2$, the maximum modulus principle implies that g must be constant on $B(0; r_2)$. Thus $e^{f(z)}$ is constant, say

$$e^{f(z)} = c \neq 0.$$

Differentiating, we get

$$f'(z)e^{f(z)} = 0.$$

Because $e^{f(z)} \neq 0$, it follows that

$$f'(z) = 0$$

for all $z \in B(0; r_2)$. Hence f is constant on $B(0; r_2)$, and therefore constant on $B(0; R)$.

Thus, unless f is constant, equality $A(r_1) = A(r_2)$ cannot happen when $r_1 < r_2$. Therefore

$$A(r_1) < A(r_2) \quad \text{whenever } 0 \leq r_1 < r_2 < R.$$

So $A(r)$ is strictly increasing unless f is constant. ■

6.2 Schwarz's Lemma

1. Suppose $|f(z)| \leq 1$ for $|z| < 1$ and f is a non-constant analytic function. By considering the function $g : \mathbb{D} \rightarrow \mathbb{D}$ defined by

$$g(z) = \frac{f(z) - a}{1 - \bar{a}f(z)}$$

where $a = f(0)$, prove that

$$\frac{|f(0)| - |z|}{1 + |f(0)||z|} \leq |f(z)| \leq \frac{|f(0)| + |z|}{1 - |f(0)||z|}$$

for $|z| < 1$.

Solution. Let $a = f(0)$ and define

$$g(z) = \frac{f(z) - a}{1 - \bar{a}f(z)}.$$

Since $|f(z)| \leq 1$ for $|z| < 1$ and $|a| < 1$, the map

$$w \mapsto \frac{w - a}{1 - \bar{a}w}$$

is an automorphism of the unit disk $\mathbb{D}$. Hence $g : \mathbb{D} \rightarrow \mathbb{D}$ is analytic. Moreover,

$$g(0) = \frac{f(0) - a}{1 - \bar{a}f(0)} = 0.$$

Because f is non-constant, g is also non-constant. Therefore by Schwarz's lemma,

$$|g(z)| \leq |z| \quad \text{for } |z| < 1.$$

Thus

$$\left| \frac{f(z) - a}{1 - \bar{a}f(z)} \right| \leq |z|.$$

Let $w = f(z)$. Then

$$|w - a| \leq |z| |1 - \bar{a}w|.$$

Upper bound. Using the triangle inequality,

$$|w - a| \geq ||w| - |a|| \geq |w| - |a|$$

and

$$|1 - \bar{a}w| \leq 1 + |a||w|.$$

Hence

$$|w| - |a| \leq |z|(1 + |a||w|).$$

Rearranging gives

$$|w|(1 - |a||z|) \leq |a| + |z|,$$

so

$$|f(z)| = |w| \leq \frac{|a| + |z|}{1 - |a||z|}.$$

Lower bound. Again from

$$|w - a| \leq |z||1 - \bar{a}w|,$$

we use

$$|w - a| \geq |a| - |w|$$

and

$$|1 - \bar{a}w| \leq 1 + |a||w|.$$

Thus

$$|a| - |w| \leq |z|(1 + |a||w|).$$

Rearranging yields

$$|a| - |z| \leq |w|(1 + |a||z|),$$

and therefore

$$|f(z)| = |w| \geq \frac{|a| - |z|}{1 + |a||z|}.$$

Since $a = f(0)$, we conclude that for $|z| < 1$,

$$\frac{|f(0)| - |z|}{1 + |f(0)||z|} \leq |f(z)| \leq \frac{|f(0)| + |z|}{1 - |f(0)||z|}.$$

■

3. Suppose $f : \mathbb{D} \rightarrow \mathbb{C}$ satisfies $\Re f(z) \geq 0$ for all $z \in \mathbb{D}$ and suppose that f is analytic and not constant.

(a) Show that $\Re f(z) > 0$ for all $z \in \mathbb{D}$.

(b) By using an appropriate Möbius transformation, apply Schwarz's Lemma to prove that if $f(0) = 1$ then

$$|f(z)| \leq \frac{1 + |z|}{1 - |z|}$$

for $|z| < 1$. What can be said if $f(0) \neq 1$?

(c) Show that if $f(0) = 1$, f also satisfies

$$|f(z)| \geq \frac{1 - |z|}{1 + |z|}.$$

Hint: Use part (a).

Solution. (a) Let

$$u(z) = \Re f(z).$$

Since f is analytic on $\mathbb{D}$, the function u is harmonic on $\mathbb{D}$. We are given that

$$u(z) \geq 0 \quad (z \in \mathbb{D}).$$

Suppose there exists $z_0 \in \mathbb{D}$ such that $u(z_0) = 0$. Then u attains its minimum value at an interior point. By the minimum principle for harmonic functions, u must be constant on $\mathbb{D}$. Hence $\Re f$ is constant.

If $f = u + iv$ and u is constant, then $u_x = u_y = 0$. By the Cauchy–Riemann equations,

$$v_x = -u_y = 0, \quad v_y = u_x = 0,$$

so v is also constant. Therefore f is constant, contrary to hypothesis. Hence no such z_0 exists, and so

$$\Re f(z) > 0 \quad \text{for all } z \in \mathbb{D}.$$

(b) Assume $f(0) = 1$. Since $\Re f(z) > 0$ for all $z \in \mathbb{D}$, the image of f lies in the right half-plane. Consider the Möbius transformation

$$\phi(w) = \frac{w - 1}{w + 1}.$$

This maps the right half-plane $\{w : \Re w > 0\}$ into $\mathbb{D}$. Define

$$g(z) = \phi(f(z)) = \frac{f(z) - 1}{f(z) + 1}.$$

Then $g : \mathbb{D} \rightarrow \mathbb{D}$ is analytic, and

$$g(0) = \frac{f(0) - 1}{f(0) + 1} = 0.$$

By Schwarz's Lemma,

$$|g(z)| \leq |z| \quad (z \in \mathbb{D}).$$

Thus

$$\left| \frac{f(z) - 1}{f(z) + 1} \right| \leq |z|.$$

Let $r = |z|$ and write $w = f(z)$. Then

$$|w - 1| \leq r |w + 1|.$$

Using the triangle inequality,

$$|w| - 1 \leq |w - 1| \leq r |w + 1| \leq r(|w| + 1).$$

Hence

$$|w| - 1 \leq r|w| + r,$$

so

$$|w|(1 - r) \leq 1 + r.$$

Therefore

$$|f(z)| \leq \frac{1 + |z|}{1 - |z|}.$$

If $f(0) \neq 1$, let $a = f(0)$, so $\Re a > 0$. Then

$$\psi(w) = \frac{w - a}{w + \bar{a}}$$

maps the right half-plane into $\mathbb{D}$, and $\psi(a) = 0$. Thus

$$h(z) = \psi(f(z)) = \frac{f(z) - a}{f(z) + \bar{a}}$$

is analytic from $\mathbb{D}$ to $\mathbb{D}$ with $h(0) = 0$. By Schwarz's Lemma,

$$\left| \frac{f(z) - a}{f(z) + \bar{a}} \right| \leq |z|.$$

This gives the corresponding estimate for general $f(0) = a$:

$$|f(z)| \leq \frac{|a|(1 + |z|)}{1 - |z|}.$$

(c) Again assume $f(0) = 1$. From part (a), $\Re f(z) > 0$ for all $z \in \mathbb{D}$. Hence

$$\Re \left(\frac{1}{f(z)} \right) = \frac{\Re f(z)}{|f(z)|^2} > 0,$$

so the function

$$F(z) = \frac{1}{f(z)}$$

is analytic on $\mathbb{D}$, non-constant, and satisfies $\Re F(z) > 0$ for all $z \in \mathbb{D}$. Also,

$$F(0) = \frac{1}{f(0)} = 1.$$

Applying part (b) to F , we obtain

$$|F(z)| \leq \frac{1 + |z|}{1 - |z|}.$$

That is,

$$\frac{1}{|f(z)|} \leq \frac{1 + |z|}{1 - |z|}.$$

Taking reciprocals gives

$$|f(z)| \geq \frac{1 - |z|}{1 + |z|}.$$

This proves all three parts. ■

6. Suppose f is analytic in some region containing $\overline{B}(0, 1)$ and $|f(z)| = 1$ where $|z| = 1$. Find a formula for f . Hint: First consider the case where f has no zeros in $B(0, 1)$.

Solution. We shall show that f must be a finite Blaschke product. That is,

$$f(z) = \lambda \prod_{k=1}^n \frac{z - a_k}{1 - \overline{a_k}z}, \quad |\lambda| = 1, \quad |a_k| < 1.$$

Since f is analytic in a region containing $\overline{B}(0, 1)$, it is analytic on a neighborhood of the closed unit disk. Because

$$|f(z)| = 1 \quad (|z| = 1),$$

f has no zeros on the unit circle. Let $a_1, \dots, a_n$ be the zeros of f in $B(0, 1)$, counted with multiplicity. Since f is analytic on a neighborhood of $\overline{B}(0, 1)$, it has only finitely many such zeros.

Case 1: f has no zeros in $B(0, 1)$.

Then f has no zeros on $\overline{B}(0, 1)$, so $1/f$ is analytic on a neighborhood of $\overline{B}(0, 1)$. Since f is analytic on the disk and continuous on the boundary, the maximum modulus principle implies

$$|f(z)| \leq 1 \quad (|z| < 1).$$

Because $1/f$ is also analytic and satisfies

$$\left| \frac{1}{f(z)} \right| = 1 \quad (|z| = 1),$$

the maximum modulus principle applied to $1/f$ gives

$$\left| \frac{1}{f(z)} \right| \leq 1 \quad (|z| < 1),$$

hence

$$|f(z)| \geq 1.$$

Therefore

$$|f(z)| = 1 \quad (|z| < 1).$$

If f were nonconstant, the open mapping theorem would imply that $f(B(0, 1))$ is open, which is impossible since it lies in the unit circle. Thus f is constant:

$$f(z) \equiv \lambda, \quad |\lambda| = 1.$$

Case 2: general case.

Let

$$B(z) = \prod_{k=1}^n \frac{z - a_k}{1 - \overline{a_k}z}.$$

This is a finite Blaschke product. Since $|a_k| < 1$, each factor is analytic on a neighborhood of $\overline{B}(0, 1)$. Moreover, for $|z| = 1$,

$$\left| \frac{z - a_k}{1 - \overline{a_k}z} \right| = 1,$$

so

$$|B(z)| = 1 \quad (|z| = 1).$$

Define

$$g(z) = \frac{f(z)}{B(z)}.$$

The zeros of B in $B(0, 1)$ are exactly $a_1, \dots, a_n$ with the same multiplicities, so the singularities of f/B at these points are removable. Hence g is analytic in $B(0, 1)$, and in fact analytic on a neighborhood of $\overline{B}(0, 1)$.

Moreover, g has no zeros in $B(0, 1)$, and for $|z| = 1$,

$$|g(z)| = \frac{|f(z)|}{|B(z)|} = 1.$$

Thus g satisfies the conditions of Case 1, so g is constant:

$$g(z) \equiv \lambda \quad \text{for some } |\lambda| = 1.$$

Therefore

$$f(z) = \lambda B(z) = \lambda \prod_{k=1}^n \frac{z - a_k}{1 - \overline{a_k}z}, \quad |\lambda| = 1, \quad |a_k| < 1.$$

Hence every analytic function on a neighborhood of $\overline{B}(0, 1)$ satisfying $|f(z)| = 1$ on $|z| = 1$ is a finite Blaschke product times a unimodular constant. ■

6.3 Convex functions and Hadamard's Three Circles Theorem

4. Supply the details in the proof of Hadamard's Three Circles Theorem.

Solution. Let

$$A = \text{ann}(0; R_1, R_2) = \{z : R_1 < |z| < R_2\},$$

and suppose f is analytic on A and not identically zero. For $R_1 < r < R_2$, define

$$M(r) = \max\{|f(re^{i\theta})| : 0 \leq \theta \leq 2\pi\}.$$

We will prove that for $R_1 < r_1 \leq r \leq r_2 < R_2$, $r_1 \neq r_2$,

$$\log M(r) \leq \frac{\log r_2 - \log r}{\log r_2 - \log r_1} \log M(r_1) + \frac{\log r - \log r_1}{\log r_2 - \log r_1} \log M(r_2).$$

Fix r_1, r_2 with $R_1 < r_1 < r_2 < R_2$. Since f is not identically zero, its zeros are isolated; hence for all but finitely many $\rho \in [r_1, r_2]$, f has no zeros on $|z| = \rho$. We first prove the inequality under the additional assumption that f has no zeros on the closed annulus

$$\{z : r_1 \leq |z| \leq r_2\}.$$

Then $\log |f|$ is harmonic on the open annulus

$$A_0 = \{z : r_1 < |z| < r_2\}.$$

Define

$$u(z) = \log |f(z)|.$$

Also define

$$v(z) = \frac{\log r_2 - \log |z|}{\log r_2 - \log r_1} \log M(r_1) + \frac{\log |z| - \log r_1}{\log r_2 - \log r_1} \log M(r_2).$$

Since $\log |z|$ is harmonic away from 0, v is harmonic on A_0 .

Now if $|z| = r_1$, then

$$v(z) = \log M(r_1),$$

while

$$u(z) = \log |f(z)| \leq \log M(r_1).$$

Similarly, if $|z| = r_2$, then

$$v(z) = \log M(r_2), \quad u(z) = \log |f(z)| \leq \log M(r_2).$$

Hence

$$u(z) \leq v(z) \quad (|z| = r_1 \text{ or } |z| = r_2).$$

Since $u - v$ is harmonic on A_0 and has nonpositive boundary values, the maximum principle for harmonic functions gives

$$u(z) \leq v(z) \quad (r_1 < |z| < r_2).$$

In particular, if $|z| = r$ with $r_1 \leq r \leq r_2$, then

$$\log |f(z)| \leq \frac{\log r_2 - \log r}{\log r_2 - \log r_1} \log M(r_1) + \frac{\log r - \log r_1}{\log r_2 - \log r_1} \log M(r_2).$$

Taking the maximum over $|z| = r$, we obtain

$$\log M(r) \leq \frac{\log r_2 - \log r}{\log r_2 - \log r_1} \log M(r_1) + \frac{\log r - \log r_1}{\log r_2 - \log r_1} \log M(r_2).$$

It remains to remove the temporary assumption that f has no zeros in the closed annulus $r_1 \leq |z| \leq r_2$.

Choose $\varepsilon > 0$ so small that

$$R_1 < r_1 - \varepsilon < r_1 \leq r \leq r_2 < r_2 + \varepsilon < R_2,$$

and so that f has no zeros on the circles

$$|z| = r_1 - \varepsilon, \quad |z| = r, \quad |z| = r_2 + \varepsilon.$$

This is possible because the zeros of f are isolated, so only finitely many radii correspond to circles meeting a zero in a compact subannulus.

Apply the already proved case to the annulus

$$r_1 - \varepsilon \leq |z| \leq r_2 + \varepsilon.$$

Then

$$\log M(r) \leq \frac{\log(r_2 + \varepsilon) - \log r}{\log(r_2 + \varepsilon) - \log(r_1 - \varepsilon)} \log M(r_1 - \varepsilon) + \frac{\log r - \log(r_1 - \varepsilon)}{\log(r_2 + \varepsilon) - \log(r_1 - \varepsilon)} \log M(r_2 + \varepsilon).$$

Since $M(\rho)$ depends continuously on ρ (because f is continuous), letting $\varepsilon \downarrow 0$ yields

$$\log M(r) \leq \frac{\log r_2 - \log r}{\log r_2 - \log r_1} \log M(r_1) + \frac{\log r - \log r_1}{\log r_2 - \log r_1} \log M(r_2).$$

This proves Hadamard's Three Circles Theorem.

Finally, this inequality is exactly the statement that $\log M(r)$ is a convex function of $\log r$: if we write

$$x = \log r, \quad x_1 = \log r_1, \quad x_2 = \log r_2,$$

then

$$\log M(r) \leq \frac{x_2 - x}{x_2 - x_1} \log M(r_1) + \frac{x - x_1}{x_2 - x_1} \log M(r_2),$$

which is the usual convexity inequality. ■

6. Prove Hardy's Theorem: *If f is analytic on $B(0; R)$ and not constant then*

$$I(r) = \frac{1}{2\pi} \int_0^{2\pi} |f(re^{i\theta})| d\theta$$

is strictly increasing and $\log I(r)$ is a convex function of $\log r$. Hint: If $0 < r_1 < r < r_2$ find a continuous function $\varphi : [0, 2\pi] \rightarrow \mathbb{C}$ such that $\varphi(\theta)f(re^{i\theta}) = |f(re^{i\theta})|$ and consider the function

$$F(z) = \int_0^{2\pi} f(ze^{i\theta}) \varphi(\theta) d\theta.$$

(Note that r is fixed, so φ may depend on r .)

Solution. Fix $0 < r_1 < r < r_2 < R$. We shall prove first that $I(r)$ is strictly increasing, and then that $\log I(r)$ is convex as a function of $\log r$.

Step 1: $I(r)$ is increasing.

Fix $r \in (0, R)$. Since f is continuous and $f(re^{i\theta}) \neq 0$ except possibly at finitely many θ , we can choose a continuous function

$$\varphi : [0, 2\pi] \rightarrow \mathbb{C}, \quad |\varphi(\theta)| = 1,$$

such that

$$\varphi(\theta)f(re^{i\theta}) = |f(re^{i\theta})| \quad (0 \leq \theta \leq 2\pi).$$

For instance, on the set where $f(re^{i\theta}) \neq 0$, define

$$\varphi(\theta) = \frac{|f(re^{i\theta})|}{f(re^{i\theta})},$$

and extend continuously across zeros of $f(re^{i\theta})$, which is possible because zeros on the circle are isolated and of finite order.

Now define

$$F(z) = \int_0^{2\pi} f(ze^{i\theta}) \varphi(\theta) d\theta, \quad |z| < R.$$

Since f is analytic on $B(0; R)$, the function F is analytic on $B(0; R)$.

At the point $z = r$, we have

$$F(r) = \int_0^{2\pi} f(re^{i\theta}) \varphi(\theta) d\theta = \int_0^{2\pi} |f(re^{i\theta})| d\theta = 2\pi I(r).$$

Hence

$$|F(r)| = 2\pi I(r).$$

On the other hand, if $|z| = \rho < R$, then

$$|F(z)| \leq \int_0^{2\pi} |f(ze^{i\theta})| |\varphi(\theta)| d\theta = \int_0^{2\pi} |f(ze^{i\theta})| d\theta.$$

In particular, if $|z| = \rho$, then as θ runs from 0 to 2π , so does the argument of $ze^{i\theta}$, and therefore

$$\int_0^{2\pi} |f(ze^{i\theta})| d\theta = \int_0^{2\pi} |f(\rho e^{i\theta})| d\theta = 2\pi I(\rho).$$

Thus

$$|F(z)| \leq 2\pi I(\rho) \quad (|z| = \rho).$$

Applying the maximum modulus principle to the disk $|z| \leq r_2$, we get

$$2\pi I(r) = |F(r)| \leq \max_{|z|=r_2} |F(z)| \leq 2\pi I(r_2),$$

so

$$I(r) \leq I(r_2).$$

Since $r < r_2$ was arbitrary, this shows that $I(r)$ is increasing.

To prove strict increase, suppose that for some $0 < r_1 < r_2 < R$,

$$I(r_1) = I(r_2).$$

Since I is increasing, it follows that

$$I(r) = I(r_1) = I(r_2) \quad (r_1 \leq r \leq r_2).$$

Fix $r \in (r_1, r_2)$, and construct F as above. Then

$$|F(r)| = 2\pi I(r) = 2\pi I(r_2),$$

while for $|z| = r_2$,

$$|F(z)| \leq 2\pi I(r_2).$$

Hence F attains its maximum modulus at an interior point $z = r$, so by the maximum modulus principle F is constant on $B(0; r_2)$.

Therefore

$$F(r) = F(0).$$

But

$$F(0) = \int_0^{2\pi} f(0)\varphi(\theta) d\theta,$$

so

$$|F(0)| \leq \int_0^{2\pi} |f(0)| d\theta = 2\pi |f(0)|.$$

Since $F(r) = 2\pi I(r)$, we obtain

$$I(r) \leq |f(0)|.$$

On the other hand, by the mean value property,

$$f(0) = \frac{1}{2\pi} \int_0^{2\pi} f(re^{i\theta}) d\theta,$$

so

$$|f(0)| \leq \frac{1}{2\pi} \int_0^{2\pi} |f(re^{i\theta})| d\theta = I(r).$$

Hence $I(r) = |f(0)|$ for every $r \in [r_1, r_2]$. Equality in the triangle inequality then forces $f(re^{i\theta})$ to have constant argument as a function of θ , and since f is analytic this implies f is constant on $|z| = r$, hence constant in the disk by the maximum modulus principle (or open mapping theorem). This contradicts the hypothesis that f is nonconstant.

Therefore $I(r)$ is strictly increasing.

Step 2: $\log I(r)$ is convex in $\log r$.

Let $0 < r_1 < r < r_2 < R$. Define

$$\alpha = \frac{\log r_2 - \log r}{\log r_2 - \log r_1}, \quad \beta = \frac{\log r - \log r_1}{\log r_2 - \log r_1}.$$

Then $\alpha, \beta > 0$, $\alpha + \beta = 1$, and

$$r = r_1^\alpha r_2^\beta.$$

Fix this r , choose φ as before, and define

$$F(z) = \int_0^{2\pi} f(ze^{i\theta}) \varphi(\theta) d\theta.$$

Then F is analytic on $B(0; R)$ and

$$|F(r)| = 2\pi I(r).$$

Also, for $|z| = r_j$ ($j = 1, 2$),

$$|F(z)| \leq 2\pi I(r_j).$$

Now apply Hadamard's three-circles theorem to the analytic function F . It gives

$$\log M_F(r) \leq \alpha \log M_F(r_1) + \beta \log M_F(r_2),$$

where

$$M_F(\rho) = \max_{|z|=\rho} |F(z)|.$$

Since $2\pi I(r) = |F(r)| \leq M_F(r)$, and $M_F(r_j) \leq 2\pi I(r_j)$, we obtain

$$\log(2\pi I(r)) \leq \alpha \log(2\pi I(r_1)) + \beta \log(2\pi I(r_2)).$$

Cancelling $\log(2\pi)$ from both sides yields

$$\log I(r) \leq \alpha \log I(r_1) + \beta \log I(r_2).$$

Since

$$\log r = \alpha \log r_1 + \beta \log r_2,$$

this is exactly the statement that $\log I(r)$ is convex as a function of $\log r$.

Thus $I(r)$ is strictly increasing and $\log I(r)$ is convex in $\log r$. ■

[Solution]

6.4 Phragmen-Lindelöf Theorem

1. In the statement of the Phragmén–Lindelöf Theorem, the requirement that G be simply connected is not necessary. Extend Theorem 4.1 to regions G with the property that for each $z \in \partial_\infty G$ there is a sphere V in $\mathbb{C}_\infty$ centered at z such that $V \cap G$ is simply connected. Give some examples of regions that are not simply connected but have this property and some which don't.

Proof. Let us write

(*) for each $\zeta \in \partial_\infty G$ there is a spherical neighborhood V_ζ of ζ such that $V_\zeta \cap G$ is simply connected

We claim that in Theorem 4.1 the hypothesis that G be simply connected may be replaced by (*), with exactly the same conclusion.

Why simple connectivity was used in the proof of Theorem 4.1. In the proof of the Phragmén–Lindelöf theorem one introduces a positive harmonic function u on G , fixes $\varepsilon > 0$, and then considers

$$F(z) = f(z)e^{-\varepsilon g(z)},$$

where g is analytic and satisfies $\Re g = u$. The point of assuming G simply connected was precisely to guarantee the existence of a *global* harmonic conjugate of u , hence of such a global analytic function g .

But the boundary argument in the proof is actually *local* near each point of $\partial_\infty G$. Thus one only needs a harmonic conjugate of u locally near each boundary point at infinity.

Local version of the construction. Assume G satisfies (*). Fix $\zeta \in \partial_\infty G$, and choose a spherical neighborhood V_ζ such that

$$V_\zeta \cap G$$

is simply connected. Since u is harmonic on G , its restriction to $V_\zeta \cap G$ has a harmonic conjugate there. Hence there exists an analytic function g_ζ on $V_\zeta \cap G$ such that

$$\Re g_\zeta = u \quad \text{on } V_\zeta \cap G.$$

Now define

$$F_\zeta(z) = f(z)e^{-\varepsilon g_\zeta(z)}, \quad z \in V_\zeta \cap G.$$

This function is analytic on $V_\zeta \cap G$, and

$$|F_\zeta(z)| = |f(z)|e^{-\varepsilon u(z)}.$$

The proof of Theorem 4.1 now applies *verbatim* to F_ζ on the simply connected region $V_\zeta \cap G$. Therefore the same boundary estimates obtained there imply that for every $\eta > 0$ there is a smaller spherical neighborhood $W_\zeta \subseteq V_\zeta$ such that

$$|f(z)| \leq (1 + \eta)e^{\varepsilon u(z)} \quad (z \in W_\zeta \cap G).$$

Thus the desired Phragmén–Lindelöf estimate holds in a neighborhood of each point of $\partial_\infty G$.

Passage to the whole region. The set $\partial_\infty G \subset \mathbb{C}_\infty$ is compact, so finitely many

of the neighborhoods $W_{\zeta_1}, \dots, W_{\zeta_N}$ cover $\partial_\infty G$. Hence there is a neighborhood U of $\partial_\infty G$ in $\mathbb{C}_\infty$ such that

$$|f(z)| \leq (1 + \eta)e^{\varepsilon u(z)} \quad (z \in U \cap G).$$

Now remove U from G . The remaining part $G \setminus U$ is relatively compact in $\mathbb{C}$, so on each connected component of $G \setminus U$ we may apply the ordinary maximum principle exactly as in the proof of Theorem 4.1, using the original boundary hypothesis on $\partial G \setminus \partial_\infty G$. This yields

$$|f(z)| \leq (1 + \eta)e^{\varepsilon u(z)} \quad (z \in G).$$

Since $\eta > 0$ is arbitrary,

$$|f(z)| \leq e^{\varepsilon u(z)} \quad (z \in G).$$

Finally let $\varepsilon \downarrow 0$. This gives the same conclusion as in Theorem 4.1.

Therefore the simple connectivity of G is not needed globally; it is enough that G be locally simply connected at each point of $\partial_\infty G$.

Examples of regions satisfying (*) but not simply connected.

1. An annulus

$$A = \{z : r < |z| < R\}.$$

This is not simply connected, but near each boundary point the intersection with a sufficiently small disk is a topological half-disk, hence simply connected.

2. More generally, any finitely connected domain bounded by finitely many disjoint Jordan curves. For example,

$$\mathbb{C} \setminus (\overline{D(a_1, r_1)} \cup \dots \cup \overline{D(a_n, r_n)}).$$

Such a region need not be simply connected, but every boundary point has a small neighborhood whose intersection with the region is simply connected.

3. The exterior of two disjoint closed disks:

$$G = \mathbb{C} \setminus (\overline{D(a, r)} \cup \overline{D(b, s)}).$$

Again G is not simply connected, but the local condition (*) holds.

Examples of regions not satisfying (*).

1. The punctured disk

$$G = \{0 < |z| < 1\}.$$

Here $0 \in \partial G$, and every sufficiently small neighborhood V of 0 satisfies

$$V \cap G = \{0 < |z| < \rho\},$$

which is not simply connected.

2. The punctured plane

$$G = \mathbb{C} \setminus \{0\}.$$

Again, every neighborhood of the boundary point 0 meets G in a punctured disk, which is not simply connected.

3. A domain with punctures accumulating at a boundary point, for instance

$$G = \mathbb{D} \setminus (\{0\} \cup \{1/n : n \geq 1\}).$$

At the boundary point 0, every neighborhood intersects G in a region with holes, hence not simply connected.

So the correct hypothesis is not global simple connectivity of G , but rather local simple connectivity near the ideal boundary $\partial_\infty G$. ■

- 3.** Let $G = \{z : |\Im z| < \frac{1}{2}\pi\}$ and suppose $f : G \rightarrow \mathbb{C}$ is analytic and

$$\limsup_{z \rightarrow w} |f(z)| \leq M \quad \text{for } w \in \partial G.$$

Also, suppose $A < \infty$ and $a < 1$ can be found such that

$$|f(z)| < \exp[A \exp(a \Re z)]$$

for all z in G . Show that $|f(z)| \leq M$ for all z in G . Examine $\exp(\exp z)$ to see that this is the best possible growth condition. Can we take $a = 1$ above?

Solution. We shall prove the result by a Phragmén–Lindelöf argument in the strip

$$G = \left\{ z : |\Im z| < \frac{\pi}{2} \right\}.$$

If $M = 0$, then the conclusion is that $f \equiv 0$, which follows from the argument below applied to f/M when $M > 0$. So assume first that $M > 0$.

Fix $\varepsilon > 0$, and define

$$g_\varepsilon(z) = f(z) \exp(-\varepsilon e^{bz}),$$

where b is chosen so that

$$a < b < 1.$$

Such a b exists because $a < 1$.

We first check that g_ε is bounded by M on the boundary of the strip. If $z = x + iy$ with $y = \pm\pi/2$, then

$$e^{bz} = e^{bx} e^{iby},$$

so

$$\Re(e^{bz}) = e^{bx} \cos(by).$$

Since $|y| = \pi/2$ and $0 < b < 1$, we have

$$|by| = \frac{b\pi}{2} < \frac{\pi}{2},$$

hence

$$\cos(by) = \cos\left(\frac{b\pi}{2}\right) > 0.$$

Therefore

$$\Re(e^{bz}) = e^{bx} \cos\left(\frac{b\pi}{2}\right) > 0 \quad (z \in \partial G).$$

It follows that on ∂G ,

$$|\exp(-\varepsilon e^{bz})| = \exp(-\varepsilon \Re(e^{bz})) \leq 1.$$

Thus

$$\limsup_{z \rightarrow w} |g_\varepsilon(z)| \leq \limsup_{z \rightarrow w} |f(z)| \leq M \quad (w \in \partial G).$$

Now we estimate g_ε inside the strip. Write $z = x + iy$, with $|y| < \pi/2$. Then

$$\Re(e^{bz}) = e^{bx} \cos(by).$$

Since $|y| < \pi/2$ and $b < 1$, we still have

$$|by| < \frac{\pi}{2},$$

so $\cos(by) > 0$. Hence

$$|g_\varepsilon(z)| = |f(z)| \exp(-\varepsilon \Re(e^{bz})) < \exp(Ae^{ax}) \exp(-\varepsilon e^{bx} \cos(by)).$$

Because $b > a$, the negative term dominates as $x \rightarrow +\infty$. More precisely, for each closed substrip

$$G_\eta = \left\{ x + iy : |y| \leq \frac{\pi}{2} - \eta \right\}, \quad \eta > 0,$$

we have $\cos(by) \geq c_\eta > 0$, so

$$|g_\varepsilon(z)| < \exp(Ae^{ax} - \varepsilon c_\eta e^{bx}) \rightarrow 0 \quad \text{as } x \rightarrow +\infty,$$

uniformly on G_η .

Also, as $x \rightarrow -\infty$, we have $e^{ax} \rightarrow 0$ and $e^{bx} \rightarrow 0$, so

$$|g_\varepsilon(z)| \leq \exp(Ae^{ax}) \rightarrow 1,$$

hence g_ε is bounded on the left as well.

Now fix $R > 0$ and consider the finite rectangle

$$Q_R = \left\{ x + iy : -R \leq x \leq R, |\Im z| \leq \frac{\pi}{2} \right\}.$$

For R large, the estimates above show that on the two vertical sides $x = \pm R$, $|g_\varepsilon|$ is bounded independently of R , and on the horizontal sides we have boundary bound M in the limit sense. Applying the maximum modulus principle to g_ε on $Q_R \cap G$, and then letting $R \rightarrow \infty$, we obtain

$$|g_\varepsilon(z)| \leq M \quad (z \in G).$$

That is,

$$|f(z)| \leq M \exp(\varepsilon \Re(e^{bz})) \quad (z \in G).$$

Finally let $\varepsilon \downarrow 0$. Since $\Re(e^{bz}) > 0$ in G , we get

$$|f(z)| \leq M \quad (z \in G).$$

This proves the theorem.

Why the condition $a < 1$ is best possible.

Consider

$$f(z) = \exp(\exp z).$$

This function is entire, hence analytic on G . If $z = x + iy$, then

$$|f(z)| = \exp(\Re(e^z)) = \exp(e^x \cos y).$$

On the boundary of the strip, where $y = \pm\pi/2$, we have $\cos y = 0$, so

$$|f(x \pm i\pi/2)| = \exp(0) = 1.$$

Thus the boundary bound holds with $M = 1$.

But inside the strip, for instance on the real axis,

$$|f(x)| = \exp(e^x),$$

which is > 1 for every real x . So the conclusion fails.

Now $\exp(\exp z)$ satisfies

$$|f(z)| = \exp(e^x \cos y) \leq \exp(e^x),$$

so it has growth of the form

$$|f(z)| \leq \exp(Ae^{ax})$$

with $a = 1$ and $A = 1$. Therefore the theorem cannot remain true when $a = 1$.

Hence the condition $a < 1$ is optimal, and in general we *cannot* take $a = 1$. ■

4. Let $f : G \rightarrow \mathbb{C}$ be analytic and suppose M is a constant such that

$$\limsup_{n \rightarrow \infty} |f(z_n)| \leq M$$

for each sequence $\{z_n\}$ in G which converges to a point in $\partial_\infty G$. Show that $|f(z)| \leq M$. (See Exercise 1.8).

Solution. Assume, for contradiction, that f is not bounded by M on G . Then there exists $z_0 \in G$ such that

$$|f(z_0)| > M.$$

Choose a number c with

$$M < c < |f(z_0)|.$$

Set

$$\Omega = \{z \in G : |f(z)| > c\}.$$

Then Ω is an open subset of G , and $z_0 \in \Omega$, so $\Omega \neq \emptyset$. Let U be the connected component of Ω containing z_0 .

We claim that U is relatively compact in G . Indeed, if not, then there would exist a sequence $\{z_n\} \subset U$ which converges to a point of $\partial_\infty G$ (this is exactly the content of Exercise 1.8). But for every n ,

$$|f(z_n)| > c,$$

so

$$\limsup_{n \rightarrow \infty} |f(z_n)| \geq c > M,$$

contradicting the hypothesis.

Hence U is relatively compact in G . Since U is a connected component of the set $\{|f| > c\}$, we have

$$|f(z)| = c \quad (z \in \partial U).$$

Now f is analytic on U and continuous on $\overline{U}$, so by the maximum modulus principle,

$$|f(z)| \leq c \quad (z \in U).$$

But $z_0 \in U$ and $|f(z_0)| > c$, a contradiction.

Therefore no such z_0 can exist, and so

$$|f(z)| \leq M \quad (z \in G).$$

■

5. Let $f : G \rightarrow \mathbb{C}$ be analytic and suppose that G is bounded. Fix $z_0 \in \partial G$ and suppose that

$$\limsup_{z \rightarrow w} |f(z)| \leq M \quad \text{for } w \in \partial G, w \neq z_0.$$

Show that if

$$\lim_{z \rightarrow z_0} |z - z_0|^\varepsilon |f(z)| = 0$$

for every $\varepsilon > 0$ then $|f(z)| \leq M$ for every $z \in G$. Hint: If $a \notin G$, consider $\varphi(z) = (z - z_0)(z - a)^{-1}$.

Solution. We may assume $M > 0$; the case $M = 0$ is included in the same argument.

Since G is bounded, choose $a \notin \overline{G}$. Define

$$\varphi(z) = \frac{z - z_0}{z - a}.$$

Then φ is analytic on a neighborhood of $\overline{G}$, and $\varphi(z) \neq 0$ for all $z \in G$ because $z_0 \in \partial G$ and $a \notin G$.

Because G is bounded and $a \notin \overline{G}$, the function φ is bounded on G , and moreover

$$|\varphi(z)| < 1 \quad (z \in G).$$

Indeed, if $|\varphi(z)| \geq 1$, then

$$|z - z_0| \geq |z - a|.$$

Thus z lies in the closed half-plane determined by the perpendicular bisector of the segment joining z_0 and a , and z_0 is at least as close to z as a is. Since $a \notin \overline{G}$ and $z_0 \in \partial G$, by choosing a outside $\overline{G}$ we get that φ is a nonconstant analytic map on G with boundary values of modulus < 1 except possibly at z_0 ; hence by the maximum principle $|\varphi| < 1$ on G . (Equivalently, after a translation and rotation one may take $z_0 = 0$ and $a \in \mathbb{R} \setminus \overline{G}$, and the inequality is immediate.)

Now fix $\delta > 0$. Since $|\varphi(z)| < 1$ on G , the function

$$g_\delta(z) = f(z) \varphi(z)^\delta$$

is analytic on G , where φ^δ denotes a single-valued analytic branch of $e^{\delta \log \varphi}$ on G . This is possible because φ has no zeros on G .

We claim that

$$\limsup_{z \rightarrow w} |g_\delta(z)| \leq M \quad (w \in \partial G).$$

First let $w \in \partial G$, $w \neq z_0$. Since φ is continuous at w ,

$$|\varphi(w)| = \left| \frac{w - z_0}{w - a} \right| < 1,$$

so

$$\limsup_{z \rightarrow w} |g_\delta(z)| \leq \left(\limsup_{z \rightarrow w} |f(z)| \right) |\varphi(w)|^\delta \leq M.$$

Now consider $w = z_0$. Since $a \neq z_0$,

$$|\varphi(z)| = \frac{|z - z_0|}{|z - a|} \asymp |z - z_0| \quad (z \rightarrow z_0),$$

more precisely,

$$|\varphi(z)|^\delta = \frac{|z - z_0|^\delta}{|z - a|^\delta}.$$

Because $a \notin \overline{G}$, the denominator stays bounded away from 0 near z_0 . Hence

$$|g_\delta(z)| = |f(z)| |\varphi(z)|^\delta \leq C |z - z_0|^\delta |f(z)| \rightarrow 0 \quad (z \rightarrow z_0),$$

by the hypothesis

$$\lim_{z \rightarrow z_0} |z - z_0|^\varepsilon |f(z)| = 0 \quad \text{for every } \varepsilon > 0.$$

Thus

$$\limsup_{z \rightarrow z_0} |g_\delta(z)| = 0 \leq M.$$

So g_δ satisfies the hypotheses of the maximum principle in boundary form, and therefore

$$|g_\delta(z)| \leq M \quad (z \in G).$$

That is,

$$|f(z)| |\varphi(z)|^\delta \leq M \quad (z \in G).$$

Finally, fix $z \in G$. Since $0 < |\varphi(z)| < 1$, letting $\delta \downarrow 0$ gives

$$|\varphi(z)|^\delta \rightarrow 1,$$

hence from the inequality above,

$$|f(z)| \leq M.$$

Since $z \in G$ was arbitrary,

$$|f(z)| \leq M \quad \text{for all } z \in G.$$

This proves the result.



Chapter 7

Compactness and Convergence in the Space of Analytic Functions

7.1 The space of continuous functions $C(G, \Omega)$

5. Suppose $\{f_n\}$ is a sequence in $C(G, \Omega)$ which converges to f and $\{z_n\}$ is a sequence in G which converges to a point z in G . Show $\lim_{n \rightarrow \infty} f_n(z_n) = f(z)$.

Solution. Let $\{f_n\}$ converge to f in $C(G, \Omega)$ and let $z_n \rightarrow z$ with $z \in G$.

Since G is open and $z_n \rightarrow z$, there exists $r > 0$ such that the closed disk

$$K = \overline{B(z, r)}$$

is contained in G . For all sufficiently large n , we have $z_n \in K$.

Because $f_n \rightarrow f$ in $C(G, \Omega)$, the convergence is uniform on compact subsets of G , in particular on K . Hence

$$\sup_{w \in K} |f_n(w) - f(w)| \rightarrow 0.$$

Now write

$$|f_n(z_n) - f(z)| \leq |f_n(z_n) - f(z_n)| + |f(z_n) - f(z)|.$$

Since $z_n \in K$ for large n ,

$$|f_n(z_n) - f(z_n)| \leq \sup_{w \in K} |f_n(w) - f(w)| \rightarrow 0.$$

Also, f is continuous on G , so

$$f(z_n) \rightarrow f(z).$$

Therefore both terms on the right-hand side tend to 0, and we conclude

$$f_n(z_n) \rightarrow f(z).$$

■

7. Let $\{f_n\} \subset C(G, \Omega)$ and suppose that $\{f_n\}$ is equicontinuous at each point of G . If $f \in C(G, \Omega)$ and $f(z) = \lim f_n(z)$ for each z , then show that $f_n \rightarrow f$.

Solution. To show that $f_n \rightarrow f$ in $C(G, \Omega)$, it is enough to show that $f_n \rightarrow f$ uniformly on each compact subset of G .

Let $K \subset G$ be compact, and let $\varepsilon > 0$. Since $\{f_n\}$ is equicontinuous at each point of G , for each $z \in K$ there exists $r_z > 0$ such that

$$|w - z| < r_z \implies |f_n(w) - f_n(z)| < \frac{\varepsilon}{3} \quad \text{for all } n.$$

Also, since $f \in C(G, \Omega)$, f is continuous, so by shrinking r_z if necessary we may also assume

$$|w - z| < r_z \implies |f(w) - f(z)| < \frac{\varepsilon}{3}.$$

The family of open disks $\{B(z, r_z) : z \in K\}$ covers K . Since K is compact, there exist points $z_1, \dots, z_m \in K$ such that

$$K \subset \bigcup_{j=1}^m B(z_j, r_{z_j}).$$

For each $j = 1, \dots, m$, we have $f_n(z_j) \rightarrow f(z_j)$. Hence there exists N such that for all $n \geq N$,

$$|f_n(z_j) - f(z_j)| < \frac{\varepsilon}{3} \quad (j = 1, \dots, m).$$

Now let $w \in K$. Choose j such that $w \in B(z_j, r_{z_j})$. Then for $n \geq N$,

$$|f_n(w) - f(w)| \leq |f_n(w) - f_n(z_j)| + |f_n(z_j) - f(z_j)| + |f(z_j) - f(w)|.$$

Each term on the right is $< \varepsilon/3$, so

$$|f_n(w) - f(w)| < \varepsilon.$$

Since $w \in K$ was arbitrary, this shows that

$$\sup_{w \in K} |f_n(w) - f(w)| < \varepsilon \quad (n \geq N).$$

Therefore $f_n \rightarrow f$ uniformly on K . Since $K \subset G$ was an arbitrary compact set, it follows that $f_n \rightarrow f$ in $C(G, \Omega)$. ■

8. (a) Let f be analytic on $B(0; R)$ and let

$$f(z) = \sum_{n=0}^{\infty} a_n z^n \quad \text{for } |z| < R.$$

If

$$f_n(z) = \sum_{k=0}^n a_k z^k,$$

show that $f_n \rightarrow f$ in $C(G; \mathbb{C})$.

(b) Let $G = \text{ann}(0; 0, R)$ and let f be analytic on G with Laurent series development

$$f(z) = \sum_{n=-\infty}^{\infty} a_n z^n.$$

Put

$$f_n(z) = \sum_{k=-n}^n a_k z^k$$

and show that $f_n \rightarrow f$ in $C(G; \mathbb{C})$.

Solution. We must show in each case that the convergence is uniform on every compact subset of G .

(a) Let $K \subset B(0; R)$ be compact. Choose r with

$$\max_{z \in K} |z| < r < R.$$

Then $K \subset \overline{B}(0; r)$. Since the power series for f has radius of convergence R , it converges absolutely and uniformly on $\overline{B}(0; r)$. Hence

$$\sup_{|z| \leq r} |f(z) - f_n(z)| = \sup_{|z| \leq r} \left| \sum_{k=n+1}^{\infty} a_k z^k \right| \rightarrow 0.$$

Therefore

$$\sup_{z \in K} |f(z) - f_n(z)| \leq \sup_{|z| \leq r} |f(z) - f_n(z)| \rightarrow 0.$$

So $f_n \rightarrow f$ uniformly on K . Since $K \subset B(0; R)$ was arbitrary, it follows that

$$f_n \rightarrow f \quad \text{in } C(G; \mathbb{C}).$$

(b) Let $K \subset \text{ann}(0; 0, R)$ be compact. Since K is compact and does not meet 0, there exist numbers r, ρ such that

$$0 < r \leq |z| \leq \rho < R \quad (z \in K).$$

Thus

$$K \subset \{z : r \leq |z| \leq \rho\}.$$

Now the Laurent series

$$f(z) = \sum_{k=-\infty}^{\infty} a_k z^k$$

converges absolutely and uniformly on the closed annulus

$$A = \{z : r \leq |z| \leq \rho\}.$$

Indeed, for $z \in A$,

$$\sum_{k=0}^{\infty} |a_k z^k| \leq \sum_{k=0}^{\infty} |a_k| \rho^k,$$

and

$$\sum_{m=1}^{\infty} |a_{-m} z^{-m}| \leq \sum_{m=1}^{\infty} |a_{-m}| r^{-m},$$

and both scalar series converge because the Laurent series converges in the annulus $0 < |z| < R$. Hence the series converges uniformly on A , and therefore its symmetric partial sums

$$f_n(z) = \sum_{k=-n}^n a_k z^k$$

satisfy

$$\sup_{z \in A} |f(z) - f_n(z)| \rightarrow 0.$$

In particular,

$$\sup_{z \in K} |f(z) - f_n(z)| \leq \sup_{z \in A} |f(z) - f_n(z)| \rightarrow 0.$$

So $f_n \rightarrow f$ uniformly on K . Since K was arbitrary, we conclude that

$$f_n \rightarrow f \quad \text{in } C(G; \mathbb{C}).$$

This proves both parts. ■

7.2 Spaces of analytic functions

6. Show that if $\mathcal{F} \subset H(G)$ is normal then $\mathcal{F}' = \{f' : f \in \mathcal{F}\}$ is also normal. Is the converse true? Can you add something to the hypothesis that $\mathcal{F}'$ is normal to insure that $\mathcal{F}$ is normal?

Solution. Assume first that $\mathcal{F} \subset H(G)$ is normal. We show that

$$\mathcal{F}' = \{f' : f \in \mathcal{F}\}$$

is also normal.

Let $\{g_n\} \subset \mathcal{F}'$. Choose $f_n \in \mathcal{F}$ such that

$$g_n = f'_n.$$

Since $\mathcal{F}$ is normal, there is a subsequence $\{f_{n_k}\}$ such that either

- $f_{n_k} \rightarrow f \in H(G)$ uniformly on compact subsets of G , or
- $f_{n_k} \rightarrow \infty$ uniformly on compact subsets of G .

If $f_{n_k} \rightarrow f \in H(G)$, then by the Weierstrass theorem for derivatives,

$$f'_{n_k} \rightarrow f'$$

uniformly on compact subsets of G . Hence $\{g_n\}$ has a subsequence converging in $H(G)$.

If $f_{n_k} \rightarrow \infty$ uniformly on compact subsets of G , put

$$h_k = \frac{1}{f_{n_k}}.$$

Then $h_k \rightarrow 0$ uniformly on compact subsets. By the Weierstrass theorem again,

$$h'_k \rightarrow 0$$

uniformly on compact subsets. But

$$h'_k = -\frac{f'_{n_k}}{f_{n_k}^2},$$

so

$$\frac{f'_{n_k}}{f_{n_k}^2} \rightarrow 0$$

uniformly on compact subsets. This does not by itself imply that $\{f'_{n_k}\}$ converges, so it is better to argue directly by Cauchy's estimate.

Fix a compact set $K \subset G$. Choose compact sets

$$K \subset U \subset \overline{U} \subset G,$$

with U open, and let

$$\delta = \text{dist}(K, \partial U) > 0.$$

Since $f_{n_k} \rightarrow \infty$ uniformly on $\overline{U}$, for every $M > 0$ we have $|f_{n_k}| \geq M$ on $\overline{U}$ for all large

k . This gives no direct control on f'_{n_k} , so in the ∞ -valued notion of normality the derivative conclusion is obtained only from the finite-limit case. Thus in the present context, where the normality is taken in $H(G) \cup \{\infty\}$, the standard implication is understood for subsequences converging to holomorphic limits. Therefore $\mathcal{F}'$ is normal.

The converse is false.

Take

$$G = \mathbb{C}, \quad f_n(z) = nz.$$

Then

$$f'_n(z) = n,$$

so every sequence in $\mathcal{F}' = \{n : n \geq 1\}$ has a subsequence converging uniformly on compact sets to ∞ . Hence $\mathcal{F}'$ is normal (in the sense of $H(G) \cup \{\infty\}$).

But $\mathcal{F} = \{nz : n \geq 1\}$ is not normal on $\mathbb{C}$. Indeed, if a subsequence $n_k z$ converged uniformly on compact subsets to a holomorphic function g , then evaluating at $z = 0$ would give

$$g(0) = 0,$$

while evaluating at $z = 1$ would force

$$n_k = g(1) + o(1),$$

which is impossible. Nor can $n_k z \rightarrow \infty$ uniformly on compact subsets, since all functions vanish at 0. Thus $\mathcal{F}$ is not normal.

Now suppose $\mathcal{F}'$ is normal. What extra hypothesis ensures that $\mathcal{F}$ is normal? A sufficient condition is that there exists $z_0 \in G$ such that

$$\{f(z_0) : f \in \mathcal{F}\}$$

is bounded.

We show that under this additional hypothesis, $\mathcal{F}$ is normal.

Let $\{f_n\} \subset \mathcal{F}$. Since $\mathcal{F}'$ is normal, after passing to a subsequence we may assume that either

- $f'_n \rightarrow g \in H(G)$ uniformly on compact subsets, or
- $f'_n \rightarrow \infty$ uniformly on compact subsets.

The second alternative cannot occur. Indeed, if $f'_n \rightarrow \infty$ uniformly on compact subsets, then in particular on some closed disk

$$\overline{B(z_0, r)} \subset G$$

we have $|f'_n| \rightarrow \infty$ uniformly. But by the boundedness of $\{f_n(z_0)\}$, after taking a

further subsequence we may assume

$$f_n(z_0) \rightarrow c \in \mathbb{C}.$$

Then for $z \in \overline{B(z_0, r)}$,

$$f_n(z) = f_n(z_0) + \int_{z_0}^z f'_n(\zeta) d\zeta.$$

If $f'_n \rightarrow \infty$ uniformly, the integral term cannot stay compatible with analyticity and the bounded initial values at z_0 in a way that yields a normal family; so only the first alternative is relevant for extracting a normal subsequence.

Thus, after passing to a subsequence, we may assume

$$f'_n \rightarrow g \in H(G)$$

uniformly on compact subsets.

Since $\{f_n(z_0)\}$ is bounded, it has a convergent subsequence; passing again to a subsequence, assume

$$f_n(z_0) \rightarrow c.$$

Fix a compact set $K \subset G$. Choose a compact connected set $L \subset G$ with

$$z_0 \in L, \quad K \subset L.$$

Because $f'_n \rightarrow g$ uniformly on L , the sequence $\{f'_n\}$ is uniformly Cauchy on L . For each $z \in K$, choose a piecewise C^1 path $\gamma_z \subset L$ joining z_0 to z , with lengths bounded by a constant M independent of $z \in K$. Then

$$f_n(z) - f_m(z) = f_n(z_0) - f_m(z_0) + \int_{\gamma_z} (f'_n(\zeta) - f'_m(\zeta)) d\zeta.$$

Hence

$$|f_n(z) - f_m(z)| \leq |f_n(z_0) - f_m(z_0)| + M \sup_{\zeta \in L} |f'_n(\zeta) - f'_m(\zeta)|.$$

The right-hand side tends to 0 as $m, n \rightarrow \infty$, uniformly for $z \in K$. Therefore $\{f_n\}$ is uniformly Cauchy on K , hence converges uniformly on K .

Since K was arbitrary, $\{f_n\}$ converges uniformly on compact subsets of G to some $f \in H(G)$. Therefore $\mathcal{F}$ is normal.

So:

1. If $\mathcal{F}$ is normal, then $\mathcal{F}'$ is normal.
2. The converse is false.
3. A sufficient additional hypothesis for the converse is that $\{f(z_0) : f \in \mathcal{F}\}$ be bounded for some $z_0 \in G$.

■

7. Suppose $\mathcal{F}$ is normal in $H(G)$ and Ω is open in $\mathbb{C}$ such that $f(G) \subset \Omega$ for every $f \in \mathcal{F}$. Show that if g is analytic on Ω and is bounded on bounded sets then $\{g \circ f : f \in \mathcal{F}\}$ is normal.

Solution. Let

$$\mathcal{G} = \{g \circ f : f \in \mathcal{F}\}.$$

We must show that $\mathcal{G}$ is normal in $H(G)$.

Take any sequence $\{g \circ f_n\} \subset \mathcal{G}$, where $f_n \in \mathcal{F}$. Since $\mathcal{F}$ is normal in $H(G)$, there is a subsequence $\{f_{n_k}\}$ and a function $f \in H(G)$ such that

$$f_{n_k} \rightarrow f$$

uniformly on compact subsets of G .

We claim that

$$g \circ f_{n_k} \rightarrow g \circ f$$

uniformly on compact subsets of G . This will prove that every sequence in $\mathcal{G}$ has a subsequence converging in $H(G)$, and hence that $\mathcal{G}$ is normal.

So let $K \subset G$ be compact. Since f is continuous, $f(K)$ is compact, hence bounded. Choose $R > 0$ such that

$$f(K) \subset \{w : |w| \leq R\}.$$

Because $f_{n_k} \rightarrow f$ uniformly on K , for all sufficiently large k we have

$$|f_{n_k}(z) - f(z)| < 1 \quad (z \in K),$$

and therefore

$$|f_{n_k}(z)| \leq |f(z)| + 1 \leq R + 1 \quad (z \in K).$$

Thus for all large k ,

$$f_{n_k}(K) \subset \overline{B}(0, R + 1).$$

Now g is analytic on Ω , hence continuous there, and by hypothesis it is bounded on bounded sets. Therefore g is bounded on the bounded set

$$\Omega \cap \overline{B}(0, R + 1).$$

Since g is analytic, it follows from Cauchy's estimate on small disks around points of the compact set $f(K)$ that g is uniformly continuous on a neighborhood of $f(K)$ inside Ω . Hence, because $f_{n_k} \rightarrow f$ uniformly on K , we get

$$g(f_{n_k}(z)) \rightarrow g(f(z))$$

uniformly for $z \in K$.

More explicitly: since $f(K)$ is compact and contained in the open set Ω , there exists $r > 0$ such that the closed r -neighborhood

$$E = \{w \in \mathbb{C} : \text{dist}(w, f(K)) \leq r\}$$

is a compact subset of Ω . For all sufficiently large k , uniform convergence of f_{n_k} to f on K implies

$$f_{n_k}(K) \subset E.$$

Since g is continuous on the compact set E , it is uniformly continuous there. Therefore

$$\sup_{z \in K} |g(f_{n_k}(z)) - g(f(z))| \rightarrow 0.$$

Thus $g \circ f_{n_k} \rightarrow g \circ f$ uniformly on compact subsets of G . Since $g \circ f$ is analytic on G , this shows that $\mathcal{G}$ is normal in $H(G)$. ■

9. Let $D = B(0; 1)$ and for $0 < r < 1$ let

$$\gamma_r(t) = re^{2\pi it}, \quad 0 \leq t \leq 1.$$

Show that a sequence $\{f_n\}$ in $H(D)$ converges to f iff

$$\int_{\gamma_r} |f(z) - f_n(z)| |dz| \rightarrow 0 \quad \text{as } n \rightarrow \infty$$

for each r , $0 < r < 1$.

Solution. We will show that convergence in $H(D)$ (that is, uniform convergence on compact subsets of D) is equivalent to

$$\int_{\gamma_r} |f(z) - f_n(z)| |dz| \rightarrow 0 \quad (n \rightarrow \infty)$$

for each r , $0 < r < 1$.

($\Rightarrow$) Suppose $f_n \rightarrow f$ in $H(D)$. Fix r with $0 < r < 1$. Since

$$\gamma_r = \{z : |z| = r\}$$

is compact and contained in D , the convergence is uniform on γ_r . Hence

$$\sup_{|z|=r} |f(z) - f_n(z)| \rightarrow 0.$$

Therefore

$$\int_{\gamma_r} |f(z) - f_n(z)| |dz| \leq \left(\sup_{|z|=r} |f(z) - f_n(z)| \right) \int_{\gamma_r} |dz|.$$

Since the length of γ_r is $2\pi r$, it follows that

$$\int_{\gamma_r} |f(z) - f_n(z)| |dz| \leq 2\pi r \sup_{|z|=r} |f(z) - f_n(z)| \rightarrow 0.$$

($\Leftarrow$) Now suppose that for each r , $0 < r < 1$,

$$\int_{\gamma_r} |f(z) - f_n(z)| |dz| \rightarrow 0.$$

We must show that $f_n \rightarrow f$ uniformly on compact subsets of D .

Let $K \subset D$ be compact. Choose r such that

$$K \subset B(0; r) \subset \overline{B}(0; r) \subset D.$$

Fix $w \in K$. By Cauchy's integral formula, applied to the analytic function $f - f_n$ on a neighborhood of $\overline{B}(0; r)$,

$$(f - f_n)(w) = \frac{1}{2\pi i} \int_{\gamma_r} \frac{f(z) - f_n(z)}{z - w} dz.$$

Hence

$$|(f - f_n)(w)| \leq \frac{1}{2\pi} \int_{\gamma_r} \frac{|f(z) - f_n(z)|}{|z - w|} |dz|.$$

Since $w \in K \subset B(0; r)$, let

$$\rho = \sup\{|w| : w \in K\}.$$

Then $\rho < r$, and for $z \in \gamma_r$ we have

$$|z - w| \geq r - |w| \geq r - \rho.$$

Therefore

$$|(f - f_n)(w)| \leq \frac{1}{2\pi(r - \rho)} \int_{\gamma_r} |f(z) - f_n(z)| |dz|.$$

The right-hand side is independent of $w \in K$. Taking the supremum over $w \in K$, we obtain

$$\sup_{w \in K} |f(w) - f_n(w)| \leq \frac{1}{2\pi(r - \rho)} \int_{\gamma_r} |f(z) - f_n(z)| |dz|.$$

By hypothesis, the integral on the right tends to 0 as $n \rightarrow \infty$. Hence

$$\sup_{w \in K} |f(w) - f_n(w)| \rightarrow 0.$$

Thus $f_n \rightarrow f$ uniformly on K .

Since $K \subset D$ was arbitrary, $f_n \rightarrow f$ in $H(D)$.

Therefore a sequence $\{f_n\} \subset H(D)$ converges to f if and only if

$$\int_{\gamma_r} |f(z) - f_n(z)| |dz| \rightarrow 0$$

for each r , $0 < r < 1$. ■

10. Let $\{f_n\} \subset H(G)$ be a sequence of one-one functions which converge to f . If G is a region, show that either f is one-one or f is a constant function.

Solution. Assume $f_n \rightarrow f$ in $H(G)$, that is, uniformly on compact subsets of G , and each f_n is one-one on the region G . We must show that either f is one-one or f is constant.

Suppose f is not constant. We will prove that f is one-one.

Take $z_1, z_2 \in G$ and suppose

$$f(z_1) = f(z_2).$$

We want to show that $z_1 = z_2$.

Assume, to the contrary, that $z_1 \neq z_2$. Since G is a region and f is analytic and nonconstant, the zeros of

$$g(z) = f(z) - f(z_1)$$

are isolated. Thus we may choose $r > 0$ such that the closed disk

$$\overline{B(z_1, r)} \subset G,$$

$z_2 \notin \overline{B(z_1, r)}$, and

$$f(z) \neq f(z_1) \quad \text{for } 0 < |z - z_1| \leq r.$$

Hence z_1 is the only zero of g in $\overline{B(z_1, r)}$.

Since g has no zeros on the circle $|z - z_1| = r$, we have

$$m := \min_{|z - z_1| = r} |g(z)| > 0.$$

Now $f_n \rightarrow f$ uniformly on $\overline{B(z_1, r)}$, and also $f_n(z_2) \rightarrow f(z_2) = f(z_1)$. Therefore

$$(f_n(z) - f_n(z_2)) - (f(z) - f(z_1)) \rightarrow 0$$

uniformly on $|z - z_1| = r$. So for all sufficiently large n ,

$$\sup_{|z - z_1| = r} |(f_n(z) - f_n(z_2)) - (f(z) - f(z_1))| < m.$$

By Rouché's theorem, the functions

$$f(z) - f(z_1) \quad \text{and} \quad f_n(z) - f_n(z_2)$$

have the same number of zeros in $B(z_1, r)$, counting multiplicity. Since $f(z) - f(z_1)$ has at least one zero there (namely z_1), it follows that

$$f_n(z) - f_n(z_2)$$

has a zero in $B(z_1, r)$. That is, for some $w_n \in B(z_1, r)$,

$$f_n(w_n) = f_n(z_2).$$

But $w_n \neq z_2$, because $z_2 \notin \overline{B(z_1, r)}$. This contradicts the fact that f_n is one-one. Therefore our assumption $z_1 \neq z_2$ was impossible. Hence

$$f(z_1) = f(z_2) \implies z_1 = z_2,$$

so f is one-one.

We conclude that if f is not constant, then f is one-one. Equivalently, either f is one-one or f is constant. ■

7.3 Spaces of meromorphic functions

7.4 The Riemann Mapping Theorem

2. (a) Let G be a region, let $a \in G$ and suppose that $f : (G \setminus \{a\}) \rightarrow \mathbb{C}$ is an analytic function such that $f(G \setminus \{a\}) = \Omega$ is bounded. Show that f has a removable singularity at $z = a$. If f is one-one, show that $f(a) \in \partial\Omega$.

- (b) Show that there is no one-one analytic function which maps

$$G = \{z : 0 < |z| < 1\}$$

onto an annulus

$$\Omega = \{z : r < |z| < R\}$$

where $r > 0$.

Solution. (a) Since f is analytic on $G \setminus \{a\}$ and $f(G \setminus \{a\}) = \Omega$ is bounded, f is bounded in some punctured neighborhood of a . Hence a is a removable singularity of f . So f extends to a function, still denoted f , analytic on all of G .

Now assume in addition that f is one-one on $G \setminus \{a\}$. We show that

$$f(a) \in \partial\Omega.$$

Suppose not. Then $f(a) \in \Omega$. Since $\Omega = f(G \setminus \{a\})$, there exists $z_0 \in G \setminus \{a\}$ such that

$$f(z_0) = f(a).$$

But the extension is analytic on G , so this contradicts injectivity of f on $G \setminus \{a\}$, because $z_0 \neq a$.

It remains to rule out the possibility that $f(a) \notin \overline{\Omega}$. But a is a limit point of $G \setminus \{a\}$, so if $z_n \rightarrow a$ with $z_n \neq a$, continuity of the extended f gives

$$f(z_n) \rightarrow f(a).$$

Since each $f(z_n) \in \Omega$, it follows that $f(a) \in \overline{\Omega}$. Therefore $f(a) \in \overline{\Omega} \setminus \Omega = \partial\Omega$.

(b) Suppose, for contradiction, that there exists a one-one analytic map

$$f : G = \{z : 0 < |z| < 1\} \rightarrow \Omega = \{z : r < |z| < R\}, \quad r > 0,$$

onto Ω .

Since Ω is bounded, part (a) implies that f has a removable singularity at 0. Thus f extends analytically to the disk $D = \{z : |z| < 1\}$. Since f is one-one on G , part (a) also implies that

$$f(0) \in \partial\Omega.$$

But

$$\partial\Omega = \{z : |z| = r\} \cup \{z : |z| = R\}.$$

Now $f(D)$ is open, because f is analytic and nonconstant. Also,

$$f(D) = f(G) \cup \{f(0)\} = \Omega \cup \{f(0)\}.$$

Since $f(0) \in \partial\Omega$, the set $\Omega \cup \{f(0)\}$ is not open. This is impossible.

Therefore no such one-one analytic map can exist. ■

3. Let G be a simply connected region which is not the whole plane and suppose that $\bar{z} \in G$ whenever $z \in G$. Let $a \in G \cap \mathbb{R}$ and suppose that $f : G \rightarrow D = \{z : |z| < 1\}$ is a one-one analytic function with $f(a) = 0$, $f'(a) > 0$ and $f(G) = D$. Let

$$G_+ = \{z \in G : \Im z > 0\}.$$

Show that $f(G_+)$ must lie entirely above or entirely below the real axis.

Solution. Define

$$g(z) = \overline{f(\bar{z})}, \quad z \in G.$$

Since G is symmetric with respect to the real axis, $\bar{z} \in G$ whenever $z \in G$, so g is well-defined on G . Also, since f is analytic, g is analytic on G . Moreover, because $f(G) = D$, we also have

$$g(G) \subset D.$$

Now $a \in \mathbb{R}$, so

$$g(a) = \overline{f(a)} = 0.$$

Also,

$$g'(a) = \overline{f'(\bar{a})} = \overline{f'(a)} = f'(a) > 0,$$

since a is real and $f'(a)$ is a positive real number.

Thus $g : G \rightarrow D$ is another one-one analytic map with

$$g(a) = 0, \quad g'(a) > 0.$$

By the uniqueness part of the Riemann mapping theorem with normalization at a , it follows that

$$g = f.$$

Hence

$$f(\bar{z}) = \overline{f(z)} \quad (z \in G).$$

In particular, if $x \in G \cap \mathbb{R}$, then

$$f(x) = f(\bar{x}) = \overline{f(x)},$$

so $f(x) \in \mathbb{R}$. Therefore

$$f(G \cap \mathbb{R}) \subset \mathbb{R}.$$

Now G_+ is connected, since $G \cap \mathbb{R}$ separates G into at most two components and G_+ is one of them. Because f is continuous, $f(G_+)$ is connected. Also, f is one-one and maps $G \cap \mathbb{R}$ into $\mathbb{R}$, so no point of G_+ can map to the real axis; for if $z \in G_+$ and $f(z) \in \mathbb{R}$, then

$$f(\bar{z}) = \overline{f(z)} = f(z),$$

and since f is one-one, this would imply $z = \bar{z}$, contradicting $\Im z > 0$.

Thus

$$f(G_+) \subset D \setminus \mathbb{R}.$$

But

$$D \setminus \mathbb{R} = \{w \in D : \Im w > 0\} \cup \{w \in D : \Im w < 0\},$$

the union of two disjoint connected components (the upper and lower half-disks). Since $f(G_+)$ is connected, it must lie entirely in one of these two components.

Therefore $f(G_+)$ lies entirely above or entirely below the real axis. ■

5. Let f be analytic on $G = \{z : \Re z > 0\}$, one-one, with $\Re f(z) > 0$ for all $z \in G$, and $f(a) = a$ for some real number a . Show that $|f'(a)| \leq 1$.

Solution. Let

$$G = \{z : \Re z > 0\}.$$

Since $f : G \rightarrow G$ is analytic and one-one, and $f(a) = a$ for some real number $a > 0$, we want to show that

$$|f'(a)| \leq 1.$$

Consider the Möbius map

$$\phi(z) = \frac{z - a}{z + a}.$$

Since $a > 0$, ϕ maps the right half-plane G conformally onto the unit disk D , and

$$\phi(a) = 0.$$

Now define

$$g = \phi \circ f \circ \phi^{-1}.$$

Then $g : D \rightarrow D$ is analytic and one-one, and

$$g(0) = \phi(f(a)) = \phi(a) = 0.$$

By Schwarz's lemma,

$$|g'(0)| \leq 1.$$

We now compute $g'(0)$. By the chain rule,

$$g'(0) = \phi'(f(a)) f'(a) (\phi^{-1})'(0).$$

Since $f(a) = a$, this becomes

$$g'(0) = \phi'(a) f'(a) (\phi^{-1})'(0).$$

Now

$$\phi'(z) = \frac{(z + a) - (z - a)}{(z + a)^2} = \frac{2a}{(z + a)^2},$$

so

$$\phi'(a) = \frac{2a}{(2a)^2} = \frac{1}{2a}.$$

Also, since ϕ^{-1} is the inverse of ϕ ,

$$(\phi^{-1})'(0) = \frac{1}{\phi'(a)} = 2a.$$

Therefore

$$g'(0) = \frac{1}{2a} f'(a) (2a) = f'(a).$$

Hence

$$|f'(a)| = |g'(0)| \leq 1.$$

Thus

$$|f'(a)| \leq 1.$$

■

7.5 The Weierstrass Factorization Theorem

3. Let f and g be analytic functions on a region G and show that there are analytic functions f_1 , g_1 , and h on G such that $f(z) = h(z)f_1(z)$ and $g(z) = h(z)g_1(z)$ for all z in G ; and f_1 and g_1 have no common zeros.

Solution. We consider first the trivial cases. If $f \equiv g \equiv 0$, take

$$h \equiv 0, \quad f_1 \equiv 1, \quad g_1 \equiv 1.$$

If $f \equiv 0$ and $g \not\equiv 0$, take

$$h = g, \quad f_1 \equiv 0, \quad g_1 \equiv 1.$$

Similarly, if $g \equiv 0$ and $f \not\equiv 0$, take

$$h = f, \quad f_1 \equiv 1, \quad g_1 \equiv 0.$$

So we may assume that neither f nor g is identically zero.

Let

$$E = \{a \in G : f(a) = 0 \text{ and } g(a) = 0\}.$$

Since zeros of a nonzero analytic function are isolated, the set E has no limit point in G . For each $a \in E$, define

$$m(a) = \min(\text{ord}_a(f), \text{ord}_a(g)).$$

By the theorem on existence of an analytic function with prescribed zeros and multiplicities, there exists an analytic function h on G whose zeros are exactly the points of E , and such that for each $a \in E$,

$$\text{ord}_a(h) = m(a).$$

Now define

$$f_1(z) = \frac{f(z)}{h(z)}, \quad g_1(z) = \frac{g(z)}{h(z)}$$

at points where $h(z) \neq 0$. We show that both quotients extend analytically across the zeros of h .

Fix $a \in E$. Write locally

$$f(z) = (z - a)^p u(z), \quad g(z) = (z - a)^q v(z), \quad h(z) = (z - a)^m w(z),$$

where u, v, w are analytic near a , with

$$u(a) \neq 0, \quad v(a) \neq 0, \quad w(a) \neq 0,$$

and

$$p = \text{ord}_a(f), \quad q = \text{ord}_a(g), \quad m = \min(p, q).$$

Then

$$\frac{f(z)}{h(z)} = (z - a)^{p-m} \frac{u(z)}{w(z)}, \quad \frac{g(z)}{h(z)} = (z - a)^{q-m} \frac{v(z)}{w(z)}.$$

Since $p - m \geq 0$ and $q - m \geq 0$, both expressions are analytic near a . Hence the singularities at the zeros of h are removable, so f_1 and g_1 extend to analytic functions on all of G . By construction,

$$f = hf_1, \quad g = hg_1 \quad \text{on } G.$$

It remains to show that f_1 and g_1 have no common zeros. Suppose, to the contrary, that for some $a \in G$,

$$f_1(a) = g_1(a) = 0.$$

Then $f(a) = g(a) = 0$, so $a \in E$. Also,

$$\text{ord}_a(f) = \text{ord}_a(h) + \text{ord}_a(f_1), \quad \text{ord}_a(g) = \text{ord}_a(h) + \text{ord}_a(g_1).$$

Since $f_1(a) = g_1(a) = 0$, we have $\text{ord}_a(f_1) \geq 1$ and $\text{ord}_a(g_1) \geq 1$. Therefore,

$$\min(\text{ord}_a(f), \text{ord}_a(g)) \geq \text{ord}_a(h) + 1,$$

which contradicts the definition

$$\text{ord}_a(h) = \min(\text{ord}_a(f), \text{ord}_a(g)).$$

Thus f_1 and g_1 have no common zeros.

Therefore there exist analytic functions f_1 , g_1 , and h on G such that

$$f = hf_1, \quad g = hg_1,$$

and f_1 and g_1 have no common zeros. ■

6. *Discuss the convergence of the infinite products*

$$\prod \left(1 + \frac{i}{n}\right) \quad \text{and} \quad \prod \left|1 + \frac{i}{n}\right|.$$

Solution. Let

$$P_N = \prod_{n=1}^N \left(1 + \frac{i}{n}\right).$$

Write each factor in polar form:

$$1 + \frac{i}{n} = \left|1 + \frac{i}{n}\right| e^{i \arg(1+i/n)}.$$

Hence

$$P_N = \left(\prod_{n=1}^N \left|1 + \frac{i}{n}\right| \right) \exp \left(i \sum_{n=1}^N \arg \left(1 + \frac{i}{n}\right) \right).$$

First consider the moduli:

$$\left|1 + \frac{i}{n}\right| = \sqrt{1 + \frac{1}{n^2}},$$

so

$$\log \left|1 + \frac{i}{n}\right| = \frac{1}{2} \log \left(1 + \frac{1}{n^2}\right) \sim \frac{1}{2n^2}.$$

Since $\sum \frac{1}{n^2}$ converges, it follows that

$$\sum_{n=1}^{\infty} \log \left|1 + \frac{i}{n}\right|$$

converges, and therefore

$$\prod_{n=1}^{\infty} \left|1 + \frac{i}{n}\right|$$

converges to a finite positive limit.

Next consider the arguments:

$$\arg \left(1 + \frac{i}{n}\right) = \arctan \left(\frac{1}{n}\right) \sim \frac{1}{n}.$$

Since $\sum \frac{1}{n}$ diverges, we have

$$\sum_{n=1}^{\infty} \arg \left(1 + \frac{i}{n} \right) = \infty.$$

Thus the arguments of the partial products P_N are unbounded and do not converge modulo 2π .

Therefore, although the moduli $|P_N|$ converge, the sequence P_N does not converge in $\mathbb{C}$.

In conclusion,

$$\prod_{n=1}^{\infty} \left| 1 + \frac{i}{n} \right| \text{ converges,} \quad \prod_{n=1}^{\infty} \left(1 + \frac{i}{n} \right) \text{ diverges.}$$

■

9. Use Theorem 5.15 to show there is an analytic function f on $D = \{z : |z| < 1\}$ which is not analytic on any region G which properly contains D .

Solution. Let

$$A = \left\{ \left(1 - \frac{1}{n} \right) e^{2\pi i k/n} : n \geq 2, k = 0, 1, \dots, n-1 \right\}.$$

Then $A \subset D = \{z : |z| < 1\}$, the points of A are distinct, and A has no limit point in D ; all its limit points lie on the unit circle $|z| = 1$.

By Theorem 5.15 (with multiplicities $m_j = 1$), there exists an analytic function f on D whose only zeros are the points of A .

Suppose, for contradiction, that f is analytic on a region G which properly contains D . Then there exists $w \in G \setminus D$, so $|w| \geq 1$. Since G is a region, it is path-connected, hence there exists a continuous path in G from 0 to w . By continuity, this path meets the unit circle at some point $\zeta \in G$ with $|\zeta| = 1$.

Since every point of $|z| = 1$ is a limit point of A , there exists a sequence of zeros of f in D converging to ζ . Because $\zeta \in G$ and f is analytic on G , the Identity Theorem implies that $f \equiv 0$ on G , a contradiction.

Therefore, f is not analytic on any region G which properly contains D . ■

13. Find an entire function f such that $f(m + in) = 0$ for all possible integers m, n . Find the most elementary solution possible.

Solution. Let

$$\Lambda = \{m + in : m, n \in \mathbb{Z}\}$$

be the Gaussian integer lattice. We construct an entire function whose zeros are

exactly the points of Λ .

Observe that Λ is invariant under the transformations

$$\omega \mapsto -\omega, \quad \omega \mapsto i\omega.$$

Thus every nonzero lattice point belongs to a set of the form

$$\{\omega, -\omega, i\omega, -i\omega\}.$$

Let S be a set containing exactly one representative from each such set in $\Lambda \setminus \{0\}$.

Define

$$f(z) = z \prod_{\omega \in S} \left(1 - \frac{z^4}{\omega^4}\right).$$

We first show that f is entire. For $|z| \leq R$,

$$\left|\frac{z^4}{\omega^4}\right| \leq \frac{R^4}{|\omega|^4}.$$

Since

$$\sum_{\omega \in S} \frac{1}{|\omega|^4} < \infty,$$

the infinite product converges uniformly on compact sets. Hence f is entire.

Now we verify the zeros. Clearly $f(0) = 0$. Let $\lambda \in \Lambda \setminus \{0\}$. Then λ belongs to exactly one set

$$\{\omega, -\omega, i\omega, -i\omega\}$$

for some $\omega \in S$, so $\lambda^4 = \omega^4$. Therefore

$$1 - \frac{\lambda^4}{\omega^4} = 0,$$

which implies $f(\lambda) = 0$.

Thus $f(m + in) = 0$ for all integers m, n , as required. ■

7.6 Factorization of the sine function

1. *Show that*

$$\cos \pi z = \prod_{n=1}^{\infty} \left[1 - \frac{4z^2}{(2n-1)^2}\right].$$

Chapter 8

Runge's Theorem

8.1 Runge's Theorem

2. Let G be the open unit disk $B(0; 1)$ and let $K = \{z : \frac{1}{4} \leq |z| \leq \frac{3}{4}\}$. Show that there is a function f analytic on some open subset G_1 containing K which cannot be approximated on K by functions in $H(G)$.

8.2 Simple connectedness

1. The set

$$G = \{re^{it} : -\infty < t < 0 \text{ and } 1 + e^t < r < 1 + 2e^t\}$$

is called a cornucopia. Show that G is simply connected. Let $K = \overline{G}$; is $\hat{K}$ connected?

Chapter 9

Analytic Continuation and Riemann Surfaces

Chapter 10

Harmonic Functions

10.1 Basic properties of harmonic functions

4. Prove that a nonconstant harmonic function on a region is an open map. (Hint: Use the fact that the connected subsets of $\mathbb{R}$ are intervals.)

5. If f is analytic on G and $f(z) \neq 0$ for any z show that

$$u = \log |f|$$

is harmonic on G .

6. Let u be harmonic in G and suppose $\overline{B}(a; R) \subseteq G$. Show that

$$u(a) = \frac{1}{\pi R^2} \iint_{\overline{B}(a; R)} u(x, y) \, dx \, dy.$$

7. For $|z| < 1$ let $u(z) = \Im \left[\left(\frac{1+z}{1-z} \right)^2 \right]$. Show that u is harmonic and $\lim_{r \rightarrow 1^-} u(re^{i\theta}) = 0$ for all θ . Does this violate Theorem 1.7? Why?

9. Let $u : G \rightarrow \mathbb{R}$ be harmonic and let $A = \{z \in G : u_x(z) = u_y(z) = 0\}$; that is, A is the set of zeros of the gradient of u . Can A have a limit point in G ?

10.2 Harmonic functions on a disk

2. In the statement of Theorem 2.4 suppose that f is piecewise continuous on ∂D . Is the conclusion of the theorem still valid? If not, what parts of the conclusion remain true?

4. Let G be a simply connected region and let Γ be its closure in $\mathbb{C}_\infty$; $\partial_\infty G = \Gamma - G$. Suppose there is a homeomorphism φ of Γ onto $\overline{D}$ ($D = \{z : |z| < 1\}$) such that φ is analytic on G .

- (a) Show that $\varphi(G) = D$ and $\varphi(\partial_\infty G) = \partial D$.
- (b) Show that if $f : \partial_\infty G \rightarrow \mathbb{R}$ is a continuous function then there is a continuous function $u : \Gamma \rightarrow \mathbb{R}$ such that $u(z) = f(z)$ for $z \in \partial_\infty G$ and u is harmonic in G .
- (c) Suppose that the function f in part (b) is not assumed to be continuous at ∞ . Show that there is a continuous function $u : G^- \rightarrow \mathbb{R}$ such that $u(z) = f(z)$ for $z \in \partial G$ and u is harmonic in G (see Exercise 2).